

Body Action Model

5.4

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1 Module Index	1
1.1 Modules	1
2 Namespace Index	3
2.1 Namespace List	3
3 Hierarchical Index	5
3.1 Class Hierarchy	5
4 Data Structure Index	7
4.1 Data Structures	7
5 File Index	9
5.1 File List	9
6 Module Documentation	13
6.1 Models	13
6.1.1 Detailed Description	13
6.2 Dynamics	13
6.2.1 Detailed Description	13
6.3 BodyAction	13
6.3.1 Detailed Description	15
7 Namespace Documentation	17
7.1 jeod Namespace Reference	17
7.1.1 Detailed Description	18
8 Data Structure Documentation	19
8.1 jeod::BodyAction Class Reference	19
8.1.1 Detailed Description	20
8.1.2 Constructor & Destructor Documentation	21
8.1.2.1 BodyAction() [1/2]	21
8.1.2.2 ~BodyAction()	21
8.1.2.3 BodyAction() [2/2]	21
8.1.3 Member Function Documentation	21
8.1.3.1 apply()	21
8.1.3.2 get_identifier()	22
8.1.3.3 get_subject_dyn_body()	22
8.1.3.4 initialize()	22
8.1.3.5 is_ready()	23
8.1.3.6 is_same_subject_body()	23
8.1.3.7 is_subject_dyn_body()	23
8.1.3.8 JEOD_CLASS_ESTABLISH_FRIENDS2()	24
8.1.3.9 operator=()	24
8.1.3.10 set_subject_body() [1/2]	24

8.1.3.11 set_subject_body() [2/2]	24
8.1.3.12 shutdown()	24
8.1.3.13 validate_body_inputs()	25
8.1.3.14 validate_name()	25
8.1.4 Field Documentation	25
8.1.4.1 action_identifier	25
8.1.4.2 action_name	26
8.1.4.3 active	26
8.1.4.4 dyn_subject	27
8.1.4.5 mass_subject	27
8.1.4.6 terminate_on_error	27
8.2 jeod::BodyActionMessages Class Reference	28
8.2.1 Detailed Description	28
8.2.2 Constructor & Destructor Documentation	28
8.2.2.1 BodyActionMessages() [1/2]	29
8.2.2.2 BodyActionMessages() [2/2]	29
8.2.3 Member Function Documentation	29
8.2.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()	29
8.2.3.2 operator=()	29
8.2.4 Field Documentation	29
8.2.4.1 fatal_error	29
8.2.4.2 illegal_value	30
8.2.4.3 invalid_name	30
8.2.4.4 invalid_object	30
8.2.4.5 not_performed	31
8.2.4.6 null_pointer	31
8.2.4.7 trace	31
8.3 jeod::BodyAttach Class Reference	32
8.3.1 Detailed Description	33
8.3.2 Constructor & Destructor Documentation	33
8.3.2.1 BodyAttach() [1/2]	33
8.3.2.2 ~BodyAttach()	33
8.3.2.3 BodyAttach() [2/2]	33
8.3.3 Member Function Documentation	33
8.3.3.1 apply()	33
8.3.3.2 initialize()	34
8.3.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()	34
8.3.3.4 operator=()	35
8.3.3.5 set_parent_body() [1/2]	35
8.3.3.6 set_parent_body() [2/2]	35
8.3.3.7 set_parent_frame()	35
8.3.4 Field Documentation	36

8.3.4.1 dyn_parent	36
8.3.4.2 mass_parent	36
8.3.4.3 ref_parent	36
8.3.4.4 succeeded	37
8.4 jeod::BodyAttachAligned Class Reference	37
8.4.1 Detailed Description	38
8.4.2 Constructor & Destructor Documentation	38
8.4.2.1 BodyAttachAligned() [1/2]	38
8.4.2.2 ~BodyAttachAligned()	38
8.4.2.3 BodyAttachAligned() [2/2]	38
8.4.3 Member Function Documentation	38
8.4.3.1 apply()	38
8.4.3.2 initialize()	39
8.4.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()	39
8.4.3.4 operator=()	39
8.4.4 Field Documentation	39
8.4.4.1 parent_point_name	40
8.4.4.2 subject_point_name	40
8.5 jeod::BodyAttachMatrix Class Reference	40
8.5.1 Detailed Description	41
8.5.2 Constructor & Destructor Documentation	41
8.5.2.1 BodyAttachMatrix() [1/2]	41
8.5.2.2 ~BodyAttachMatrix()	42
8.5.2.3 BodyAttachMatrix() [2/2]	42
8.5.3 Member Function Documentation	42
8.5.3.1 apply()	42
8.5.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	42
8.5.3.3 operator=()	42
8.5.4 Field Documentation	43
8.5.4.1 offset_pstr_cstr_pstr	43
8.5.4.2 pstr_cstr	43
8.6 jeod::BodyDetach Class Reference	43
8.6.1 Detailed Description	44
8.6.2 Constructor & Destructor Documentation	44
8.6.2.1 BodyDetach() [1/2]	44
8.6.2.2 ~BodyDetach()	45
8.6.2.3 BodyDetach() [2/2]	45
8.6.3 Member Function Documentation	45
8.6.3.1 apply()	45
8.6.3.2 is_ready()	45
8.6.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()	46
8.6.3.4 operator=()	46

8.7 jeod::BodyDetachSpecific Class Reference	46
8.7.1 Detailed Description	47
8.7.2 Constructor & Destructor Documentation	47
8.7.2.1 BodyDetachSpecific() [1/2]	47
8.7.2.2 ~BodyDetachSpecific()	47
8.7.2.3 BodyDetachSpecific() [2/2]	48
8.7.3 Member Function Documentation	48
8.7.3.1 apply()	48
8.7.3.2 initialize()	48
8.7.3.3 is_ready()	49
8.7.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()	49
8.7.3.5 operator=()	49
8.7.3.6 set_detach_from_body() [1/2]	49
8.7.3.7 set_detach_from_body() [2/2]	50
8.7.4 Field Documentation	50
8.7.4.1 dyn_detach_from	50
8.7.4.2 mass_detach_from	50
8.8 jeod::BodyReattach Class Reference	51
8.8.1 Detailed Description	51
8.8.2 Constructor & Destructor Documentation	52
8.8.2.1 BodyReattach() [1/2]	52
8.8.2.2 ~BodyReattach()	52
8.8.2.3 BodyReattach() [2/2]	52
8.8.3 Member Function Documentation	52
8.8.3.1 apply()	52
8.8.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	53
8.8.3.3 operator=()	53
8.8.4 Field Documentation	53
8.8.4.1 offset_pstr_cstr_pstr	53
8.8.4.2 pstr_cstr	53
8.9 jeod::DynBodyFrameSwitch Class Reference	54
8.9.1 Detailed Description	55
8.9.2 Member Enumeration Documentation	55
8.9.2.1 SwitchSense	55
8.9.3 Constructor & Destructor Documentation	55
8.9.3.1 DynBodyFrameSwitch() [1/2]	55
8.9.3.2 ~DynBodyFrameSwitch()	55
8.9.3.3 DynBodyFrameSwitch() [2/2]	56
8.9.4 Member Function Documentation	56
8.9.4.1 apply()	56
8.9.4.2 initialize()	56
8.9.4.3 is_ready()	57

8.9.4.4 JEOD_CLASS_ESTABLISH_FRIENDS2()	57
8.9.4.5 operator=()	57
8.9.5 Field Documentation	57
8.9.5.1 integ_frame	57
8.9.5.2 integ_frame_name	58
8.9.5.3 sort_grav_controls	58
8.9.5.4 switch_distance	58
8.9.5.5 switch_sense	58
8.10 jeod::DynBodyInit Class Reference	59
8.10.1 Detailed Description	60
8.10.2 Constructor & Destructor Documentation	60
8.10.2.1 DynBodyInit() [1/2]	61
8.10.2.2 ~DynBodyInit()	61
8.10.2.3 DynBodyInit() [2/2]	61
8.10.3 Member Function Documentation	61
8.10.3.1 apply()	61
8.10.3.2 apply_user_inputs()	62
8.10.3.3 compute_rotational_state()	62
8.10.3.4 compute_translational_state()	62
8.10.3.5 find_body_frame()	62
8.10.3.6 find_dyn_body()	63
8.10.3.7 find_planet()	63
8.10.3.8 find_ref_frame()	64
8.10.3.9 initialize()	64
8.10.3.10 initializes_what()	65
8.10.3.11 is_ready()	66
8.10.3.12 JEOD_CLASS_ESTABLISH_FRIENDS2()	66
8.10.3.13 operator=()	66
8.10.3.14 report_failure()	66
8.10.4 Field Documentation	67
8.10.4.1 ang_velocity	67
8.10.4.2 body_frame_id	67
8.10.4.3 body_ref_frame	68
8.10.4.4 orientation	68
8.10.4.5 position	68
8.10.4.6 rate_in_parent	68
8.10.4.7 reference_ref_frame	69
8.10.4.8 reference_ref_frame_name	69
8.10.4.9 reverse_sense	69
8.10.4.10 state	70
8.10.4.11 subscribed_frame	70
8.10.4.12 velocity	70

8.11 jeod::DynBodyInitLvLhRotState Class Reference	71
8.11.1 Detailed Description	71
8.11.2 Constructor & Destructor Documentation	71
8.11.2.1 DynBodyInitLvLhRotState() [1/2]	72
8.11.2.2 ~DynBodyInitLvLhRotState()	72
8.11.2.3 DynBodyInitLvLhRotState() [2/2]	72
8.11.3 Member Function Documentation	72
8.11.3.1 initialize()	72
8.11.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	73
8.11.3.3 operator=()	73
8.12 jeod::DynBodyInitLvLhState Class Reference	73
8.12.1 Detailed Description	74
8.12.2 Constructor & Destructor Documentation	74
8.12.2.1 DynBodyInitLvLhState() [1/2]	74
8.12.2.2 ~DynBodyInitLvLhState()	74
8.12.2.3 DynBodyInitLvLhState() [2/2]	75
8.12.3 Member Function Documentation	75
8.12.3.1 apply()	75
8.12.3.2 initialize()	75
8.12.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()	76
8.12.3.4 operator=()	76
8.12.3.5 set_lvLh_frame_object()	76
8.12.4 Field Documentation	76
8.12.4.1 lvLh_object_ptr	76
8.12.4.2 lvLh_type	77
8.13 jeod::DynBodyInitLvLhTransState Class Reference	77
8.13.1 Detailed Description	78
8.13.2 Constructor & Destructor Documentation	78
8.13.2.1 DynBodyInitLvLhTransState() [1/2]	78
8.13.2.2 ~DynBodyInitLvLhTransState()	78
8.13.2.3 DynBodyInitLvLhTransState() [2/2]	78
8.13.3 Member Function Documentation	78
8.13.3.1 initialize()	78
8.13.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	79
8.13.3.3 operator=()	79
8.14 jeod::DynBodyInitNedRotState Class Reference	79
8.14.1 Detailed Description	80
8.14.2 Constructor & Destructor Documentation	80
8.14.2.1 DynBodyInitNedRotState() [1/2]	80
8.14.2.2 ~DynBodyInitNedRotState()	80
8.14.2.3 DynBodyInitNedRotState() [2/2]	80
8.14.3 Member Function Documentation	81

8.14.3.1 initialize()	81
8.14.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	82
8.14.3.3 operator=()	82
8.15 jeod::DynBodyInitNedState Class Reference	82
8.15.1 Detailed Description	83
8.15.2 Constructor & Destructor Documentation	83
8.15.2.1 DynBodyInitNedState() [1/2]	84
8.15.2.2 ~DynBodyInitNedState()	84
8.15.2.3 DynBodyInitNedState() [2/2]	84
8.15.3 Member Function Documentation	84
8.15.3.1 apply()	84
8.15.3.2 initialize()	85
8.15.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()	85
8.15.3.4 operator=()	85
8.15.3.5 set_use_alt_pfix()	85
8.15.4 Field Documentation	86
8.15.4.1 altlatlong_type	86
8.15.4.2 pfix_ptr	86
8.15.4.3 ref_point	86
8.15.4.4 use_alt_pfix	87
8.16 jeod::DynBodyInitNedTransState Class Reference	87
8.16.1 Detailed Description	88
8.16.2 Constructor & Destructor Documentation	88
8.16.2.1 DynBodyInitNedTransState() [1/2]	88
8.16.2.2 ~DynBodyInitNedTransState()	88
8.16.2.3 DynBodyInitNedTransState() [2/2]	88
8.16.3 Member Function Documentation	88
8.16.3.1 initialize()	88
8.16.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()	89
8.16.3.3 operator=()	89
8.17 jeod::DynBodyInitOrbit Class Reference	89
8.17.1 Detailed Description	91
8.17.2 Member Enumeration Documentation	91
8.17.2.1 OrbitalSet	91
8.17.3 Constructor & Destructor Documentation	92
8.17.3.1 DynBodyInitOrbit() [1/2]	92
8.17.3.2 ~DynBodyInitOrbit()	92
8.17.3.3 DynBodyInitOrbit() [2/2]	92
8.17.4 Member Function Documentation	92
8.17.4.1 apply()	92
8.17.4.2 initialize()	93
8.17.4.3 JEOD_CLASS_ESTABLISH_FRIENDS2()	93

8.17.4.4 operator=()	93
8.17.5 Field Documentation	93
8.17.5.1 alt_apoapsis	94
8.17.5.2 alt_periapsis	94
8.17.5.3 arg_latitude	94
8.17.5.4 arg_periapsis	94
8.17.5.5 ascending_node	95
8.17.5.6 eccentricity	95
8.17.5.7 inclination	95
8.17.5.8 mean_anomaly	95
8.17.5.9 orb_radius	96
8.17.5.10 orbit_frame	96
8.17.5.11 orbit_frame_name	96
8.17.5.12 planet	96
8.17.5.13 planet_name	97
8.17.5.14 radial_vel	97
8.17.5.15 semi_latus_rectum	97
8.17.5.16 semi_major_axis	97
8.17.5.17 set	98
8.17.5.18 time_periapsis	98
8.17.5.19 true_anomaly	98
8.18 jeod::DynBodyInitPlanetDerived Class Reference	99
8.18.1 Detailed Description	100
8.18.2 Constructor & Destructor Documentation	100
8.18.2.1 DynBodyInitPlanetDerived() [1/2]	100
8.18.2.2 ~DynBodyInitPlanetDerived()	100
8.18.2.3 DynBodyInitPlanetDerived() [2/2]	100
8.18.3 Member Function Documentation	100
8.18.3.1 apply()	100
8.18.3.2 initialize()	101
8.18.3.3 is_ready()	101
8.18.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()	102
8.18.3.5 operator=()	102
8.18.4 Field Documentation	102
8.18.4.1 body_is_required	102
8.18.4.2 ref_body	102
8.18.4.3 ref_body_name	103
8.18.4.4 required_items	103
8.19 jeod::DynBodyInitRotState Class Reference	103
8.19.1 Detailed Description	104
8.19.2 Member Enumeration Documentation	104
8.19.2.1 StateItems	104

8.19.3 Constructor & Destructor Documentation	105
8.19.3.1 DynBodyInitRotState() [1/2]	105
8.19.3.2 ~DynBodyInitRotState()	105
8.19.3.3 DynBodyInitRotState() [2/2]	105
8.19.4 Member Function Documentation	105
8.19.4.1 apply()	105
8.19.4.2 initialize()	106
8.19.4.3 initializes_what()	106
8.19.4.4 is_ready()	107
8.19.4.5 JEOD_CLASS_ESTABLISH_FRIENDS2()	107
8.19.4.6 operator=()	107
8.19.5 Field Documentation	107
8.19.5.1 state_items	107
8.20 jeod::DynBodyInitTransState Class Reference	108
8.20.1 Detailed Description	109
8.20.2 Member Enumeration Documentation	109
8.20.2.1 StateItems	109
8.20.3 Constructor & Destructor Documentation	109
8.20.3.1 DynBodyInitTransState() [1/2]	109
8.20.3.2 ~DynBodyInitTransState()	109
8.20.3.3 DynBodyInitTransState() [2/2]	109
8.20.4 Member Function Documentation	110
8.20.4.1 apply()	110
8.20.4.2 initialize()	110
8.20.4.3 initializes_what()	111
8.20.4.4 is_ready()	111
8.20.4.5 JEOD_CLASS_ESTABLISH_FRIENDS2()	111
8.20.4.6 operator=()	112
8.20.5 Field Documentation	112
8.20.5.1 state_items	112
8.21 jeod::DynBodyInitWrtPlanet Class Reference	112
8.21.1 Detailed Description	113
8.21.2 Constructor & Destructor Documentation	113
8.21.2.1 DynBodyInitWrtPlanet() [1/2]	113
8.21.2.2 ~DynBodyInitWrtPlanet()	114
8.21.2.3 DynBodyInitWrtPlanet() [2/2]	114
8.21.3 Member Function Documentation	114
8.21.3.1 apply()	114
8.21.3.2 initialize()	114
8.21.3.3 initializes_what()	115
8.21.3.4 is_ready()	115
8.21.3.5 JEOD_CLASS_ESTABLISH_FRIENDS2()	116

8.21.3.6 operator=()	116
8.21.4 Field Documentation	116
8.21.4.1 planet	116
8.21.4.2 planet_name	116
8.21.4.3 set_items	117
8.22 jeod::MassBodyInit Class Reference	117
8.22.1 Detailed Description	118
8.22.2 Constructor & Destructor Documentation	118
8.22.2.1 MassBodyInit() [1/2]	118
8.22.2.2 ~MassBodyInit()	118
8.22.2.3 MassBodyInit() [2/2]	119
8.22.3 Member Function Documentation	119
8.22.3.1 allocate_points()	119
8.22.3.2 apply()	119
8.22.3.3 get_mass_point()	120
8.22.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()	120
8.22.3.5 operator=()	120
8.22.4 Field Documentation	120
8.22.4.1 points	120
8.22.4.2 properties	121
9 File Documentation	123
9.1 body_action.cc File Reference	123
9.1.1 Detailed Description	123
9.2 body_action.hh File Reference	123
9.2.1 Detailed Description	124
9.3 body_action_messages.cc File Reference	124
9.3.1 Detailed Description	124
9.3.2 Macro Definition Documentation	124
9.3.2.1 MAKE_BODYACTION_MESSAGE_CODE	124
9.4 body_action_messages.hh File Reference	125
9.4.1 Detailed Description	125
9.5 body_attach.cc File Reference	125
9.5.1 Detailed Description	125
9.6 body_attach.hh File Reference	126
9.6.1 Detailed Description	126
9.7 body_attach_aligned.cc File Reference	126
9.7.1 Detailed Description	127
9.8 body_attach_aligned.hh File Reference	127
9.8.1 Detailed Description	127
9.9 body_attach_matrix.cc File Reference	127
9.9.1 Detailed Description	127

9.10 body_attach_matrix.hh File Reference	128
9.10.1 Detailed Description	128
9.11 body_detach.cc File Reference	128
9.11.1 Detailed Description	128
9.12 body_detach.hh File Reference	129
9.12.1 Detailed Description	129
9.13 body_detach_specific.cc File Reference	129
9.13.1 Detailed Description	129
9.14 body_detach_specific.hh File Reference	130
9.14.1 Detailed Description	130
9.15 body_reattach.cc File Reference	130
9.15.1 Detailed Description	130
9.16 body_reattach.hh File Reference	131
9.16.1 Detailed Description	131
9.17 class_declarations.hh File Reference	131
9.17.1 Detailed Description	131
9.18 dyn_body_frame_switch.cc File Reference	132
9.18.1 Detailed Description	132
9.19 dyn_body_frame_switch.hh File Reference	132
9.19.1 Detailed Description	133
9.20 dyn_body_init.cc File Reference	133
9.20.1 Detailed Description	133
9.21 dyn_body_init.hh File Reference	133
9.21.1 Detailed Description	134
9.22 dyn_body_init_lvlh_rot_state.cc File Reference	134
9.22.1 Detailed Description	134
9.23 dyn_body_init_lvlh_rot_state.hh File Reference	134
9.23.1 Detailed Description	135
9.24 dyn_body_init_lvlh_state.cc File Reference	135
9.24.1 Detailed Description	135
9.25 dyn_body_init_lvlh_state.hh File Reference	135
9.25.1 Detailed Description	136
9.26 dyn_body_init_lvlh_trans_state.cc File Reference	136
9.26.1 Detailed Description	136
9.27 dyn_body_init_lvlh_trans_state.hh File Reference	136
9.27.1 Detailed Description	137
9.28 dyn_body_init_ned_rot_state.cc File Reference	137
9.28.1 Detailed Description	137
9.29 dyn_body_init_ned_rot_state.hh File Reference	137
9.29.1 Detailed Description	138
9.30 dyn_body_init_ned_state.cc File Reference	138
9.30.1 Detailed Description	138

9.31 dyn_body_init_ned_state.hh File Reference	139
9.31.1 Detailed Description	139
9.32 dyn_body_init_ned_trans_state.cc File Reference	139
9.32.1 Detailed Description	140
9.33 dyn_body_init_ned_trans_state.hh File Reference	140
9.33.1 Detailed Description	140
9.34 dyn_body_init_orbit.cc File Reference	140
9.34.1 Detailed Description	141
9.35 dyn_body_init_orbit.hh File Reference	141
9.35.1 Detailed Description	141
9.36 dyn_body_init_planet_derived.cc File Reference	141
9.36.1 Detailed Description	142
9.37 dyn_body_init_planet_derived.hh File Reference	142
9.37.1 Detailed Description	142
9.38 dyn_body_init_rot_state.cc File Reference	142
9.38.1 Detailed Description	143
9.39 dyn_body_init_rot_state.hh File Reference	143
9.39.1 Detailed Description	143
9.40 dyn_body_init_trans_state.cc File Reference	143
9.40.1 Detailed Description	144
9.41 dyn_body_init_trans_state.hh File Reference	144
9.41.1 Detailed Description	144
9.42 dyn_body_init_wrt_planet.cc File Reference	144
9.42.1 Detailed Description	145
9.43 dyn_body_init_wrt_planet.hh File Reference	145
9.43.1 Detailed Description	145
9.44 mass_body_init.cc File Reference	145
9.44.1 Detailed Description	146
9.45 mass_body_init.hh File Reference	146
9.45.1 Detailed Description	146

Index	147
--------------	------------

Chapter 1

Module Index

1.1 Modules

Here is a list of all modules:

Models	13
Dynamics	13
BodyAction	13

Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

jeod	Namespace jeod	17
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Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

jeod::BodyAction	19
jeod::BodyAttach	32
jeod::BodyAttachAligned	37
jeod::BodyAttachMatrix	40
jeod::BodyDetach	43
jeod::BodyDetachSpecific	46
jeod::BodyReattach	51
jeod::DynBodyFrameSwitch	54
jeod::DynBodyInit	59
jeod::DynBodyInitRotState	103
jeod::DynBodyInitTransState	108
jeod::DynBodyInitOrbit	89
jeod::DynBodyInitWrtPlanet	112
jeod::DynBodyInitPlanetDerived	99
jeod::DynBodyInitLvLhState	73
jeod::DynBodyInitLvLhRotState	71
jeod::DynBodyInitLvLhTransState	77
jeod::DynBodyInitNedState	82
jeod::DynBodyInitNedRotState	79
jeod::DynBodyInitNedTransState	87
jeod::MassBodyInit	117
jeod::BodyActionMessages	28

Chapter 4

Data Structure Index

4.1 Data Structures

Here are the data structures with brief descriptions:

jeod::BodyAction	
BodyAction is the base class for the BodyAction model	19
jeod::BodyActionMessages	
Specifies the message IDs used in the BodyAction model	28
jeod::BodyAttach	
Provides the basic ability to attach one MassBody to another	32
jeod::BodyAttachAligned	
Attaches a pair of MassBody objects at a pair of MassPoints	37
jeod::BodyAttachMatrix	
Attaches a pair of MassBody objects using the offset+matrix attach mechanism	40
jeod::BodyDetach	
Provides the basic ability to detach one MassBody from another	43
jeod::BodyDetachSpecific	
Causes the subject body to detach from a specific body by severing the link immediately spawning from the detach_from body	46
jeod::BodyReattach	
Alters the nature of an existing attachment	51
jeod::DynBodyFrameSwitch	
Switch a DynBody 's integration frame to a specified frame when the body switches to that integration frame's sphere of influence	54
jeod::DynBodyInit	
Base class for initialize the state of a DynBody	59
jeod::DynBodyInitLvlhRotState	
Initialize a vehicle's rotational state with respect to some vehicle's LVLH frame	71
jeod::DynBodyInitLvlhState	
Initialize selected aspects of a vehicle's state with respect to some vehicle's LVLH frame	73
jeod::DynBodyInitLvlhTransState	
Initialize a vehicle's translational state with respect to some other vehicle's LVLH frame	77
jeod::DynBodyInitNedRotState	
Initialize a vehicle's rotational state wrt some vehicle's North-East-Down frame	79
jeod::DynBodyInitNedState	
Initialize selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet	82
jeod::DynBodyInitNedTransState	
Initialize a vehicle's translational state wrt some vehicle's North-East-Down frame	87

jeod::DynBodyInitOrbit	
Initialize a vehicle's translational state given an orbital specification	89
jeod::DynBodyInitPlanetDerived	
(Initialize selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet	99
jeod::DynBodyInitRotState	
Initialize aspects of a vehicle's rotational state	103
jeod::DynBodyInitTransState	
Initialize aspects of a vehicle's translational state	108
jeod::DynBodyInitWrtPlanet	
Initialize selected aspects of a vehicle's state with respect to some frame based on the planet .	112
jeod::MassBodyInit	
Base class for initializing a MassBody	117

Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

body_action.cc	
Define methods for the BodyAction class	123
body_action.hh	
Define the class BodyAction, the base class used for performing actions on a MassBody or DynBody object	123
body_action_messages.cc	
Implement the class BodyActionMessages	124
body_action_messages.hh	
Define the class BodyActionMessages, the class that specifies the message IDs used in the BodyAction model	125
body_attach.cc	
Define methods for the mass body initialization class	125
body_attach.hh	
Define the class MassBodyAttach, the base class used for attaching a pair of MassBody objects to one another	126
body_attach_aligned.cc	
Define methods for the mass body initialization class	126
body_attach_aligned.hh	
Define the class BodyAttachAligned, which causes one MassBody to be attached to another at a pair of MassPoints	127
body_attach_matrix.cc	
Define methods for the mass body initialization class	127
body_attach_matrix.hh	
Define the class MassBodyAttachMatrix, which causes one MassBody to be attached given a transformation	128
body_detach.cc	
Define methods for the MassBodyDetach class	128
body_detach.hh	
Define the class MassBodyDetach, the base class used for detaching one MassBody object from one another	129
body_detach_specific.cc	
Define methods for the BodyDetachSpecific class	129
body_detach_specific.hh	
Define the class MassBodyDetachSpecific, the class used for detaching one MassBody object from another specified MassBody	130

body_reattach.cc	Define methods for the mass body initialization class	130
body_reattach.hh	Define the class MassBodyReattach, which causes one MassBody to be reattached given a transformation	131
class_declarations.hh	Forward declarations of classes defined in dyn_body_init_XXX.hh files	131
dyn_body_frame_switch.cc	Define methods for the class DynBodyFrameSwitch	132
dyn_body_frame_switch.hh	Define the class DynBodyFrameSwitch, the BodyAction derived class used for switch a DynBody's integration frame	132
dyn_body_init.cc	Define methods for the base body initialization class	133
dyn_body_init.hh	Define the class DynBodyInit, the base class used for initializing the state of a DynBody object	133
dyn_body_init_lvlh_rot_state.cc	Define methods for DynBodyInitLvLhRotState	134
dyn_body_init_lvlh_rot_state.hh	Define the class DynBodyInitLvLhRotState, which initialize a vehicle's rotational state with respect to some vehicle's LVLH frame	134
dyn_body_init_lvlh_state.cc	Define methods for the DynBodyInitLvLhState class	135
dyn_body_init_lvlh_state.hh	Define the class DynBodyInitLvLhState, the base class for initializing selected aspects of a vehicle's state with respect to some vehicle's LVLH frame	135
dyn_body_init_lvlh_trans_state.cc	Define methods for DynBodyInitLvLhTransState	136
dyn_body_init_lvlh_trans_state.hh	Define the class DynBodyInitLvLhTransState, which initialize a vehicle's translational state with respect to some other vehicle's LVLH frame	136
dyn_body_init_ned_rot_state.cc	Define methods for DynBodyInitNedRotState	137
dyn_body_init_ned_rot_state.hh	Define the class DynBodyInitNedRotState, which initialize a vehicle's rotational state wrt some other vehicle's North-East-Down frame	137
dyn_body_init_ned_state.cc	Define methods for DynBodyInitNedState	138
dyn_body_init_ned_state.hh	Define the class DynBodyInitNedState, the base class for initializing selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet	139
dyn_body_init_ned_trans_state.cc	Define methods for DynBodyInitNedTransState	139
dyn_body_init_ned_trans_state.hh	Define the class DynBodyInitNedTransState, which initialize a vehicle's translational state wrt some other vehicle's North-East-Down frame	140
dyn_body_init_orbit.cc	Define classes for items represented in some ephemeris model	140
dyn_body_init_orbit.hh	Define the class DynBodyInitOrbit, which initializes a vehicle in in some orbit	141
dyn_body_init_planet_derived.cc	Define methods for the DynBodyInitPlanetDerived class	141
dyn_body_init_planet_derived.hh	Define the class DynBodyInitPlanetDerived, the base class for initializing selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet	142

dyn_body_init_rot_state.cc	
Define methods for DynBodyInitRotState	142
dyn_body_init_rot_state.hh	
Define the class DynBodyInitRotState that initialize aspects of a vehicle's rotational state	143
dyn_body_init_trans_state.cc	
Define methods for DynBodyInitTransState	143
dyn_body_init_trans_state.hh	
Define the class DynBodyInitTransState that initialize aspects of a vehicle's translational state . .	144
dyn_body_init_wrt_planet.cc	
Define methods for the DynBodyInitWrtPlanet class	144
dyn_body_init_wrt_planet.hh	
Define the class DynBodyInitWrtPlanet, the base class for initializing selected aspects of a vehicle's state with respect to some state that is connected to a planet in some way	145
mass_body_init.cc	
Define methods for the mass body initialization class	145
mass_body_init.hh	
Define the class MassBodyInit, the base class used for initializing the core mass properties of a MassBody object	146

Chapter 6

Module Documentation

6.1 Models

Modules

- [Dynamics](#)

6.1.1 Detailed Description

6.2 Dynamics

Modules

- [BodyAction](#)

6.2.1 Detailed Description

6.3 BodyAction

Files

- file [body_action.hh](#)
Define the class `BodyAction`, the base class used for performing actions on a `MassBody` or `DynBody` object.
- file [body_action_messages.hh](#)
Define the class `BodyActionMessages`, the class that specifies the message IDs used in the `BodyAction` model.
- file [body_attach.hh](#)
Define the class `MassBodyAttach`, the base class used for attaching a pair of `MassBody` objects to one another.
- file [body_attach_aligned.hh](#)
Define the class `BodyAttachAligned`, which causes one `MassBody` to be attached to another at a pair of `MassPoints`.
- file [body_attach_matrix.hh](#)
Define the class `MassBodyAttachMatrix`, which causes one `MassBody` to be attached given a transformation.
- file [body_detach.hh](#)

- Define the class *MassBodyDetach*, the base class used for detaching one *MassBody* object from one another.

 - file [body_detach_specific.hh](#)

Define the class *MassBodyDetachSpecific*, the class used for detaching one *MassBody* object from another specified *MassBody*.
 - file [body_reattach.hh](#)

Define the class *MassBodyReattach*, which causes one *MassBody* to be reattached given a transformation.
 - file [class_declarations.hh](#)

Forward declarations of classes defined in *dyn_body_init_XXX.hh* files.
 - file [dyn_body_frame_switch.hh](#)

Define the class *DynBodyFrameSwitch*, the *BodyAction* derived class used for switch a *DynBody*'s integration frame.
 - file [dyn_body_init.hh](#)

Define the class *DynBodyInit*, the base class used for initializing the state of a *DynBody* object.
 - file [dyn_body_init_lvhl_rot_state.hh](#)

Define the class *DynBodyInitLvhlRotState*, which initialize a vehicle's rotational state with respect to some vehicle's LVLH frame.
 - file [dyn_body_init_lvhl_state.hh](#)

Define the class *DynBodyInitLvhlState*, the base class for initializing selected aspects of a vehicle's state with respect to some vehicle's LVLH frame.
 - file [dyn_body_init_lvhl_trans_state.hh](#)

Define the class *DynBodyInitLvhlTransState*, which initialize a vehicle's translational state with respect to some other vehicle's LVLH frame.
 - file [dyn_body_init_ned_rot_state.hh](#)

Define the class *DynBodyInitNedRotState*, which initialize a vehicle's rotational state wrt some other vehicle's North-East-Down frame.
 - file [dyn_body_init_ned_state.hh](#)

Define the class *DynBodyInitNedState*, the base class for initializing selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet.
 - file [dyn_body_init_ned_trans_state.hh](#)

Define the class *DynBodyInitNedTransState*, which initialize a vehicle's translational state wrt some other vehicle's North-East-Down frame.
 - file [dyn_body_init_orbit.hh](#)

Define the class *DynBodyInitOrbit*, which initializes a vehicle in in some orbit.
 - file [dyn_body_init_planet_derived.hh](#)

Define the class *DynBodyInitPlanetDerived*, the base class for initializing selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet.
 - file [dyn_body_init_rot_state.hh](#)

Define the class *DynBodyInitRotState* that initialize aspects of a vehicle's rotational state.
 - file [dyn_body_init_trans_state.hh](#)

Define the class *DynBodyInitTransState* that initialize aspects of a vehicle's translational state.
 - file [dyn_body_init_wrt_planet.hh](#)

Define the class *DynBodyInitWrtPlanet*, the base class for initializing selected aspects of a vehicle's state with respect to some state that is connected to a planet in some way.
 - file [mass_body_init.hh](#)

Define the class *MassBodyInit*, the base class used for initializing the core mass properties of a *MassBody* object.
 - file [body_action.cc](#)

Define methods for the *BodyAction* class.
 - file [body_action_messages.cc](#)

Implement the class *BodyActionMessages*.
 - file [body_attach.cc](#)

Define methods for the mass body initialization class.
 - file [body_attach_aligned.cc](#)

Define methods for the mass body initialization class.
 - file [body_attach_matrix.cc](#)

- Define methods for the mass body initialization class.*

 - file [body_detach.cc](#)

Define methods for the MassBodyDetach class.
- file [body_detach_specific.cc](#)

Define methods for the BodyDetachSpecific class.
- file [body_reattach.cc](#)

Define methods for the mass body initialization class.
- file [dyn_body_frame_switch.cc](#)

Define methods for the class DynBodyFrameSwitch.
- file [dyn_body_init.cc](#)

Define methods for the base body initialization class.
- file [dyn_body_init_lvih_rot_state.cc](#)

Define methods for DynBodyInitLvihRotState.
- file [dyn_body_init_lvih_state.cc](#)

Define methods for the DynBodyInitLvihState class.
- file [dyn_body_init_lvih_trans_state.cc](#)

Define methods for DynBodyInitLvihTransState.
- file [dyn_body_init_ned_rot_state.cc](#)

Define methods for DynBodyInitNedRotState.
- file [dyn_body_init_ned_state.cc](#)

Define methods for DynBodyInitNedState.
- file [dyn_body_init_ned_trans_state.cc](#)

Define methods for DynBodyInitNedTransState.
- file [dyn_body_init_orbit.cc](#)

Define classes for items represented in some ephemeris model.
- file [dyn_body_init_planet_derived.cc](#)

Define methods for the DynBodyInitPlanetDerived class.
- file [dyn_body_init_rot_state.cc](#)

Define methods for DynBodyInitRotState.
- file [dyn_body_init_trans_state.cc](#)

Define methods for DynBodyInitTransState.
- file [dyn_body_init_wrt_planet.cc](#)

Define methods for the DynBodyInitWrtPlanet class.
- file [mass_body_init.cc](#)

Define methods for the mass body initialization class.

6.3.1 Detailed Description

Chapter 7

Namespace Documentation

7.1 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [BodyAction](#)
BodyAction is the base class for the BodyAction model.
- class [BodyActionMessages](#)
Specifies the message IDs used in the BodyAction model.
- class [BodyAttach](#)
Provides the basic ability to attach one MassBody to another.
- class [BodyAttachAligned](#)
Attaches a pair of MassBody objects at a pair of MassPoints.
- class [BodyAttachMatrix](#)
Attaches a pair of MassBody objects using the offset+matrix attach mechanism.
- class [BodyDetach](#)
Provides the basic ability to detach one MassBody from another.
- class [BodyDetachSpecific](#)
Causes the subject body to detach from a specific body by severing the link immediately spawning from the detach←→_from body.
- class [BodyReattach](#)
Alters the nature of an existing attachment.
- class [DynBodyFrameSwitch](#)
Switch a DynBody's integration frame to a specified frame when the body switches to that integration frame's sphere of influence.
- class [DynBodyInit](#)
Base class for initialize the state of a DynBody.
- class [DynBodyInitLvlhRotState](#)
Initialize a vehicle's rotational state with respect to some vehicle's LVLH frame.
- class [DynBodyInitLvlhState](#)
Initialize selected aspects of a vehicle's state with respect to some vehicle's LVLH frame.
- class [DynBodyInitLvlhTransState](#)
initialize a vehicle's translational state with respect to some other vehicle's LVLH frame.

- class [DynBodyInitNedRotState](#)
Initialize a vehicle's rotational state wrt some vehicle's North-East-Down frame.
- class [DynBodyInitNedState](#)
Initialize selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet.
- class [DynBodyInitNedTransState](#)
Initialize a vehicle's translational state wrt some vehicle's North-East-Down frame.
- class [DynBodyInitOrbit](#)
Initialize a vehicle's translational state given an orbital specification.
- class [DynBodyInitPlanetDerived](#)
(Initialize selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet.
- class [DynBodyInitRotState](#)
Initialize aspects of a vehicle's rotational state.
- class [DynBodyInitTransState](#)
Initialize aspects of a vehicle's translational state.
- class [DynBodyInitWrtPlanet](#)
Initialize selected aspects of a vehicle's state with respect to some frame based on the planet.
- class [MassBodyInit](#)
Base class for initializing a MassBody.

7.1.1 Detailed Description

Namespace jeod.

Chapter 8

Data Structure Documentation

8.1 jeod::BodyAction Class Reference

[BodyAction](#) is the base class for the [BodyAction](#) model.

```
#include <body_action.hh>
```

Inheritance diagram for jeod::BodyAction:



Public Member Functions

- void [set_subject_body](#) (MassBody &mass_body_in)
Set the subject mass body of this action.
- void [set_subject_dyn_body](#) (DynBody &dyn_body_in)
Set the subject dyn body of this action.
- bool [is_same_subject_body](#) (MassBody &mass_body_in)
Test the input mass body against the subject body and return true if they are the same body.
- bool [is_subject_dyn_body](#) ()
Check if the subject is a DynBody.
- DynBody * [get_subject_dyn_body](#) ()
- [BodyAction](#) ()=default
- virtual [~BodyAction](#) ()=default
- [BodyAction](#) (const [BodyAction](#) &)=delete
- [BodyAction](#) & operator= (const [BodyAction](#) &)=delete
- virtual void [shutdown](#) ()
Release resources allocated by a [BodyAction](#) object.
- const std::string & [get_identifier](#) () const
Accessor for action_identifier.
- virtual void [initialize](#) (DynManager &dyn_manager)
Begin initialization of a [BodyAction](#).
- virtual bool [is_ready](#) ()
In general, determine if the initializer is ready to be applied.
- virtual void [apply](#) (DynManager &dyn_manager)
Complete initialization.

Data Fields

- bool `active` {true}
Controls when the action is performed.
- bool `terminate_on_error` {true}
Indicates whether errors encountered while performing the action are to terminate the simulation.
- std::string `action_name` {""}
An identifier for this action.

Protected Member Functions

- virtual bool `validate_body_inputs` (DynBody *&dyn_body_in, MassBody *&mass_body_in, const std::string &body_base_name, bool allow_failure=false)
- void `validate_name` (const std::string &variable_value, const std::string &variable_name, const std::string &variable_type)
Ensure that a string is not trivially invalid.

Protected Attributes

- MassBody * `mass_subject` {}
The MassBody of the body that is the subject of this action.
- DynBody * `dyn_subject` {}
The DynBody of the body that is the subject of this action.
- std::string `action_identifier` {""}
An identifier for this action, constructed from the class name and the action name at initialization time.

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2` (jeod, `BodyAction`)

8.1.1 Detailed Description

`BodyAction` is the base class for the `BodyAction` model.

A `BodyAction` instance that performs some operation on a MassBody object. The simulation Dynamics Manager object manages a collection of `BodyAction` objects for the purpose of initializing MassBody objects and later, for performing asynchronous actions on them.

The `BodyAction` model hinges on three methods:

- `initialize()` The `initialize()` method initializes the `BodyAction`. This method does not and must not operate on the subject of the action. All derived classes must forward the `initialize()` call to the immediate parent class and then perform class-dependent object initializations.
- `is_ready()` The `is_ready` method indicates whether the action is ready to be applied. For example, an action that initializes the translation state of a vehicle relative to some other vehicle cannot do its job until that other vehicle's translational state is set. The `is_ready()` method for such an action should return false until the other vehicle's translational state has been set.
- `apply()` The `apply()` method applies the action – it does something to the subject of the action. All derived classes must perform class-dependent actions and then must forward the `apply()` call to the immediate parent class.

Definition at line 107 of file `body_action.hh`.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 BodyAction() [1/2]

```
jeod::BodyAction::BodyAction ( ) [default]
```

8.1.2.2 ~BodyAction()

```
virtual jeod::BodyAction::~~BodyAction ( ) [virtual], [default]
```

8.1.2.3 BodyAction() [2/2]

```
jeod::BodyAction::BodyAction (
    const BodyAction & ) [delete]
```

8.1.3 Member Function Documentation

8.1.3.1 apply()

```
void jeod::BodyAction::apply (
    DynManager & dyn_manager ) [virtual]
```

Complete initialization.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Jeod manager
----------------	--------------------	--------------

Reimplemented in [jeod::MassBodyInit](#), [jeod::DynBodyInitWrtPlanet](#), [jeod::DynBodyInitTransState](#), [jeod::DynBodyInitRotState](#), [jeod::DynBodyInitPlanetDerived](#), [jeod::DynBodyInitOrbit](#), [jeod::DynBodyInitNedState](#), [jeod::DynBodyInitLvlhState](#), [jeod::DynBodyInit](#), [jeod::DynBodyFrameSwitch](#), [jeod::BodyReattach](#), [jeod::BodyDetachSpecific](#), [jeod::BodyDetach](#), [jeod::BodyAttachMatrix](#), [jeod::BodyAttachAligned](#), and [jeod::BodyAttach](#).

Definition at line 87 of file `body_action.cc`.

References `shutdown()`.

Referenced by [jeod::BodyAttach::apply\(\)](#), [jeod::BodyDetach::apply\(\)](#), [jeod::BodyDetachSpecific::apply\(\)](#), [jeod::BodyReattach::apply\(\)](#), [jeod::DynBodyFrameSwitch::apply\(\)](#), [jeod::DynBodyInit::apply\(\)](#), and [jeod::MassBodyInit::apply\(\)](#).

8.1.3.2 `get_identifier()`

```
const std::string & jeod::BodyAction::get_identifier ( ) const [inline]
```

Accessor for `action_identifier`.

Returns

Action identifier

Definition at line 241 of file `body_action.hh`.

8.1.3.3 `get_subject_dyn_body()`

```
DynBody * jeod::BodyAction::get_subject_dyn_body ( )
```

Definition at line 209 of file `body_action.cc`.

References `dyn_subject`, and `mass_subject`.

Referenced by `jeod::DynBodyInitLvHRotState::initialize()`.

8.1.3.4 `initialize()`

```
void jeod::BodyAction::initialize (
    DynManager & dyn_manager ) [virtual]
```

Begin initialization of a [BodyAction](#).

The initialize method for all subclasses of [BodyAction](#) *must* pass the initialize call to their immediate parent class, which in turn must do the same, eventually invoking this method.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Dynamics manager
----------------------	---------------------------------	------------------

Reimplemented in [jeod::DynBodyInitWrtPlanet](#), [jeod::DynBodyInitTransState](#), [jeod::DynBodyInitRotState](#), [jeod::DynBodyInitPlanetDerived](#), [jeod::DynBodyInitOrbit](#), [jeod::DynBodyInitNedTransState](#), [jeod::DynBodyInitNedState](#), [jeod::DynBodyInitNedRotState](#), [jeod::DynBodyInitLvHTransState](#), [jeod::DynBodyInitLvHState](#), [jeod::DynBodyInitLvHRotState](#), [jeod::DynBodyInit](#), [jeod::DynBodyFrameSwitch](#), [jeod::BodyDetachSpecific](#), [jeod::BodyAttachAligned](#), and [jeod::BodyAttach](#).

Definition at line 66 of file `body_action.cc`.

References `action_identifier`, `action_name`, `dyn_subject`, `mass_subject`, and `validate_body_inputs()`.

Referenced by `jeod::BodyAttach::initialize()`, `jeod::BodyDetachSpecific::initialize()`, `jeod::DynBodyFrameSwitch::initialize()`, and `jeod::DynBodyInit::initialize()`.

8.1.3.5 is_ready()

```
bool jeod::BodyAction::is_ready ( ) [virtual]
```

In general, determine if the initializer is ready to be applied.

This base class method simply queries the active flag. Subclasses should override this default method.

Returns

Can initializer run?

Reimplemented in [jeod::DynBodyInitWrtPlanet](#), [jeod::DynBodyInitTransState](#), [jeod::DynBodyInitRotState](#), [jeod::DynBodyInitPlanetDerived](#), [jeod::DynBodyInit](#), [jeod::DynBodyFrameSwitch](#), [jeod::BodyDetachSpecific](#), and [jeod::BodyDetach](#).

Definition at line 101 of file `body_action.cc`.

References `active`.

Referenced by `jeod::DynBodyFrameSwitch::is_ready()`, and `jeod::DynBodyInit::is_ready()`.

8.1.3.6 is_same_subject_body()

```
bool jeod::BodyAction::is_same_subject_body (
    MassBody & mass_body_in )
```

Test the input mass body against the subject body and return true if they are the same body.

Definition at line 181 of file `body_action.cc`.

References `dyn_subject`, and `mass_subject`.

8.1.3.7 is_subject_dyn_body()

```
bool jeod::BodyAction::is_subject_dyn_body ( )
```

Check if the subject is a `DynBody`.

Definition at line 193 of file `body_action.cc`.

References `dyn_subject`, and `mass_subject`.

8.1.3.8 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::BodyAction::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    BodyAction ) [private]
```

8.1.3.9 operator=()

```
BodyAction& jeod::BodyAction::operator= (
    const BodyAction & ) [delete]
```

8.1.3.10 set_subject_body() [1/2]

```
void jeod::BodyAction::set_subject_body (
    DynBody & dyn_body_in )
```

Set the subject dyn body of this action.

Resets mass_subject to null

Definition at line 112 of file body_action.cc.

References dyn_subject, and mass_subject.

8.1.3.11 set_subject_body() [2/2]

```
void jeod::BodyAction::set_subject_body (
    MassBody & mass_body_in )
```

Set the subject mass body of this action.

Resets dyn_subject to null

Definition at line 106 of file body_action.cc.

References dyn_subject, and mass_subject.

8.1.3.12 shutdown()

```
void jeod::BodyAction::shutdown ( ) [virtual]
```

Release resources allocated by a [BodyAction](#) object.

Definition at line 54 of file body_action.cc.

Referenced by [apply\(\)](#).

8.1.3.13 validate_body_inputs()

```
bool jeod::BodyAction::validate_body_inputs (
    DynBody *& dyn_body_in,
    MassBody *& mass_body_in,
    const std::string & body_base_name,
    bool allow_failure = false ) [protected], [virtual]
```

Definition at line 118 of file body_action.cc.

References `action_identifier`, `jeod::BodyActionMessages::fatal_error`, and `jeod::BodyActionMessages::null_pointer`.

Referenced by `initialize()`, `jeod::BodyAttach::initialize()`, and `jeod::BodyDetachSpecific::initialize()`.

8.1.3.14 validate_name()

```
void jeod::BodyAction::validate_name (
    const std::string & variable_value,
    const std::string & variable_name,
    const std::string & variable_type ) [protected]
```

Ensure that a string is not trivially invalid.

Parameters

in	<i>variable_value</i>	String to be checked
in	<i>variable_name</i>	For error reporting
in	<i>variable_type</i>	For error reporting

Definition at line 231 of file body_action.cc.

References `action_identifier`, and `jeod::BodyActionMessages::invalid_name`.

Referenced by `jeod::DynBodyInit::find_body_frame()`, `jeod::DynBodyInit::find_dyn_body()`, `jeod::DynBodyInit::find_planet()`, `jeod::DynBodyInit::find_ref_frame()`, and `jeod::DynBodyInitOrbit::initialize()`.

8.1.4 Field Documentation

8.1.4.1 action_identifier

```
std::string jeod::BodyAction::action_identifier {""} [protected]
```

An identifier for this action, constructed from the class name and the action name at initialization time.

This is used for generating error and debug messages. `trick_units(-)`

Definition at line 187 of file body_action.hh.

Referenced by `jeod::BodyAttach::apply()`, `jeod::BodyDetach::apply()`, `jeod::BodyDetachSpecific::apply()`, `jeod::BodyReattach::apply()`, `jeod::DynBodyFrameSwitch::apply()`, `jeod::DynBodyInit::apply()`, `jeod::DynBodyInitNedState::apply()`, `jeod::DynBodyInitOrbit::apply()`, `jeod::MassBodyInit::apply()`, `jeod::DynBodyInit::find_body_frame()`, `jeod::DynBodyInit::find_dyn_body()`, `jeod::DynBodyInit::find_planet()`, `jeod::DynBodyInit::find_ref_frame()`, `initialize()`, `jeod::BodyAttach::initialize()`, `jeod::BodyAttachAligned::initialize()`, `jeod::DynBodyFrameSwitch::initialize()`, `jeod::DynBodyInit::initialize()`, `jeod::DynBodyInitLvlhRotState::initialize()`, `jeod::DynBodyInitLvlhTransState::initialize()`, `jeod::DynBodyInitNedRotState::initialize()`, `jeod::DynBodyInitNedTransState::initialize()`, `jeod::DynBodyInitOrbit::initialize()`, `jeod::DynBodyInitRotState::initialize()`, `jeod::DynBodyInitTransState::initialize()`, `jeod::DynBodyInitRotState::is_ready()`, `jeod::DynBodyInitTransState::is_ready()`, `jeod::DynBodyInit::report_failure()`, `validate_body_inputs()`, and `validate_name()`.

8.1.4.2 action_name

```
std::string jeod::BodyAction::action_name { "" }
```

An identifier for this action.

This can be left as empty (default value). The `action_name` is used only when an error is detected. The generated error message identifies the action name if supplied. The intent is to generate an error message that helps the user pinpoint the source of the error. `trick_units(-)`

Definition at line 163 of file body_action.hh.

Referenced by `initialize()`.

8.1.4.3 active

```
bool jeod::BodyAction::active { true }
```

Controls when the action is performed.

The action will be performed when the action is activated via this flag and when all other prerequisites for the action have been satisfied. The default value for this flag is class-dependent, set in various constructors. The default is true for actions that can reasonably be performed during initialization time and false for actions that are most likely performed while the simulation is running. `trick_units(-)`

Definition at line 142 of file body_action.hh.

Referenced by `jeod::BodyDetach::BodyDetach()`, `jeod::BodyDetachSpecific::BodyDetachSpecific()`, `jeod::BodyReattach::BodyReattach()`, `is_ready()`, `jeod::BodyDetach::is_ready()`, and `jeod::BodyDetachSpecific::is_ready()`.

8.1.4.4 dyn_subject

```
DynBody* jeod::BodyAction::dyn_subject {} [protected]
```

The DynBody of the body that is the subject of this action.

This or the subject pointer must be supplied. Actions on the body are performed by the apply methods of specific class derived from the [BodyAction](#) class. `trick_units(-)`

Definition at line 180 of file `body_action.hh`.

Referenced by `jeod::BodyAttach::apply()`, `jeod::BodyAttachAligned::apply()`, `jeod::BodyAttachMatrix::apply()`, `jeod::BodyDetach::apply()`, `jeod::BodyDetachSpecific::apply()`, `jeod::DynBodyFrameSwitch::apply()`, `jeod::DynBodyInit::apply()`, `jeod::DynBodyInit::apply_user_inputs()`, `get_subject_dyn_body()`, `initialize()`, `jeod::DynBodyFrameSwitch::initialize()`, `jeod::DynBodyInit::initialize()`, `jeod::DynBodyFrameSwitch::is_ready()`, `is_same_subject_dyn_body()`, `is_subject_dyn_body()`, and `set_subject_body()`.

8.1.4.5 mass_subject

```
MassBody* jeod::BodyAction::mass_subject {} [protected]
```

The MassBody of the body that is the subject of this action.

This or the `dyn_subject` pointer must be supplied. Actions on the body are performed by the apply methods of specific class derived from the [BodyAction](#) class. `trick_units(-)`

Definition at line 172 of file `body_action.hh`.

Referenced by `jeod::BodyAttach::apply()`, `jeod::BodyAttachAligned::apply()`, `jeod::BodyAttachMatrix::apply()`, `jeod::BodyDetach::apply()`, `jeod::BodyDetachSpecific::apply()`, `jeod::BodyReattach::apply()`, `jeod::MassBodyInit::apply()`, `get_subject_dyn_body()`, `initialize()`, `jeod::DynBodyFrameSwitch::initialize()`, `jeod::DynBodyInit::initialize()`, `is_same_subject_dyn_body()`, `is_subject_dyn_body()`, and `set_subject_body()`.

8.1.4.6 terminate_on_error

```
bool jeod::BodyAction::terminate_on_error {true}
```

Indicates whether errors encountered while performing the action are to terminate the simulation.

Several of the low-level methods used to perform the action do not terminate the simulation on encountering an error condition. They instead leave states unchanged and return an error indicator. This flag, if set, causes the simulation to be terminated when such an error condition occurs. The default value for this flag is true, set in the constructor. `trick_units(-)`

Definition at line 154 of file `body_action.hh`.

Referenced by `jeod::BodyAttach::apply()`, `jeod::BodyDetach::apply()`, `jeod::BodyDetachSpecific::apply()`, and `jeod::BodyReattach::apply()`.

The documentation for this class was generated from the following files:

- [body_action.hh](#)
- [body_action.cc](#)

8.2 jeod::BodyActionMessages Class Reference

Specifies the message IDs used in the [BodyAction](#) model.

```
#include <body_action_messages.hh>
```

Public Member Functions

- [BodyActionMessages](#) ()=delete
- [BodyActionMessages](#) (const [BodyActionMessages](#) &)=delete
- [BodyActionMessages](#) & operator= (const [BodyActionMessages](#) &)=delete

Static Public Attributes

- static const char * [fatal_error](#) = "dynamics/body_action/" "fatal_error"
Issued when performing an action results in an error return from the method performing the action.
- static const char * [illegal_value](#) = "dynamics/body_action/" "illegal_value"
Issued when a simple type (e.g.
- static const char * [invalid_name](#) = "dynamics/body_action/" "invalid_name"
Issued when a name is invalid (NULL, empty, or does not name an object of the specified type).
- static const char * [invalid_object](#) = "dynamics/body_action/" "invalid_object"
Issued when a pointer points to an object of the wrong type.
- static const char * [null_pointer](#) = "dynamics/body_action/" "null_pointer"
Error issued when a pointer is required but was not provided.
- static const char * [not_performed](#) = "dynamics/body_action/" "not_performed"
Issued when a [BodyAction](#) cannot be run.
- static const char * [trace](#) = "dynamics/body_action/" "trace"
Debug message issued to trace [BodyAction](#) actions.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [BodyActionMessages](#))

8.2.1 Detailed Description

Specifies the message IDs used in the [BodyAction](#) model.

Assumptions and Limitations

- This is a complete catalog of all messages sent by the [BodyAction](#) model.
- This is not an exhaustive list of all the things that can go awry.

Definition at line 81 of file [body_action_messages.hh](#).

8.2.2 Constructor & Destructor Documentation

8.2.2.1 BodyActionMessages() [1/2]

```
jeod::BodyActionMessages::BodyActionMessages ( ) [delete]
```

8.2.2.2 BodyActionMessages() [2/2]

```
jeod::BodyActionMessages::BodyActionMessages (
    const BodyActionMessages & ) [delete]
```

8.2.3 Member Function Documentation

8.2.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::BodyActionMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    BodyActionMessages ) [private]
```

8.2.3.2 operator=()

```
BodyActionMessages& jeod::BodyActionMessages::operator= (
    const BodyActionMessages & ) [delete]
```

8.2.4 Field Documentation

8.2.4.1 fatal_error

```
char const * jeod::BodyActionMessages::fatal_error = "dynamics/body_action/" "fatal_error"
[static]
```

Issued when performing an action results in an error return from the method performing the action.

trick_units(-)

Definition at line 90 of file body_action_messages.hh.

Referenced by jeod::BodyAttach::apply(), jeod::BodyDetach::apply(), jeod::BodyDetachSpecific::apply(), jeod::BodyReattach::apply(), and jeod::BodyAction::validate_body_inputs().

8.2.4.2 illegal_value

```
char const * jeod::BodyActionMessages::illegal_value = "dynamics/body_action/" "illegal_value"
[static]
```

Issued when a simple type (e.g.

an enum) has an illegal value. `trick_units(-)`

Definition at line 95 of file `body_action_messages.hh`.

Referenced by `jeod::DynBodyInitLvHState::apply()`, `jeod::DynBodyInitNedState::apply()`, `jeod::DynBodyInitOrbit::apply()`, `jeod::DynBodyInitLvHRotState::initialize()`, `jeod::DynBodyInitLvHTransState::initialize()`, `jeod::DynBodyInitNedRotState::initialize()`, `jeod::DynBodyInitNedTransState::initialize()`, `jeod::DynBodyInitOrbit::initialize()`, `jeod::DynBodyInitRotState::initialize()`, and `jeod::DynBodyInitTransState::initialize()`.

8.2.4.3 invalid_name

```
char const * jeod::BodyActionMessages::invalid_name = "dynamics/body_action/" "invalid_name"
[static]
```

Issued when a name is invalid (NULL, empty, or does not name an object of the specified type).

`trick_units(-)`

Definition at line 101 of file `body_action_messages.hh`.

Referenced by `jeod::DynBodyInit::compute_rotational_state()`, `jeod::DynBodyInit::compute_translational_state()`, `jeod::DynBodyInit::find_body_frame()`, `jeod::DynBodyInit::find_dyn_body()`, `jeod::DynBodyInit::find_planet()`, `jeod::DynBodyInit::find_ref_frame()`, `jeod::BodyAttachAligned::initialize()`, `jeod::DynBodyFrameSwitch::initialize()`, `jeod::DynBodyInitOrbit::initialize()`, and `jeod::BodyAction::validate_name()`.

8.2.4.4 invalid_object

```
char const * jeod::BodyActionMessages::invalid_object = "dynamics/body_action/" "invalid_
object" [static]
```

Issued when a pointer points to an object of the wrong type.

`trick_units(-)`

Definition at line 106 of file `body_action_messages.hh`.

Referenced by `jeod::BodyReattach::apply()`, `jeod::DynBodyFrameSwitch::initialize()`, `jeod::DynBodyInit::initialize()`, `jeod::DynBodyInitOrbit::initialize()`, `jeod::DynBodyInitRotState::is_ready()`, and `jeod::DynBodyInitTransState::is_ready()`.

8.2.4.5 not_performed

```
char const * jeod::BodyActionMessages::not_performed = "dynamics/body_action/" "not_performed"
[static]
```

Issued when a [BodyAction](#) cannot be run.

trick_units(-)

Definition at line 116 of file body_action_messages.hh.

Referenced by jeod::BodyAttach::apply(), jeod::BodyDetach::apply(), jeod::BodyDetachSpecific::apply(), and jeod::DynBodyInit::report_failure().

8.2.4.6 null_pointer

```
char const * jeod::BodyActionMessages::null_pointer = "dynamics/body_action/" "null_pointer"
[static]
```

Error issued when a pointer is required but was not provided.

trick_units(-)

Definition at line 111 of file body_action_messages.hh.

Referenced by jeod::BodyAttach::initialize(), jeod::DynBodyInitLvlhRotState::initialize(), and jeod::BodyAction::validate_body_inputs().

8.2.4.7 trace

```
char const * jeod::BodyActionMessages::trace = "dynamics/body_action/" "trace" [static]
```

Debug message issued to trace [BodyAction](#) actions.

trick_units(-)

Definition at line 121 of file body_action_messages.hh.

Referenced by jeod::BodyAttach::apply(), jeod::BodyDetach::apply(), jeod::BodyDetachSpecific::apply(), jeod::BodyReattach::apply(), jeod::DynBodyFrameSwitch::apply(), jeod::DynBodyInit::apply(), and jeod::MassBodyInit::apply().

The documentation for this class was generated from the following files:

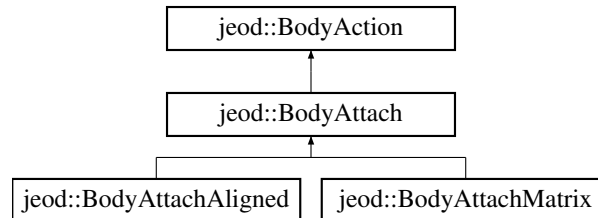
- [body_action_messages.hh](#)
- [body_action_messages.cc](#)

8.3 jeod::BodyAttach Class Reference

Provides the basic ability to attach one MassBody to another.

```
#include <body_attach.hh>
```

Inheritance diagram for jeod::BodyAttach:



Public Member Functions

- void [set_parent_body](#) (MassBody &mass_body_in)
Set the parent mass body of this action.
- void [set_parent_body](#) (DynBody &dyn_body_in)
Set the parent dyn body of this action.
- void [set_parent_frame](#) (RefFrame &ref_parent_in)
Set the parent ref frame of this action.
- [BodyAttach](#) ()=default
- [~BodyAttach](#) () override=default
- [BodyAttach](#) (const [BodyAttach](#) &)=delete
- [BodyAttach](#) & [operator=](#) (const [BodyAttach](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize a MassBodyAttach.
- void [apply](#) (DynManager &dyn_manager) override
A derived class presumably has performed the attachment, which may not have worked, and forwarded the apply call to this method.

Data Fields

- bool [succeeded](#) {}
Did the attachment succeed?

Protected Attributes

- MassBody * [mass_parent](#) {}
The MassBody corresponding to which the subject body is to be attached, directly if the subject body is a root body, and indirectly by attaching the subject body's root body to the parent body otherwise.
- DynBody * [dyn_parent](#) {}
The DynBody corresponding to which the subject body is to be attached, directly if the subject body is a root body, and indirectly by attaching the subject body's root body to the parent body otherwise.
- RefFrame * [ref_parent](#) {}
The RefFrame corresponding to which the subject body is to be attached, directly if the subject body is a root body, and indirectly by attaching the subject body's root body to the parent RefFrame otherwise.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [BodyAttach](#))

Additional Inherited Members

8.3.1 Detailed Description

Provides the basic ability to attach one MassBody to another.

This can be either an initialization or asynchronous [BodyAction](#). The action will be performed when the sim user or some simulation job enables the active flag.

MassBodyAttach actions that are ready at simulation initialization time are run as a part of the initialization process, sandwiched between initializing mass properties and initializing state. Attach actions that are not ready at initialization time remain in the pending actions queue until the active flag is set.

Definition at line 94 of file body_attach.hh.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 BodyAttach() [1/2]

```
jeod::BodyAttach::BodyAttach ( ) [default]
```

8.3.2.2 ~BodyAttach()

```
jeod::BodyAttach::~~BodyAttach ( ) [override], [default]
```

8.3.2.3 BodyAttach() [2/2]

```
jeod::BodyAttach::BodyAttach (
    const BodyAttach & ) [delete]
```

8.3.3 Member Function Documentation

8.3.3.1 apply()

```
void jeod::BodyAttach::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

A derived class presumably has performed the attachment, which may not have worked, and forwarded the apply call to this method.

This method acts on the status from that child class attachment.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Jeod manager
----------------	--------------------	--------------

Reimplemented from [jeod::BodyAction](#).

Reimplemented in [jeod::BodyAttachMatrix](#), and [jeod::BodyAttachAligned](#).

Definition at line 101 of file `body_attach.cc`.

References [jeod::BodyAction::action_identifier](#), [jeod::BodyAction::apply\(\)](#), [dyn_parent](#), [jeod::BodyAction::dyn_subject](#), [jeod::BodyActionMessages::fatal_error](#), [mass_parent](#), [jeod::BodyAction::mass_subject](#), [jeod::BodyActionMessages::not_performed](#), [ref_parent](#), [succeeded](#), [jeod::BodyAction::terminate_on_error](#), and [jeod::BodyActionMessages::trace](#).

Referenced by [jeod::BodyAttachAligned::apply\(\)](#), and [jeod::BodyAttachMatrix::apply\(\)](#).

8.3.3.2 initialize()

```
void jeod::BodyAttach::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize a MassBodyAttach.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::BodyAction](#).

Reimplemented in [jeod::BodyAttachAligned](#).

Definition at line 52 of file `body_attach.cc`.

References [jeod::BodyAction::action_identifier](#), [dyn_parent](#), [jeod::BodyAction::initialize\(\)](#), [mass_parent](#), [jeod::BodyActionMessages::null_pointer](#), [ref_parent](#), and [jeod::BodyAction::validate_body_inputs\(\)](#).

Referenced by [jeod::BodyAttachAligned::initialize\(\)](#).

8.3.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::BodyAttach::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    BodyAttach ) [private]
```


8.3.3.4 operator=()

```
BodyAttach& jeod::BodyAttach::operator= (
    const BodyAttach & ) [delete]
```

8.3.3.5 set_parent_body() [1/2]

```
void jeod::BodyAttach::set_parent_body (
    DynBody & dyn_body_in )
```

Set the parent dyn body of this action.

Resets mass_parent, frame_parent to null

Definition at line 81 of file body_attach.cc.

References dyn_parent, mass_parent, and ref_parent.

8.3.3.6 set_parent_body() [2/2]

```
void jeod::BodyAttach::set_parent_body (
    MassBody & mass_body_in )
```

Set the parent mass body of this action.

Resets dyn_parent, frame_parent to null

Definition at line 74 of file body_attach.cc.

References dyn_parent, mass_parent, and ref_parent.

8.3.3.7 set_parent_frame()

```
void jeod::BodyAttach::set_parent_frame (
    RefFrame & ref_parent_in )
```

Set the parent ref frame of this action.

Resets mass_parent, dyn_parent to null

Definition at line 88 of file body_attach.cc.

References dyn_parent, mass_parent, and ref_parent.

8.3.4 Field Documentation

8.3.4.1 dyn_parent

```
DynBody* jeod::BodyAttach::dyn_parent {} [protected]
```

The DynBody corresponding to which the subject body is to be attached, directly if the subject body is a root body, and indirectly by attaching the subject body's root body to the parent body otherwise.

This pointer is one of the 3 possible pointers that must be supplied. `trick_units(-)`

Definition at line 134 of file `body_attach.hh`.

Referenced by `apply()`, `jeod::BodyAttachAligned::apply()`, `jeod::BodyAttachMatrix::apply()`, `initialize()`, `set_parent↔_body()`, and `set_parent_frame()`.

8.3.4.2 mass_parent

```
MassBody* jeod::BodyAttach::mass_parent {} [protected]
```

The MassBody corresponding to which the subject body is to be attached, directly if the subject body is a root body, and indirectly by attaching the subject body's root body to the parent body otherwise.

This pointer is one of the 3 possible pointers that must be supplied. `trick_units(-)`

Definition at line 126 of file `body_attach.hh`.

Referenced by `apply()`, `jeod::BodyAttachAligned::apply()`, `jeod::BodyAttachMatrix::apply()`, `initialize()`, `set_parent↔_body()`, and `set_parent_frame()`.

8.3.4.3 ref_parent

```
RefFrame* jeod::BodyAttach::ref_parent {} [protected]
```

The RefFrame corresponding to which the subject body is to be attached, directly if the subject body is a root body, and indirectly by attaching the subject body's root body to the parent RefFrame otherwise.

This pointer is one of the 3 possible pointers that must be supplied. `trick_units(-)`

Definition at line 142 of file `body_attach.hh`.

Referenced by `apply()`, `jeod::BodyAttachAligned::apply()`, `jeod::BodyAttachMatrix::apply()`, `initialize()`, `set_parent↔_body()`, and `set_parent_frame()`.

8.3.4.4 succeeded

```
bool jeod::BodyAttach::succeeded {}
```

Did the attachment succeed?

trick_units(-)

Definition at line 117 of file body_attach.hh.

Referenced by `apply()`, `jeod::BodyAttachAligned::apply()`, and `jeod::BodyAttachMatrix::apply()`.

The documentation for this class was generated from the following files:

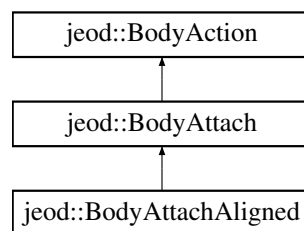
- [body_attach.hh](#)
- [body_attach.cc](#)

8.4 jeod::BodyAttachAligned Class Reference

Attaches a pair of MassBody objects at a pair of MassPoints.

```
#include <body_attach_aligned.hh>
```

Inheritance diagram for jeod::BodyAttachAligned:



Public Member Functions

- [BodyAttachAligned](#) ()=default
- [~BodyAttachAligned](#) () override=default
- [BodyAttachAligned](#) (const [BodyAttachAligned](#) &)=delete
- [BodyAttachAligned](#) & [operator=](#) (const [BodyAttachAligned](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize a MassBodyAttach.
- void [apply](#) (DynManager &dyn_manager) override
Initialize the core mass properties of the subject MassBody.

Data Fields

- std::string [subject_point_name](#) {""}
The name of the mass point on the subject mass body to be attached to to the parent_point_name mass point on the parent mass body.
- std::string [parent_point_name](#) {""}
The name of the mass point on the parent mass body to be attached to to the mass pointed named subject_point_name on the subject mass body.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [BodyAttachAligned](#))

Additional Inherited Members

8.4.1 Detailed Description

Attaches a pair of MassBody objects at a pair of MassPoints.

When the action is ready, the attachment proceeds as follows:

- The points indicated by the subject and parent mass point names will be coincident after attachment is complete.
- The orientation between the two reference frames associated with the two attach points is a 180 degree yaw.

Definition at line 86 of file `body_attach_aligned.hh`.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 BodyAttachAligned() [1/2]

```
jeod::BodyAttachAligned::BodyAttachAligned ( ) [default]
```

8.4.2.2 ~BodyAttachAligned()

```
jeod::BodyAttachAligned::~~BodyAttachAligned ( ) [override], [default]
```

8.4.2.3 BodyAttachAligned() [2/2]

```
jeod::BodyAttachAligned::BodyAttachAligned (
    const BodyAttachAligned & ) [delete]
```

8.4.3 Member Function Documentation

8.4.3.1 apply()

```
void jeod::BodyAttachAligned::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the core mass properties of the subject MassBody.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Jeod manager
----------------	--------------------	--------------

Reimplemented from [jeod::BodyAttach](#).

Definition at line 87 of file `body_attach_aligned.cc`.

References `jeod::BodyAttach::apply()`, `jeod::BodyAttach::dyn_parent`, `jeod::BodyAction::dyn_subject`, `jeod::BodyAttach::mass_parent`, `jeod::BodyAction::mass_subject`, `parent_point_name`, `jeod::BodyAttach::ref_parent`, `subject_point_name`, and `jeod::BodyAttach::succeeded`.

8.4.3.2 initialize()

```
void jeod::BodyAttachAligned::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize a MassBodyAttach.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::BodyAttach](#).

Definition at line 54 of file `body_attach_aligned.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyAttach::initialize()`, `jeod::BodyActionMessages::invalid_name`, `parent_point_name`, and `subject_point_name`.

8.4.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::BodyAttachAligned::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    BodyAttachAligned ) [private]
```

8.4.3.4 operator=()

```
BodyAttachAligned& jeod::BodyAttachAligned::operator= (
    const BodyAttachAligned & ) [delete]
```

8.4.4 Field Documentation

8.4.4.1 parent_point_name

```
std::string jeod::BodyAttachAligned::parent_point_name {""}
```

The name of the mass point on the parent mass body to be attached to to the mass pointed named `subject_point_name` on the subject mass body.

The supplied name can omit the parent mass body name dot prefix if desired. `trick_units(-)`

Definition at line 103 of file `body_attach_aligned.hh`.

Referenced by `apply()`, and `initialize()`.

8.4.4.2 subject_point_name

```
std::string jeod::BodyAttachAligned::subject_point_name {""}
```

The name of the mass point on the subject mass body to be attached to to the `parent_point_name` mass point on the parent mass body.

The supplied name can omit the subject mass body name dot prefix if desired. `trick_units(-)`

Definition at line 96 of file `body_attach_aligned.hh`.

Referenced by `apply()`, and `initialize()`.

The documentation for this class was generated from the following files:

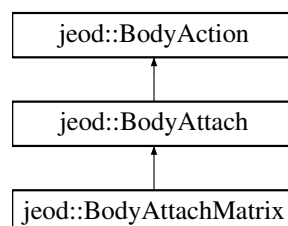
- [body_attach_aligned.hh](#)
- [body_attach_aligned.cc](#)

8.5 jeod::BodyAttachMatrix Class Reference

Attaches a pair of `MassBody` objects using the offset+matrix attach mechanism.

```
#include <body_attach_matrix.hh>
```

Inheritance diagram for `jeod::BodyAttachMatrix`:



Public Member Functions

- [BodyAttachMatrix](#) ()=default
- [~BodyAttachMatrix](#) () override=default
- [BodyAttachMatrix](#) (const [BodyAttachMatrix](#) &)=delete
- [BodyAttachMatrix](#) & [operator=](#) (const [BodyAttachMatrix](#) &)=delete
- void [apply](#) (DynManager &dyn_manager) override

Initialize the core mass properties of the subject MassBody.

Data Fields

- double [offset_pstr_cstr_pstr](#) [3] {}
Location of this body's structural origin with respect to the new parent body's structural origin (or generic reference frame), specified in structural coordinates of the new parent body.
- Orientation [pstr_cstr](#)
Orientation of child's structural frame with respect to that of the new parent; sense is parent-to-child.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [BodyAttachMatrix](#))

Additional Inherited Members

8.5.1 Detailed Description

Attaches a pair of MassBody objects using the offset+matrix attach mechanism.

When the action is ready, the attachment is made such that:

- The displacement between the origins of the parent and subject bodies' structural frames is that given by the [offset_pstr_cstr_pstr](#) data member.
- The orientation between these two reference frames's axes is that given by the [pstr_cstr](#) data member.

Definition at line 86 of file [body_attach_matrix.hh](#).

8.5.2 Constructor & Destructor Documentation

8.5.2.1 [BodyAttachMatrix](#)() [1/2]

```
jeod::BodyAttachMatrix::BodyAttachMatrix ( ) [default]
```

8.5.2.2 ~BodyAttachMatrix()

```
jeod::BodyAttachMatrix::~~BodyAttachMatrix ( ) [override], [default]
```

8.5.2.3 BodyAttachMatrix() [2/2]

```
jeod::BodyAttachMatrix::BodyAttachMatrix (
    const BodyAttachMatrix & ) [delete]
```

8.5.3 Member Function Documentation

8.5.3.1 apply()

```
void jeod::BodyAttachMatrix::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the core mass properties of the subject MassBody.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Jeod manager
----------------	--------------------	--------------

Reimplemented from [jeod::BodyAttach](#).

Definition at line 50 of file `body_attach_matrix.cc`.

References [jeod::BodyAttach::apply\(\)](#), [jeod::BodyAttach::dyn_parent](#), [jeod::BodyAction::dyn_subject](#), [jeod::BodyAttach::mass_parent](#), [jeod::BodyAction::mass_subject](#), [offset_pstr_cstr_pstr](#), [pstr_cstr](#), [jeod::BodyAttach::ref_parent](#), and [jeod::BodyAttach::succeeded](#).

8.5.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::BodyAttachMatrix::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    BodyAttachMatrix ) [private]
```

8.5.3.3 operator=()

```
BodyAttachMatrix& jeod::BodyAttachMatrix::operator= (
    const BodyAttachMatrix & ) [delete]
```


8.5.4 Field Documentation

8.5.4.1 offset_pstr_cstr_pstr

```
double jeod::BodyAttachMatrix::offset_pstr_cstr_pstr[3] {}
```

Location of this body's structural origin with respect to the new parent body's structural origin (or generic reference frame), specified in structural coordinates of the new parent body.

trick_units(m)

Definition at line 96 of file body_attach_matrix.hh.

Referenced by apply().

8.5.4.2 pstr_cstr

```
Orientation jeod::BodyAttachMatrix::pstr_cstr
```

Orientation of child's structural frame with respect to that of the new parent; sense is parent-to-child.

trick_units(—)

Definition at line 102 of file body_attach_matrix.hh.

Referenced by apply().

The documentation for this class was generated from the following files:

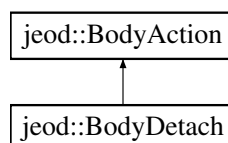
- [body_attach_matrix.hh](#)
- [body_attach_matrix.cc](#)

8.6 jeod::BodyDetach Class Reference

Provides the basic ability to detach one MassBody from another.

```
#include <body_detach.hh>
```

Inheritance diagram for jeod::BodyDetach:



Public Member Functions

- [BodyDetach](#) ()
Construct a MassBodyDetach.
- [~BodyDetach](#) () override=default
- [BodyDetach](#) (const [BodyDetach](#) &)=delete
- [BodyDetach](#) & [operator=](#) (const [BodyDetach](#) &)=delete
- void [apply](#) (DynManager &dyn_manager) override
Detach the body from its parent.
- bool [is_ready](#) () override
Queries whether the "active" flag has been set.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [BodyDetach](#))

Additional Inherited Members

8.6.1 Detailed Description

Provides the basic ability to detach one MassBody from another.

This is inherently an asynchronous [BodyAction](#). The [is_ready\(\)](#) method simply returns the action's active flag. The action will be performed when the sim user or some simulation job enables the active flag.

The basic detachment action is to cause a body to detach from its immediate parent body. Subclasses can cause bodies to detach elsewhere.

Definition at line 89 of file `body_detach.hh`.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 [BodyDetach\(\)](#) [1/2]

```
jeod::BodyDetach::BodyDetach ( )
```

Construct a MassBodyDetach.

Definition at line 49 of file `body_detach.cc`.

References `jeod::BodyAction::active`.

8.6.2.2 ~BodyDetach()

```
jeod::BodyDetach::~~BodyDetach ( ) [override], [default]
```

8.6.2.3 BodyDetach() [2/2]

```
jeod::BodyDetach::BodyDetach (
    const BodyDetach & ) [delete]
```

8.6.3 Member Function Documentation

8.6.3.1 apply()

```
void jeod::BodyDetach::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Detach the body from its parent.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Jeod manager
----------------	--------------------	--------------

Reimplemented from [jeod::BodyAction](#).

Definition at line 58 of file `body_detach.cc`.

References [jeod::BodyAction::action_identifier](#), [jeod::BodyAction::apply\(\)](#), [jeod::BodyAction::dyn_subject](#), [jeod::BodyActionMessages::fatal_error](#), [jeod::BodyAction::mass_subject](#), [jeod::BodyActionMessages::not_performed](#), [jeod::BodyAction::terminate_on_error](#), and [jeod::BodyActionMessages::trace](#).

8.6.3.2 is_ready()

```
bool jeod::BodyDetach::is_ready ( ) [override], [virtual]
```

Queries whether the "active" flag has been set.

Returns

Can detach process run?

Reimplemented from [jeod::BodyAction](#).

Definition at line 115 of file `body_detach.cc`.

References [jeod::BodyAction::active](#).

8.6.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::BodyDetach::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    BodyDetach ) [private]
```

8.6.3.4 operator=()

```
BodyDetach& jeod::BodyDetach::operator= (
    const BodyDetach & ) [delete]
```

The documentation for this class was generated from the following files:

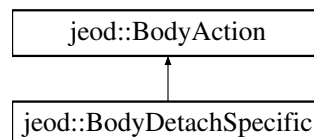
- [body_detach.hh](#)
- [body_detach.cc](#)

8.7 jeod::BodyDetachSpecific Class Reference

Causes the subject body to detach from a specific body by severing the link immediately spawning from the detach↔_from body.

```
#include <body_detach_specific.hh>
```

Inheritance diagram for jeod::BodyDetachSpecific:



Public Member Functions

- void [set_detach_from_body](#) (MassBody &mass_body_in)
Set the subject mass body of this action.
- void [set_detach_from_body](#) (DynBody &dyn_body_in)
Set the subject mass body of this action.
- [BodyDetachSpecific](#) ()
Construct a [BodyDetachSpecific](#).
- [~BodyDetachSpecific](#) () override=default
- [BodyDetachSpecific](#) (const [BodyDetachSpecific](#) &)=delete
- [BodyDetachSpecific](#) & [operator=](#) (const [BodyDetachSpecific](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize a [BodyDetachSpecific](#).
- void [apply](#) (DynManager &dyn_manager) override
Detach the body from its parent.
- bool [is_ready](#) () override
Queries whether the "active" flag has been set.

Protected Attributes

- MassBody * [mass_detach_from](#) {}
The mass body from the subject of this action is to detach.
- DynBody * [dyn_detach_from](#) {}
The dynamic body from the subject of this action is to detach.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [BodyDetachSpecific](#))

Additional Inherited Members

8.7.1 Detailed Description

Causes the subject body to detach from a specific body by severing the link immediately spawning from the detach←_from body.

This method works between two dynamic bodies (DynBody) or mass bodies (MassBody), but not mixtures of the two classes. The subject body itself is detached from its parent if and only if the specified detach_from body is the subject body's immediate parent. In the case that the detach_from body is some indirect parent, the body that detaches is the the immediate child body of the detach_from body that is along the connectivity path from the subject body to the detach_from * body. Specifying a detach_from body that is not a parent (direct or indirect) body of the subject body is an error.

Definition at line 92 of file body_detach_specific.hh.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 BodyDetachSpecific() [1/2]

```
jeod::BodyDetachSpecific::BodyDetachSpecific ( )
```

Construct a [BodyDetachSpecific](#).

Definition at line 51 of file body_detach_specific.cc.

References [jeod::BodyAction::active](#).

8.7.2.2 ~BodyDetachSpecific()

```
jeod::BodyDetachSpecific::~~BodyDetachSpecific ( ) [override], [default]
```

8.7.2.3 BodyDetachSpecific() [2/2]

```
jeod::BodyDetachSpecific::BodyDetachSpecific (
    const BodyDetachSpecific & ) [delete]
```

8.7.3 Member Function Documentation

8.7.3.1 apply()

```
void jeod::BodyDetachSpecific::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Detach the body from its parent.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::BodyAction](#).

Definition at line 72 of file `body_detach_specific.cc`.

References [jeod::BodyAction::action_identifier](#), [jeod::BodyAction::apply\(\)](#), [dyn_detach_from](#), [jeod::BodyAction::dyn_subject](#), [jeod::BodyActionMessages::fatal_error](#), [mass_detach_from](#), [jeod::BodyAction::mass_subject](#), [jeod::BodyActionMessages::not_performed](#), [jeod::BodyAction::terminate_on_error](#), and [jeod::BodyActionMessages::trace](#).

8.7.3.2 initialize()

```
void jeod::BodyDetachSpecific::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize a [BodyDetachSpecific](#).

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::BodyAction](#).

Definition at line 60 of file `body_detach_specific.cc`.

References [dyn_detach_from](#), [jeod::BodyAction::initialize\(\)](#), [mass_detach_from](#), and [jeod::BodyAction::validate_body_inputs\(\)](#).

8.7.3.3 is_ready()

```
bool jeod::BodyDetachSpecific::is_ready ( ) [override], [virtual]
```

Queries whether the "active" flag has been set.

Returns

Can detach process run?

Reimplemented from [jeod::BodyAction](#).

Definition at line 163 of file `body_detach_specific.cc`.

References `jeod::BodyAction::active`.

8.7.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::BodyDetachSpecific::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    BodyDetachSpecific ) [private]
```

8.7.3.5 operator=()

```
BodyDetachSpecific& jeod::BodyDetachSpecific::operator= (
    const BodyDetachSpecific & ) [delete]
```

8.7.3.6 set_detach_from_body() [1/2]

```
void jeod::BodyDetachSpecific::set_detach_from_body (
    DynBody & dyn_body_in )
```

Set the subject mass body of this action.

Resets `dyn_subject` to null

Definition at line 153 of file `body_detach_specific.cc`.

References `dyn_detach_from`, and `mass_detach_from`.

8.7.3.7 set_detach_from_body() [2/2]

```
void jeod::BodyDetachSpecific::set_detach_from_body (
    MassBody & mass_body_in )
```

Set the subject mass body of this action.

Resets dyn_subject to null

Definition at line 147 of file body_detach_specific.cc.

References dyn_detach_from, and mass_detach_from.

8.7.4 Field Documentation

8.7.4.1 dyn_detach_from

```
DynBody* jeod::BodyDetachSpecific::dyn_detach_from {} [protected]
```

The dynamic body from the subject of this action is to detach.

This pointer or the detach_from member must be supplied for dynamic body detachment. The detachment is performed between the mass_detach_from object and the direct descendant of the mass_detach_from object that is in the parental lineage from the subject body to the mass_detach_from body. trick_units(-)

Definition at line 126 of file body_detach_specific.hh.

Referenced by apply(), initialize(), and set_detach_from_body().

8.7.4.2 mass_detach_from

```
MassBody* jeod::BodyDetachSpecific::mass_detach_from {} [protected]
```

The mass body from the subject of this action is to detach.

This pointer must be supplied for pure MassBody detachments. The initialize method will attempt to determine if this MassBody refers to a DynBody. The detachment is performed between the mass_detach_from object and the direct descendant of the detach_from object that is in the parental lineage from the subject body to the detach_from body. trick_units(-)

Definition at line 116 of file body_detach_specific.hh.

Referenced by apply(), initialize(), and set_detach_from_body().

The documentation for this class was generated from the following files:

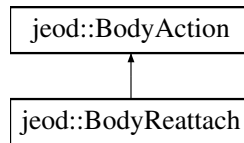
- [body_detach_specific.hh](#)
- [body_detach_specific.cc](#)

8.8 jeod::BodyReattach Class Reference

Alters the nature of an existing attachment.

```
#include <body_reattach.hh>
```

Inheritance diagram for jeod::BodyReattach:



Public Member Functions

- [BodyReattach](#) ()
Construct a MassBodyReattach.
- [~BodyReattach](#) () override=default
- [BodyReattach](#) (const [BodyReattach](#) &)=delete
- [BodyReattach](#) & [operator=](#) (const [BodyReattach](#) &)=delete
- void [apply](#) (DynManager &dyn_manager) override
Initialize the core mass properties of the subject MassBody.

Data Fields

- double [offset_pstr_cstr_pstr](#) [3] {}
Location of this body's structural origin with respect to the new parent body's structural origin, specified in structural coordinates of the new parent body.
- Orientation [pstr_cstr](#)
Orientation of child's structural frame with respect to that of the new parent; sense is parent-to-child.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [BodyReattach](#))

Additional Inherited Members

8.8.1 Detailed Description

Alters the nature of an existing attachment.

When the action is ready, the attachment is altered such that:

- The displacement between the origins of the parent and subject bodies' structural frames is that given by the [offset_pstr_cstr_pstr](#) data member.
- The orientation between these two reference frames's axes is that given by the [pstr_cstr](#) data member. Note that no parent body is specified. Reattachment does not change the attachment tree. It instead alters the physical relationships between a pair of objects that are already attached.

Definition at line 90 of file [body_reattach.hh](#).

8.8.2 Constructor & Destructor Documentation

8.8.2.1 BodyReattach() [1/2]

```
jeod::BodyReattach::BodyReattach ( )
```

Construct a MassBodyReattach.

Definition at line 48 of file body_reattach.cc.

References [jeod::BodyAction::active](#).

8.8.2.2 ~BodyReattach()

```
jeod::BodyReattach::~~BodyReattach ( ) [override], [default]
```

8.8.2.3 BodyReattach() [2/2]

```
jeod::BodyReattach::BodyReattach (
    const BodyReattach & ) [delete]
```

8.8.3 Member Function Documentation

8.8.3.1 apply()

```
void jeod::BodyReattach::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the core mass properties of the subject MassBody.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Jeod manager
----------------	--------------------	--------------

Reimplemented from [jeod::BodyAction](#).

Definition at line 57 of file body_reattach.cc.

References `jeod::BodyAction::action_identifier`, `jeod::BodyAction::apply()`, `jeod::BodyActionMessages::fatal_error`, `jeod::BodyActionMessages::invalid_object`, `jeod::BodyAction::mass_subject`, `offset_pstr_cstr_pstr`, `pstr_cstr`, `jeod::BodyAction::terminate_on_error`, and `jeod::BodyActionMessages::trace`.

8.8.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::BodyReattach::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    BodyReattach ) [private]
```

8.8.3.3 operator=()

```
BodyReattach& jeod::BodyReattach::operator= (
    const BodyReattach & ) [delete]
```

8.8.4 Field Documentation

8.8.4.1 offset_pstr_cstr_pstr

```
double jeod::BodyReattach::offset_pstr_cstr_pstr[3] {}
```

Location of this body's structural origin with respect to the new parent body's structural origin, specified in structural coordinates of the new parent body.

`trick_units(m)`

Definition at line 100 of file `body_reattach.hh`.

Referenced by `apply()`.

8.8.4.2 pstr_cstr

```
Orientation jeod::BodyReattach::pstr_cstr
```

Orientation of child's structural frame with respect to that of the new parent; sense is parent-to-child.

`trick_units(-)`

Definition at line 106 of file `body_reattach.hh`.

Referenced by `apply()`.

The documentation for this class was generated from the following files:

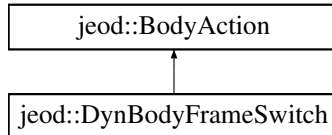
- [body_reattach.hh](#)
- [body_reattach.cc](#)

8.9 jeod::DynBodyFrameSwitch Class Reference

Switch a DynBody's integration frame to a specified frame when the body switches to that integration frame's sphere of influence.

```
#include <dyn_body_frame_switch.hh>
```

Inheritance diagram for jeod::DynBodyFrameSwitch:



Public Types

- enum [SwitchSense](#) { [SwitchOnApproach](#) = 0 , [SwitchOnDeparture](#) = 1 }
- Specifies whether the [is_ready\(\)](#) method is to look for the vehicle entering ([SwitchOnApproach](#)) the new integration frame's sphere of influence versus leaving ([SwitchOnDeparture](#)) the current integration frame's sphere of influence.*

Public Member Functions

- [DynBodyFrameSwitch](#) ()=default
- [~DynBodyFrameSwitch](#) () override=default
- [DynBodyFrameSwitch](#) (const [DynBodyFrameSwitch](#) &)=delete
- [DynBodyFrameSwitch](#) & [operator=](#) (const [DynBodyFrameSwitch](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialization a [DynBodyFrameSwitch](#) instance.
- void [apply](#) (DynManager &dyn_manager) override
Switch reference frames.
- bool [is_ready](#) () override
Determine whether it is time to switch frames.

Data Fields

- std::string [integ_frame_name](#) {}
The name of the new integration frame.
- [SwitchSense](#) [switch_sense](#) {[SwitchOnApproach](#)}
Indicates whether the switch occurs when the subject DynBody enters a sphere of influence around the new integration frame or leaves a sphere of influence around of the current integration frame.
- bool [sort_grav_controls](#) {}
If set, the body's gravitational controls are sorted in ascending acceleration magnitude.
- double [switch_distance](#) {9e99}
The radius of the sphere of influence.

Protected Attributes

- EphemerisRefFrame * [integ_frame](#) {}
The reference frame corresponding to the input [integ_frame_name](#).

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyFrameSwitch](#))

Additional Inherited Members

8.9.1 Detailed Description

Switch a DynBody's integration frame to a specified frame when the body switches to that integration frame's sphere of influence.

Definition at line 86 of file `dyn_body_frame_switch.hh`.

8.9.2 Member Enumeration Documentation

8.9.2.1 SwitchSense

```
enum jeod::DynBodyFrameSwitch::SwitchSense
```

Specifies whether the `is_ready()` method is to look for the vehicle entering (`SwitchOnApproach`) the new integration frame's sphere of influence versus leaving (`SwitchOnDeparture`) the current integration frame's sphere of influence.

Enumerator

<code>SwitchOnApproach</code>	
<code>SwitchOnDeparture</code>	

Definition at line 97 of file `dyn_body_frame_switch.hh`.

8.9.3 Constructor & Destructor Documentation

8.9.3.1 DynBodyFrameSwitch() [1/2]

```
jeod::DynBodyFrameSwitch::DynBodyFrameSwitch ( ) [default]
```

8.9.3.2 ~DynBodyFrameSwitch()

```
jeod::DynBodyFrameSwitch::~~DynBodyFrameSwitch ( ) [override], [default]
```

8.9.3.3 DynBodyFrameSwitch() [2/2]

```
jeod::DynBodyFrameSwitch::DynBodyFrameSwitch (
    const DynBodyFrameSwitch & ) [delete]
```

8.9.4 Member Function Documentation

8.9.4.1 apply()

```
void jeod::DynBodyFrameSwitch::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Switch reference frames.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Jeod manager
----------------	--------------------	--------------

Reimplemented from [jeod::BodyAction](#).

Definition at line 119 of file `dyn_body_frame_switch.cc`.

References [jeod::BodyAction::action_identifier](#), [jeod::BodyAction::apply\(\)](#), [jeod::BodyAction::dyn_subject](#), [integ↵frame](#), [integ_frame_name](#), [sort_grav_controls](#), and [jeod::BodyActionMessages::trace](#).

8.9.4.2 initialize()

```
void jeod::DynBodyFrameSwitch::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialization a [DynBodyFrameSwitch](#) instance.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::BodyAction](#).

Definition at line 57 of file `dyn_body_frame_switch.cc`.

References [jeod::BodyAction::action_identifier](#), [jeod::BodyAction::dyn_subject](#), [jeod::BodyAction::initialize\(\)](#), [integ_frame](#), [integ_frame_name](#), [jeod::BodyActionMessages::invalid_name](#), [jeod::BodyActionMessages::invalid↵_object](#), and [jeod::BodyAction::mass_subject](#).

8.9.4.3 is_ready()

```
bool jeod::DynBodyFrameSwitch::is_ready ( ) [override], [virtual]
```

Determine whether it is time to switch frames.

A frame-switch action is ready if it is activated and if the vehicle has entered/left the appropriate sphere of influence.

Returns

Can action be applied?

Reimplemented from [jeod::BodyAction](#).

Definition at line 167 of file `dyn_body_frame_switch.cc`.

References [jeod::BodyAction::dyn_subject](#), [integ_frame](#), [jeod::BodyAction::is_ready\(\)](#), [switch_distance](#), [switch_sense](#), and [SwitchOnApproach](#).

8.9.4.4 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyFrameSwitch::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyFrameSwitch ) [private]
```

8.9.4.5 operator=()

```
DynBodyFrameSwitch& jeod::DynBodyFrameSwitch::operator= (
    const DynBodyFrameSwitch & ) [delete]
```

8.9.5 Field Documentation

8.9.5.1 integ_frame

```
EphemerisRefFrame* jeod::DynBodyFrameSwitch::integ_frame {} [protected]
```

The reference frame corresponding to the input `integ_frame_name`.

`trick_io(**)`

Definition at line 143 of file `dyn_body_frame_switch.hh`.

Referenced by [apply\(\)](#), [initialize\(\)](#), and [is_ready\(\)](#).

8.9.5.2 integ_frame_name

```
std::string jeod::DynBodyFrameSwitch::integ_frame_name {""}
```

The name of the new integration frame.

This name must specify a valid valid integration frame. Failure to do so constitutes a fatal error. `trick_units(-)`

Definition at line 119 of file `dyn_body_frame_switch.hh`.

Referenced by `apply()`, and `initialize()`.

8.9.5.3 sort_grav_controls

```
bool jeod::DynBodyFrameSwitch::sort_grav_controls {}
```

If set, the body's gravitational controls are sorted in ascending acceleration magnitude.

`trick_units(-)`

Definition at line 132 of file `dyn_body_frame_switch.hh`.

Referenced by `apply()`.

8.9.5.4 switch_distance

```
double jeod::DynBodyFrameSwitch::switch_distance {9e99}
```

The radius of the sphere of influence.

`trick_units(m)`

Definition at line 137 of file `dyn_body_frame_switch.hh`.

Referenced by `is_ready()`.

8.9.5.5 switch_sense

```
SwitchSense jeod::DynBodyFrameSwitch::switch_sense {SwitchOnApproach}
```

Indicates whether the switch occurs when the subject `DynBody` enters a sphere of influence around the new integration frame or leaves a sphere sphere of influence around of the current integration frame.

`trick_units(-)`

Definition at line 126 of file `dyn_body_frame_switch.hh`.

Referenced by `is_ready()`.

The documentation for this class was generated from the following files:

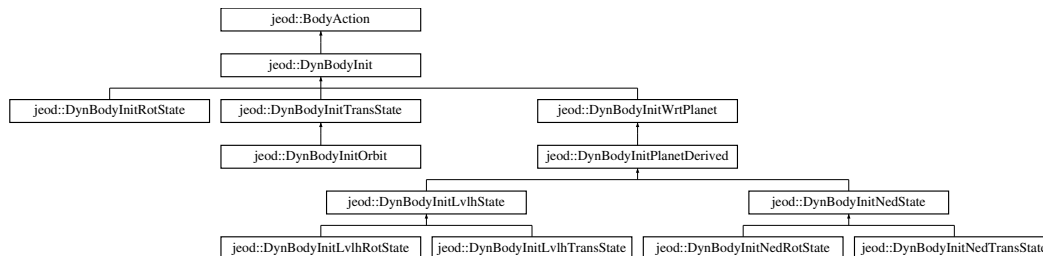
- [dyn_body_frame_switch.hh](#)
- [dyn_body_frame_switch.cc](#)

8.10 jeod::DynBodyInit Class Reference

Base class for initialize the state of a DynBody.

```
#include <dyn_body_init.hh>
```

Inheritance diagram for jeod::DynBodyInit:



Public Member Functions

- [DynBodyInit](#) ()=default
- [~DynBodyInit](#) () override=default
- [DynBodyInit](#) (const [DynBodyInit](#) &)=delete
- [DynBodyInit](#) & [operator=](#) (const [DynBodyInit](#) &)=delete
- virtual void [report_failure](#) ()
Report on an initializer that could not be processed.
- void [initialize](#) (DynManager &dyn_manager) override
Complete initialization of a [DynBodyInit](#).
- virtual RefFrameItems::Items [initializes_what](#) ()
In general, specify what state elements are to be initialized.
- bool [is_ready](#) () override
In general, determine if the initializer is ready to be applied.
- void [apply](#) (DynManager &dyn_manager) override
Complete initialization of the subject DynBody.

Data Fields

- std::string [body_frame_id](#) {}
The suffix of the frame name (i.e., the part of the name after the vehicle identifier) to which this initializer pertains.
- std::string [reference_ref_frame_name](#) {}
The name of the reference frame against which state data specified in a [DynBodyInit](#) subclass are referenced.
- RefFrameState [state](#)
Contains state information set by the initializer, which is always a subclass of [DynBodyInit](#).
- double [position](#) [3] {}
Relative position between the subject and reference reference frame origins.
- double [velocity](#) [3] {}
Relative velocity between the subject and reference reference frame origins.
- Orientation [orientation](#)
Relative orientation between the subject and reference reference frame axes.
- double [ang_velocity](#) [3] {}
Relative angular velocity between the subject and reference axes.
- bool [reverse_sense](#) {}
Indicates how the user input state items are to be interpreted.
- bool [rate_in_parent](#) {}
Indicates how the user input angular velocity is to be interpreted.

Protected Member Functions

- void [apply_user_inputs](#) ()
Compute the state wrt the reference reference frame, incorporate the user-input items to this relative state, and compute the state relative to the target frame's parent.
- void [compute_rotational_state](#) ()
This method is obsolete.
- void [compute_translational_state](#) ()
This method is obsolete.
- Planet * [find_planet](#) (const DynManager &dyn_manager, const std::string &planet_name, const std::string &variable_name)
Find the Planet with the given name, failing if not found.
- DynBody * [find_dyn_body](#) (const DynManager &dyn_manager, const std::string &dyn_body_name, const std::string &variable_name)
Find the DynBody with the given name, failing if not found.
- RefFrame * [find_ref_frame](#) (const DynManager &dyn_manager, const std::string &ref_frame_name, const std::string &variable_name)
Find the RefFrame with the given name, failing if not found.
- BodyRefFrame * [find_body_frame](#) (DynBody &frame_container, const std::string &body_frame_identifier, const std::string &variable_name)
Find the RefFrame with the given name, failing if not found.

Protected Attributes

- BodyRefFrame * [body_ref_frame](#) {}
The reference frame whose name is vehicle_name.body_frame_id.
- RefFrame * [reference_ref_frame](#) {}
The reference frame whose name is reference_ref_frame_name.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, DynBodyInit)

Private Attributes

- RefFrame * [subscribed_frame](#) {}
The subscribed-to frame (the reference_ref_frame at initialization time), cached so that this frame will be unsubscribed at application time.

8.10.1 Detailed Description

Base class for initialize the state of a DynBody.

Definition at line 88 of file dyn_body_init.hh.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 DynBodyInit() [1/2]

```
jeod::DynBodyInit::DynBodyInit ( ) [default]
```

8.10.2.2 ~DynBodyInit()

```
jeod::DynBodyInit::~~DynBodyInit ( ) [override], [default]
```

8.10.2.3 DynBodyInit() [2/2]

```
jeod::DynBodyInit::DynBodyInit (
    const DynBodyInit & ) [delete]
```

8.10.3 Member Function Documentation

8.10.3.1 apply()

```
void jeod::DynBodyInit::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Complete initialization of the subject `DynBody`.

The `apply` method for all subclasses of [DynBodyInit](#) *must* pass the `apply` call to their immediate parent class, which in turn must do the same, eventually invoking this method.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Jeod manager
----------------------	---------------------------------	--------------

Reimplemented from [jeod::BodyAction](#).

Reimplemented in [jeod::DynBodyInitWrtPlanet](#), [jeod::DynBodyInitTransState](#), [jeod::DynBodyInitRotState](#), [jeod::DynBodyInitPlanetDerived](#), [jeod::DynBodyInitOrbit](#), [jeod::DynBodyInitNedState](#), and [jeod::DynBodyInitLvlhState](#).

Definition at line 210 of file `dyn_body_init.cc`.

References [jeod::BodyAction::action_identifier](#), [jeod::BodyAction::apply\(\)](#), [body_ref_frame](#), [jeod::BodyAction::dyn↔_subject](#), [initializes_what\(\)](#), [reference_ref_frame](#), [state](#), [subscribed_frame](#), and [jeod::BodyActionMessages::trace](#).

Referenced by [jeod::DynBodyInitRotState::apply\(\)](#), [jeod::DynBodyInitTransState::apply\(\)](#), and [jeod::DynBodyInit↔WrtPlanet::apply\(\)](#).

8.10.3.2 apply_user_inputs()

```
void jeod::DynBodyInit::apply_user_inputs ( ) [protected]
```

Compute the state wrt the reference reference frame, incorporate the user-input items to this relative state, and compute the state relative to the target frame's parent.

Definition at line 273 of file dyn_body_init.cc.

References ang_velocity, body_ref_frame, jeod::BodyAction::dyn_subject, initializes_what(), orientation, position, rate_in_parent, reference_ref_frame, reverse_sense, state, and velocity.

Referenced by jeod::DynBodyInitLvHState::apply(), jeod::DynBodyInitNedState::apply(), jeod::DynBodyInitRotState::apply(), jeod::DynBodyInitTransState::apply(), compute_rotational_state(), and compute_translational_state().

8.10.3.3 compute_rotational_state()

```
void jeod::DynBodyInit::compute_rotational_state ( ) [protected]
```

This method is obsolete.

Use apply_user_inputs instead.

Definition at line 333 of file dyn_body_init.cc.

References apply_user_inputs(), and jeod::BodyActionMessages::invalid_name.

8.10.3.4 compute_translational_state()

```
void jeod::DynBodyInit::compute_translational_state ( ) [protected]
```

This method is obsolete.

Use apply_user_inputs instead.

Definition at line 353 of file dyn_body_init.cc.

References apply_user_inputs(), and jeod::BodyActionMessages::invalid_name.

8.10.3.5 find_body_frame()

```
BodyRefFrame * jeod::DynBodyInit::find_body_frame (
    DynBody & frame_container,
    const std::string & body_frame_identifier,
    const std::string & variable_name ) [protected]
```

Find the RefFrame with the given name, failing if not found.

Returns

Found BodyRefFrame

Parameters

in	<i>frame_container</i>	Body containing frame
in	<i>body_frame_identifier</i>	BodyRefFrame identifier
in	<i>variable_name</i>	For error reporting

Definition at line 491 of file `dyn_body_init.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyActionMessages::invalid_name`, and `jeod::BodyActionMessages::validate_name()`.

Referenced by `initialize()`.

8.10.3.6 find_dyn_body()

```
DynBody * jeod::DynBodyInit::find_dyn_body (
    const DynManager & dyn_manager,
    const std::string & dyn_body_name,
    const std::string & variable_name ) [protected]
```

Find the DynBody with the given name, failing if not found.

Returns

Found DynBody

Parameters

in	<i>dyn_manager</i>	Dynamics manager
in	<i>dyn_body_name</i>	DynBody name
in	<i>variable_name</i>	For error reporting

Definition at line 415 of file `dyn_body_init.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyActionMessages::invalid_name`, and `jeod::BodyActionMessages::validate_name()`.

Referenced by `jeod::DynBodyInitPlanetDerived::initialize()`.

8.10.3.7 find_planet()

```
Planet * jeod::DynBodyInit::find_planet (
    const DynManager & dyn_manager,
    const std::string & planet_name,
    const std::string & variable_name ) [protected]
```

Find the Planet with the given name, failing if not found.

Returns

Found Planet

Parameters

in	<i>dyn_manager</i>	Dynamics manager
in	<i>planet_name</i>	Planet name
in	<i>variable_name</i>	For error reporting

Definition at line 377 of file `dyn_body_init.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyActionMessages::invalid_name`, and `jeod::BodyActionMessages::validate_name()`.

Referenced by `jeod::DynBodyInitOrbit::initialize()`, and `jeod::DynBodyInitWrtPlanet::initialize()`.

8.10.3.8 find_ref_frame()

```
RefFrame * jeod::DynBodyInit::find_ref_frame (
    const DynManager & dyn_manager,
    const std::string & ref_frame_name,
    const std::string & variable_name ) [protected]
```

Find the RefFrame with the given name, failing if not found.

Returns

Found ref_frame

Parameters

in	<i>dyn_manager</i>	Dynamics manager
in	<i>ref_frame_name</i>	RefFrame name
in	<i>variable_name</i>	For error reporting

Definition at line 453 of file `dyn_body_init.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyActionMessages::invalid_name`, and `jeod::BodyActionMessages::validate_name()`.

Referenced by `initialize()`.

8.10.3.9 initialize()

```
void jeod::DynBodyInit::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Complete initialization of a [DynBodyInit](#).

The initialize method for all subclasses of [DynBodyInit](#) *must* pass the initialize call to their immediate parent class, which in turn must do the same, eventually invoking this method.

Parameters

<code>in, out</code>	<code>dyn_manager</code>	Dynamics manager
----------------------	--------------------------	------------------

Reimplemented from [jeod::BodyAction](#).

Reimplemented in [jeod::DynBodyInitWrtPlanet](#), [jeod::DynBodyInitTransState](#), [jeod::DynBodyInitRotState](#), [jeod::DynBodyInitPlanetDerived](#), [jeod::DynBodyInitOrbit](#), [jeod::DynBodyInitNedTransState](#), [jeod::DynBodyInitNedState](#), [jeod::DynBodyInitNedRotState](#), [jeod::DynBodyInitLvlhTransState](#), [jeod::DynBodyInitLvlhState](#), and [jeod::DynBodyInitLvlhRotState](#).

Definition at line 63 of file `dyn_body_init.cc`.

References `jeod::BodyAction::action_identifier`, `body_frame_id`, `body_ref_frame`, `jeod::BodyAction::dyn_subject`, `find_body_frame()`, `find_ref_frame()`, `jeod::BodyAction::initialize()`, `jeod::BodyActionMessages::invalid_object`, `jeod::BodyAction::mass_subject`, `reference_ref_frame`, `reference_ref_frame_name`, and `subscribed_frame`.

Referenced by `jeod::DynBodyInitRotState::initialize()`, `jeod::DynBodyInitTransState::initialize()`, and `jeod::DynBodyInitWrtPlanet::initialize()`.

8.10.3.10 initializes_what()

```
RefFrameItems::Items jeod::DynBodyInit::initializes_what ( ) [virtual]
```

In general, specify what state elements are to be initialized.

This method indicates that no such elements are initialized. A subclass that does something *must* override this default method.

Returns

Initialized states

Reimplemented in [jeod::DynBodyInitWrtPlanet](#), [jeod::DynBodyInitTransState](#), and [jeod::DynBodyInitRotState](#).

Definition at line 135 of file `dyn_body_init.cc`.

Referenced by `apply()`, `apply_user_inputs()`, `is_ready()`, and `report_failure()`.

8.10.3.11 is_ready()

```
bool jeod::DynBodyInit::is_ready ( ) [override], [virtual]
```

In general, determine if the initializer is ready to be applied.

This method determines whether the self-dependencies are satisfied. Dependencies on the reference reference frame are the responsibility of derived classes.

Returns

Can initializer run?

Reimplemented from [jeod::BodyAction](#).

Reimplemented in [jeod::DynBodyInitWrtPlanet](#), [jeod::DynBodyInitTransState](#), [jeod::DynBodyInitRotState](#), and [jeod::DynBodyInitPlanetDerived](#).

Definition at line 147 of file `dyn_body_init.cc`.

References `body_ref_frame`, `initializes_what()`, `jeod::BodyAction::is_ready()`, `rate_in_parent`, and `reverse_sense`.

Referenced by `jeod::DynBodyInitRotState::is_ready()`, `jeod::DynBodyInitTransState::is_ready()`, and `jeod::DynBodyInitWrtPlanet::is_ready()`.

8.10.3.12 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInit::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInit ) [private]
```

8.10.3.13 operator=()

```
DynBodyInit& jeod::DynBodyInit::operator= (
    const DynBodyInit & ) [delete]
```

8.10.3.14 report_failure()

```
void jeod::DynBodyInit::report_failure ( ) [virtual]
```

Report on an initializer that could not be processed.

Definition at line 246 of file `dyn_body_init.cc`.

References `jeod::BodyAction::action_identifier`, `body_ref_frame`, `initializes_what()`, `jeod::BodyActionMessages::not_performed`, and `reference_ref_frame`.

8.10.4 Field Documentation

8.10.4.1 ang_velocity

```
double jeod::DynBodyInit::ang_velocity[3] {}
```

Relative angular velocity between the subject and reference axes.

The flags `reverse_sense` and `rate_in_parent` give four interpretations:

- Default (both `reverse_sense` and `rate_in_parent` are false):
Angular velocity of the subject frame with respect to the reference frame, expressed in subject frame coordinates.
- `reverse_sense` is clear, `rate_in_parent` is set:
Angular velocity of the subject frame with respect to the reference frame, expressed in reference frame coordinates.
- `reverse_sense` is set, `rate_in_parent` is clear:
Angular velocity of the reference frame with respect to the subject frame, expressed in reference frame coordinates.
- Both `reverse_sense` and `rate_in_parent` are set:
Angular velocity of the reference frame with respect to the subject frame, expressed in subject frame coordinates. `trick_units(rad/s)`

Definition at line 157 of file `dyn_body_init.hh`.

Referenced by `jeod::DynBodyInitLvHState::apply()`, and `apply_user_inputs()`.

8.10.4.2 body_frame_id

```
std::string jeod::DynBodyInit::body_frame_id {""}
```

The suffix of the frame name (i.e., the part of the name after the vehicle identifier) to which this initializer pertains.

`trick_units(-)`

Definition at line 96 of file `dyn_body_init.hh`.

Referenced by `initialize()`.

8.10.4.3 body_ref_frame

```
BodyRefFrame* jeod::DynBodyInit::body_ref_frame {} [protected]
```

The reference frame whose name is vehicle_name.body_frame_id.

This is the frame to which the state is applied. trick_io(**)

Definition at line 181 of file dyn_body_init.hh.

Referenced by apply(), apply_user_inputs(), initialize(), is_ready(), and report_failure().

8.10.4.4 orientation

```
Orientation jeod::DynBodyInit::orientation
```

Relative orientation between the subject and reference reference frame axes.

The normal sense (reverse_sense is not set) is the transformation from reference to subject. The reverse meaning (reverse_sense set) is the transformation from subject to reference. trick_units(-)

Definition at line 139 of file dyn_body_init.hh.

Referenced by jeod::DynBodyInitLvlhState::apply(), and apply_user_inputs().

8.10.4.5 position

```
double jeod::DynBodyInit::position[3] {}
```

Relative position between the subject and reference reference frame origins.

The normal sense (reverse_sense is not set) is the position of the subject origin with respect to the reference origin, expressed in reference coordinates. The reverse meaning (reverse_sense set) is the position of the reference origin with respect to the subject origin, expressed in subject coordinates. trick_units(m)

Definition at line 121 of file dyn_body_init.hh.

Referenced by jeod::DynBodyInitLvlhState::apply(), jeod::DynBodyInitOrbit::apply(), and apply_user_inputs().

8.10.4.6 rate_in_parent

```
bool jeod::DynBodyInit::rate_in_parent {}
```

Indicates how the user input angular velocity is to be interpreted.

This item works in conjunction with reverse_sense. See ang_velocity for a complete description. trick_units(-)

Definition at line 174 of file dyn_body_init.hh.

Referenced by apply_user_inputs(), and is_ready().

8.10.4.7 reference_ref_frame

```
RefFrame* jeod::DynBodyInit::reference_ref_frame {} [protected]
```

The reference frame whose name is `reference_ref_frame_name`.

This is the frame against which the user state is reference. `trick_io(**)`

Definition at line 187 of file `dyn_body_init.hh`.

Referenced by `apply()`, `jeod::DynBodyInitLvIhState::apply()`, `jeod::DynBodyInitNedState::apply()`, `apply_user_inputs()`, `initialize()`, `jeod::DynBodyInitOrbit::initialize()`, `jeod::DynBodyInitWrtPlanet::initialize()`, `jeod::DynBodyInitRotState::is_ready()`, `jeod::DynBodyInitTransState::is_ready()`, and `report_failure()`.

8.10.4.8 reference_ref_frame_name

```
std::string jeod::DynBodyInit::reference_ref_frame_name {""}
```

The name of the reference frame against which state data specified in a [DynBodyInit](#) subclass are referenced.

`trick_units(-)`

Definition at line 102 of file `dyn_body_init.hh`.

Referenced by `initialize()`.

8.10.4.9 reverse_sense

```
bool jeod::DynBodyInit::reverse_sense {}
```

Indicates how the user input state items are to be interpreted.

If clear (default setting), indicates that the user input position, velocity, orientation, and angular velocity are to be interpreted in the standard JEOD parent to child sense. The meaning is reversed when this flag is set. See the descriptions of the individual state items for details. `trick_units(-)`

Definition at line 167 of file `dyn_body_init.hh`.

Referenced by `apply_user_inputs()`, and `is_ready()`.

8.10.4.10 state

```
RefFrameState jeod::DynBodyInit::state
```

Contains state information set by the initializer, which is always a subclass of [DynBodyInit](#).

The [DynBodyInit](#) apply method copies the state elements indicated by the initializer's `initializes_what` method to the frame indicated by the `frame_id` and then propagates the initialized states up/down the vehicle attachment tree. `trick_units(-)`

Definition at line 111 of file `dyn_body_init.hh`.

Referenced by `apply()`, and `apply_user_inputs()`.

8.10.4.11 subscribed_frame

```
RefFrame* jeod::DynBodyInit::subscribed_frame {} [private]
```

The subscribed-to frame (the `reference_ref_frame` at initialization time), cached so that this frame will be unsubscribed at application time.

`trick_io(**)`

Definition at line 195 of file `dyn_body_init.hh`.

Referenced by `apply()`, and `initialize()`.

8.10.4.12 velocity

```
double jeod::DynBodyInit::velocity[3] {}
```

Relative velocity between the subject and reference reference frame origins.

The normal sense (`reverse_sense` is not set) is the velocity of the subject origin with respect to the reference origin, expressed in and observed in reference coordinates. The reverse meaning (`reverse_sense` set) is the velocity of the reference origin with respect to the subject origin, expressed in and observed in subject coordinates. `trick_units(m/s)`

Definition at line 131 of file `dyn_body_init.hh`.

Referenced by `jeod::DynBodyInitLvlhState::apply()`, `jeod::DynBodyInitOrbit::apply()`, and `apply_user_inputs()`.

The documentation for this class was generated from the following files:

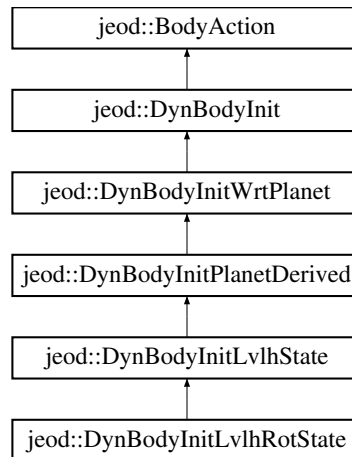
- [dyn_body_init.hh](#)
- [dyn_body_init.cc](#)

8.11 jeod::DynBodyInitLvlhRotState Class Reference

Initialize a vehicle's rotational state with respect to some vehicle's LVLH frame.

```
#include <dyn_body_init_lvlh_rot_state.hh>
```

Inheritance diagram for jeod::DynBodyInitLvlhRotState:



Public Member Functions

- [DynBodyInitLvlhRotState \(\)](#)
DynBodyInitLvlhRotState default constructor.
- [~DynBodyInitLvlhRotState \(\)](#) override=default
- [DynBodyInitLvlhRotState \(const DynBodyInitLvlhRotState &\)=delete](#)
- [DynBodyInitLvlhRotState & operator= \(const DynBodyInitLvlhRotState &\)=delete](#)
- void [initialize](#) (DynManager &dyn_manager) override
Initialize the initializer.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, DynBodyInitLvlhRotState)

Additional Inherited Members

8.11.1 Detailed Description

Initialize a vehicle's rotational state with respect to some vehicle's LVLH frame.

That some vehicle can be this vehicle itself.

Definition at line 82 of file `dyn_body_init_lvlh_rot_state.hh`.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 DynBodyInitLvLhRotState() [1/2]

```
jeod::DynBodyInitLvLhRotState::DynBodyInitLvLhRotState ( )
```

[DynBodyInitLvLhRotState](#) default constructor.

Definition at line 56 of file `dyn_body_init_lvLh_rot_state.cc`.

References `jeod::DynBodyInitWrtPlanet::set_items`.

8.11.2.2 ~DynBodyInitLvLhRotState()

```
jeod::DynBodyInitLvLhRotState::~~DynBodyInitLvLhRotState ( ) [override], [default]
```

8.11.2.3 DynBodyInitLvLhRotState() [2/2]

```
jeod::DynBodyInitLvLhRotState::DynBodyInitLvLhRotState (
    const DynBodyInitLvLhRotState & ) [delete]
```

8.11.3 Member Function Documentation

8.11.3.1 initialize()

```
void jeod::DynBodyInitLvLhRotState::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Dynamics manager
----------------------	---------------------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 65 of file `dyn_body_init_lvLh_rot_state.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyAction::get_subject_dyn_body()`, `jeod::BodyActionMessages::illegal_value`, `jeod::DynBodyInitLvLhState::initialize()`, `jeod::BodyActionMessages::null_pointer`, `jeod::DynBodyInitPlanetDerived::ref_body`, `jeod::DynBodyInitPlanetDerived::ref_body_name`, and `jeod::DynBodyInitWrtPlanet::set_items`.

8.11.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitLvlhRotState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitLvlhRotState ) [private]
```

8.11.3.3 operator=()

```
DynBodyInitLvlhRotState& jeod::DynBodyInitLvlhRotState::operator= (
    const DynBodyInitLvlhRotState & ) [delete]
```

The documentation for this class was generated from the following files:

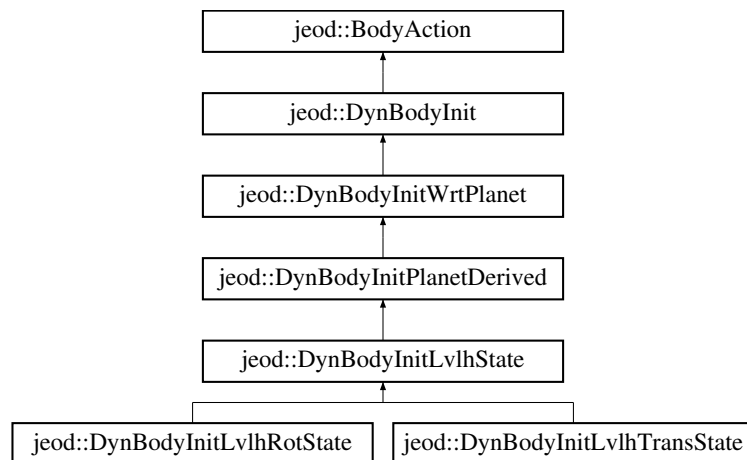
- [dyn_body_init_lvlh_rot_state.hh](#)
- [dyn_body_init_lvlh_rot_state.cc](#)

8.12 jeod::DynBodyInitLvlhState Class Reference

Initialize selected aspects of a vehicle's state with respect to some vehicle's LVLH frame.

```
#include <dyn_body_init_lvlh_state.hh>
```

Inheritance diagram for jeod::DynBodyInitLvlhState:



Public Member Functions

- [DynBodyInitLvlhState \(\)](#)
DynBodyInitLvlhState default constructor.
- [~DynBodyInitLvlhState \(\)](#) override=default
- [DynBodyInitLvlhState \(const DynBodyInitLvlhState &\)=delete](#)
- [DynBodyInitLvlhState & operator= \(const DynBodyInitLvlhState &\)=delete](#)
- void [set_lvlh_frame_object](#) (LvlhFrame &lvlh_frame_object)
Cache a pointer to a user-supplied LvlhFrame object.
- void [initialize](#) (DynManager &dyn_manager) override
Initialize the initializer.
- void [apply](#) (DynManager &dyn_manager) override
Apply the initializer: Construct the reference LVLH frame so the parent initializer can compute the vehicle's state relative to the vehicle's inertial frame.

Data Fields

- `LvlhType::Type` [lvlh_type](#) {`LvlhType::Rectilinear`}
- Indicates type of LVLH coordinates desired.*

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (`jeod`, [DynBodyInitLvlhState](#))

Private Attributes

- `LvlhFrame *` [lvlh_object_ptr](#) {}
- A pointer to an LvlhFrame which can be supplied by the user.*

Additional Inherited Members

8.12.1 Detailed Description

Initialize selected aspects of a vehicle's state with respect to some vehicle's LVLH frame.

Definition at line 84 of file `dyn_body_init_lvlh_state.hh`.

8.12.2 Constructor & Destructor Documentation

8.12.2.1 `DynBodyInitLvlhState()` [1/2]

```
jeod::DynBodyInitLvlhState::DynBodyInitLvlhState ( )
```

[DynBodyInitLvlhState](#) default constructor.

Definition at line 47 of file `dyn_body_init_lvlh_state.cc`.

References `jeod::DynBodyInitPlanetDerived::required_items`.

8.12.2.2 `~DynBodyInitLvlhState()`

```
jeod::DynBodyInitLvlhState::~~DynBodyInitLvlhState ( ) [override], [default]
```


8.12.2.3 DynBodyInitLvlhState() [2/2]

```
jeod::DynBodyInitLvlhState::DynBodyInitLvlhState (
    const DynBodyInitLvlhState & ) [delete]
```

8.12.3 Member Function Documentation

8.12.3.1 apply()

```
void jeod::DynBodyInitLvlhState::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Apply the initializer: Construct the reference LVLH frame so the parent initializer can compute the vehicle's state relative to the vehicle's inertial frame.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 88 of file `dyn_body_init_lvlh_state.cc`.

References [jeod::DynBodyInit::ang_velocity](#), [jeod::DynBodyInitPlanetDerived::apply\(\)](#), [jeod::DynBodyInit::apply_↔
user_inputs\(\)](#), [jeod::BodyActionMessages::illegal_value](#), [lvlh_object_ptr](#), [lvlh_type](#), [jeod::DynBodyInit::orientation](#), [jeod::DynBodyInitWrtPlanet::planet](#), [jeod::DynBodyInit::position](#), [jeod::DynBodyInitPlanetDerived::ref_body](#), [jeod↔
::DynBodyInit::reference_ref_frame](#), and [jeod::DynBodyInit::velocity](#).

8.12.3.2 initialize()

```
void jeod::DynBodyInitLvlhState::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Reimplemented in [jeod::DynBodyInitLvlhTransState](#).

Definition at line 65 of file `dyn_body_init_lvlh_state.cc`.

References `jeod::DynBodyInitPlanetDerived::body_is_required`, `jeod::DynBodyInitPlanetDerived::initialize()`, and `lvlh_object_ptr`.

Referenced by `jeod::DynBodyInitLvlhRotState::initialize()`, and `jeod::DynBodyInitLvlhTransState::initialize()`.

8.12.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitLvlhState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitLvlhState ) [private]
```

8.12.3.4 operator=()

```
DynBodyInitLvlhState& jeod::DynBodyInitLvlhState::operator= (
    const DynBodyInitLvlhState & ) [delete]
```

8.12.3.5 set_lvlh_frame_object()

```
void jeod::DynBodyInitLvlhState::set_lvlh_frame_object (
    LvlhFrame & lvlh_frame_object )
```

Cache a pointer to a user-supplied `LvlhFrame` object.

Parameters

in	<i>lvlh_frame_object</i>	LVLH frame object
----	--------------------------	-------------------

Definition at line 56 of file `dyn_body_init_lvlh_state.cc`.

References `lvlh_object_ptr`.

8.12.4 Field Documentation

8.12.4.1 lvlh_object_ptr

```
LvlhFrame* jeod::DynBodyInitLvlhState::lvlh_object_ptr {} [private]
```

A pointer to an `LvlhFrame` which can be supplied by the user.

`trick_units(-)`

Definition at line 98 of file `dyn_body_init_lvlh_state.hh`.

Referenced by `apply()`, `initialize()`, and `set_lvlh_frame_object()`.

8.12.4.2 lvlh_type

```
LvlhType::Type jeod::DynBodyInitLvlhState::lvlh_type {LvlhType::Rectilinear}
```

Indicates type of LVLH coordinates desired.

Default is rectilinear. `trick_units(-)`

Definition at line 92 of file `dyn_body_init_lvlh_state.hh`.

Referenced by `apply()`.

The documentation for this class was generated from the following files:

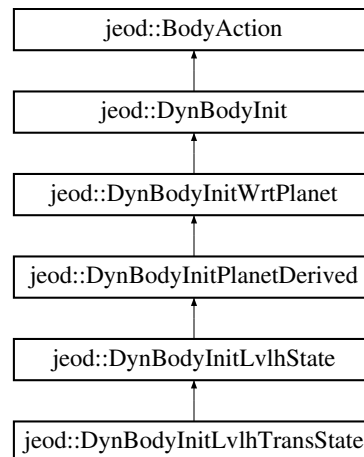
- [dyn_body_init_lvlh_state.hh](#)
- [dyn_body_init_lvlh_state.cc](#)

8.13 jeod::DynBodyInitLvlhTransState Class Reference

initialize a vehicle's translational state with respect to some other vehicle's LVLH frame.

```
#include <dyn_body_init_lvlh_trans_state.hh>
```

Inheritance diagram for `jeod::DynBodyInitLvlhTransState`:



Public Member Functions

- [DynBodyInitLvlhTransState](#) ()
DynBodyInitLvlhTransState default constructor.
- [~DynBodyInitLvlhTransState](#) () override=default
- [DynBodyInitLvlhTransState](#) (const [DynBodyInitLvlhTransState](#) &)=delete
- [DynBodyInitLvlhTransState](#) & operator= (const [DynBodyInitLvlhTransState](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize the initializer.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitLvlhTransState](#))

Additional Inherited Members

8.13.1 Detailed Description

initialize a vehicle's translational state with respect to some other vehicle's LVLH frame.

Definition at line 82 of file `dyn_body_init_lvlh_trans_state.hh`.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 [DynBodyInitLvlhTransState\(\)](#) [1/2]

```
jeod::DynBodyInitLvlhTransState::DynBodyInitLvlhTransState ( )
```

[DynBodyInitLvlhTransState](#) default constructor.

Definition at line 51 of file `dyn_body_init_lvlh_trans_state.cc`.

References `jeod::DynBodyInitWrtPlanet::set_items`.

8.13.2.2 [~DynBodyInitLvlhTransState\(\)](#)

```
jeod::DynBodyInitLvlhTransState::~~DynBodyInitLvlhTransState ( ) [override], [default]
```

8.13.2.3 [DynBodyInitLvlhTransState\(\)](#) [2/2]

```
jeod::DynBodyInitLvlhTransState::DynBodyInitLvlhTransState (
    const DynBodyInitLvlhTransState & ) [delete]
```

8.13.3 Member Function Documentation

8.13.3.1 [initialize\(\)](#)

```
void jeod::DynBodyInitLvlhTransState::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

<code>in, out</code>	<code>dyn_manager</code>	Dynamics manager
----------------------	--------------------------	------------------

Reimplemented from [jeod::DynBodyInitLvlhState](#).

Definition at line 60 of file `dyn_body_init_lvlh_trans_state.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyActionMessages::illegal_value`, `jeod::DynBodyInitLvlhState::initialize()`, and `jeod::DynBodyInitWrtPlanet::set_items`.

8.13.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitLvlhTransState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitLvlhTransState ) [private]
```

8.13.3.3 operator=()

```
DynBodyInitLvlhTransState& jeod::DynBodyInitLvlhTransState::operator= (
    const DynBodyInitLvlhTransState & ) [delete]
```

The documentation for this class was generated from the following files:

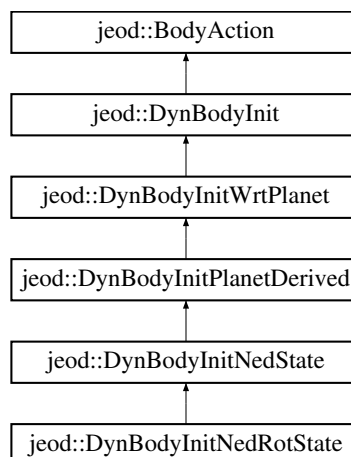
- [dyn_body_init_lvlh_trans_state.hh](#)
- [dyn_body_init_lvlh_trans_state.cc](#)

8.14 jeod::DynBodyInitNedRotState Class Reference

Initialize a vehicle's rotational state wrt some vehicle's North-East-Down frame.

```
#include <dyn_body_init_ned_rot_state.hh>
```

Inheritance diagram for `jeod::DynBodyInitNedRotState`:



Public Member Functions

- [DynBodyInitNedRotState](#) ()
DynBodyInitNedRotState default constructor.
- [~DynBodyInitNedRotState](#) () override=default
- [DynBodyInitNedRotState](#) (const [DynBodyInitNedRotState](#) &)=delete
- [DynBodyInitNedRotState](#) & [operator=](#) (const [DynBodyInitNedRotState](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize the initializer.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitNedRotState](#))

Additional Inherited Members

8.14.1 Detailed Description

Initialize a vehicle's rotational state wrt some vehicle's North-East-Down frame.

Definition at line 82 of file `dyn_body_init_ned_rot_state.hh`.

8.14.2 Constructor & Destructor Documentation

8.14.2.1 DynBodyInitNedRotState() [1/2]

```
jeod::DynBodyInitNedRotState::DynBodyInitNedRotState ( )
```

[DynBodyInitNedRotState](#) default constructor.

Definition at line 53 of file `dyn_body_init_ned_rot_state.cc`.

References `jeod::DynBodyInitWrtPlanet::set_items`.

8.14.2.2 ~DynBodyInitNedRotState()

```
jeod::DynBodyInitNedRotState::~~DynBodyInitNedRotState ( ) [override], [default]
```

8.14.2.3 DynBodyInitNedRotState() [2/2]

```
jeod::DynBodyInitNedRotState::DynBodyInitNedRotState (
    const DynBodyInitNedRotState & ) [delete]
```

8.14.3 Member Function Documentation

8.14.3.1 initialize()

```
void jeod::DynBodyInitNedRotState::initialize (  
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

<code>in, out</code>	<code>dyn_manager</code>	Dynamics manager
----------------------	--------------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 62 of file `dyn_body_init_ned_rot_state.cc`.

References [jeod::BodyAction::action_identifier](#), [jeod::BodyActionMessages::illegal_value](#), [jeod::DynBodyInitNedState::initialize\(\)](#), and [jeod::DynBodyInitWrtPlanet::set_items](#).

8.14.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitNedRotState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitNedRotState ) [private]
```

8.14.3.3 operator=()

```
DynBodyInitNedRotState& jeod::DynBodyInitNedRotState::operator= (
    const DynBodyInitNedRotState & ) [delete]
```

The documentation for this class was generated from the following files:

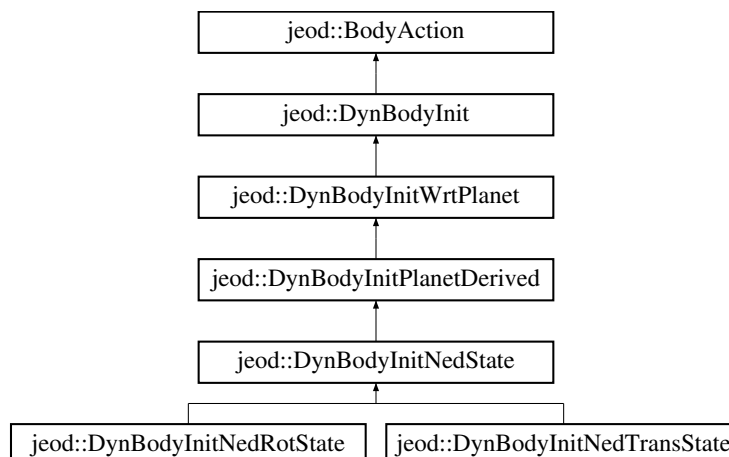
- [dyn_body_init_ned_rot_state.hh](#)
- [dyn_body_init_ned_rot_state.cc](#)

8.15 jeod::DynBodyInitNedState Class Reference

Initialize selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet.

```
#include <dyn_body_init_ned_state.hh>
```

Inheritance diagram for `jeod::DynBodyInitNedState`:



Public Member Functions

- [DynBodyInitNedState](#) ()
DynBodyInitNedState default constructor.
- [~DynBodyInitNedState](#) () override=default
- [DynBodyInitNedState](#) (const [DynBodyInitNedState](#) &)=delete
- [DynBodyInitNedState](#) & [operator=](#) (const [DynBodyInitNedState](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize the initializer.
- void [apply](#) (DynManager &dyn_manager) override
Apply the initializer.
- void [set_use_alt_pfix](#) (const bool use_alt_pfix_in)
Setter for use_alt_pfix.

Data Fields

- AltLatLongState [ref_point](#)
Reference point for the local geodetic/geocentric, used only if the reference body is NULL.
- NorthEastDown::AltLatLongType [altlatlong_type](#) {NorthEastDown::undefined}
Use spherical or elliptical coordinates?

Protected Attributes

- bool [use_alt_pfix](#) {}
Use pfix or alt_pfix flag.
- EphemerisRefFrame * [pfix_ptr](#) {}
Pointer to planet fixed frame to be used, either pfix or alt_pfix.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitNedState](#))

Additional Inherited Members

8.15.1 Detailed Description

Initialize selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet.

Definition at line 87 of file `dyn_body_init_ned_state.hh`.

8.15.2 Constructor & Destructor Documentation

8.15.2.1 DynBodyInitNedState() [1/2]

```
jeod::DynBodyInitNedState::DynBodyInitNedState ( )
```

[DynBodyInitNedState](#) default constructor.

Definition at line 59 of file `dyn_body_init_ned_state.cc`.

References `jeod::DynBodyInitPlanetDerived::body_is_required`, and `jeod::DynBodyInitPlanetDerived::required_`↔
items.

8.15.2.2 ~DynBodyInitNedState()

```
jeod::DynBodyInitNedState::~~DynBodyInitNedState ( ) [override], [default]
```

8.15.2.3 DynBodyInitNedState() [2/2]

```
jeod::DynBodyInitNedState::DynBodyInitNedState (
    const DynBodyInitNedState & ) [delete]
```

8.15.3 Member Function Documentation

8.15.3.1 apply()

```
void jeod::DynBodyInitNedState::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Apply the initializer.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Dynamics manager
----------------------	---------------------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 104 of file `dyn_body_init_ned_state.cc`.

References `jeod::BodyAction::action_identifier`, `altlatlong_type`, `jeod::DynBodyInitPlanetDerived::apply()`, `jeod::`↔
`DynBodyInit::apply_user_inputs()`, `jeod::BodyActionMessages::illegal_value`, `pfix_ptr`, `jeod::DynBodyInitWrt`↔
`Planet::planet`, `jeod::DynBodyInitWrtPlanet::planet_name`, `jeod::DynBodyInitPlanetDerived::ref_body`, `ref_point`,
`jeod::DynBodyInit::reference_ref_frame`, and `jeod::DynBodyInitWrtPlanet::set_items`.

8.15.3.2 initialize()

```
void jeod::DynBodyInitNedState::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Reimplemented in [jeod::DynBodyInitNedTransState](#).

Definition at line 77 of file `dyn_body_init_ned_state.cc`.

References [jeod::DynBodyInitPlanetDerived::body_is_required](#), [jeod::DynBodyInitPlanetDerived::initialize\(\)](#), [pfix↔_ptr](#), [jeod::DynBodyInitWrtPlanet::planet](#), [jeod::DynBodyInitPlanetDerived::ref_body_name](#), and [use_alt_pfix](#).

Referenced by [jeod::DynBodyInitNedRotState::initialize\(\)](#), and [jeod::DynBodyInitNedTransState::initialize\(\)](#).

8.15.3.3 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitNedState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitNedState ) [private]
```

8.15.3.4 operator=()

```
DynBodyInitNedState& jeod::DynBodyInitNedState::operator= (
    const DynBodyInitNedState & ) [delete]
```

8.15.3.5 set_use_alt_pfix()

```
void jeod::DynBodyInitNedState::set_use_alt_pfix (
    const bool use_alt_pfix_in )
```

Setter for `use_alt_pfix`.

Definition at line 68 of file `dyn_body_init_ned_state.cc`.

References [use_alt_pfix](#).

8.15.4 Field Documentation

8.15.4.1 altlatlong_type

```
NorthEastDown::AltLatLongType jeod::DynBodyInitNedState::altlatlong_type {NorthEastDown←
::undefined}
```

Use spherical or elliptical coordinates?

trick_units(—)

Definition at line 101 of file dyn_body_init_ned_state.hh.

Referenced by apply().

8.15.4.2 pfix_ptr

```
EphemerisRefFrame* jeod::DynBodyInitNedState::pfix_ptr {} [protected]
```

Pointer to planet fixed frame to be used, either pfix or alt_pfix.

trick_units(—)

Definition at line 113 of file dyn_body_init_ned_state.hh.

Referenced by apply(), and initialize().

8.15.4.3 ref_point

```
AltLatLongState jeod::DynBodyInitNedState::ref_point
```

Reference point for the local geodetic/geocentric, used only if the reference body is NULL.

trick_units(—)

Definition at line 96 of file dyn_body_init_ned_state.hh.

Referenced by apply().

8.15.4.4 use_alt_pfix

```
bool jeod::DynBodyInitNedState::use_alt_pfix {} [protected]
```

Use pfix or alt_pfix flag.

trick_units(—)

Definition at line 107 of file dyn_body_init_ned_state.hh.

Referenced by initialize(), and set_use_alt_pfix().

The documentation for this class was generated from the following files:

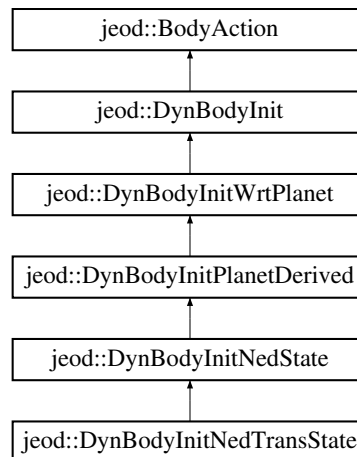
- [dyn_body_init_ned_state.hh](#)
- [dyn_body_init_ned_state.cc](#)

8.16 jeod::DynBodyInitNedTransState Class Reference

Initialize a vehicle's translational state wrt some vehicle's North-East-Down frame.

```
#include <dyn_body_init_ned_trans_state.hh>
```

Inheritance diagram for jeod::DynBodyInitNedTransState:



Public Member Functions

- [DynBodyInitNedTransState](#) ()
DynBodyInitNedTransState default constructor.
- [~DynBodyInitNedTransState](#) () override=default
- [DynBodyInitNedTransState](#) (const [DynBodyInitNedTransState](#) &)=delete
- [DynBodyInitNedTransState](#) & operator= (const [DynBodyInitNedTransState](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize the initializer.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitNedTransState](#))

Additional Inherited Members

8.16.1 Detailed Description

Initialize a vehicle's translational state wrt some vehicle's North-East-Down frame.

Definition at line 82 of file `dyn_body_init_ned_trans_state.hh`.

8.16.2 Constructor & Destructor Documentation

8.16.2.1 `DynBodyInitNedTransState()` [1/2]

```
jeod::DynBodyInitNedTransState::DynBodyInitNedTransState ( )
```

[DynBodyInitNedTransState](#) default constructor.

Definition at line 51 of file `dyn_body_init_ned_trans_state.cc`.

References `jeod::DynBodyInitWrtPlanet::set_items`.

8.16.2.2 `~DynBodyInitNedTransState()`

```
jeod::DynBodyInitNedTransState::~~DynBodyInitNedTransState ( ) [override], [default]
```

8.16.2.3 `DynBodyInitNedTransState()` [2/2]

```
jeod::DynBodyInitNedTransState::DynBodyInitNedTransState (
    const DynBodyInitNedTransState & ) [delete]
```

8.16.3 Member Function Documentation

8.16.3.1 `initialize()`

```
void jeod::DynBodyInitNedTransState::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

<code>in, out</code>	<code>dyn_manager</code>	Dynamics manager
----------------------	--------------------------	------------------

Reimplemented from [jeod::DynBodyInitNedState](#).

Definition at line 60 of file `dyn_body_init_ned_trans_state.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyActionMessages::illegal_value`, `jeod::DynBodyInitNedState::initialize()`, and `jeod::DynBodyInitWrtPlanet::set_items`.

8.16.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitNedTransState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitNedTransState ) [private]
```

8.16.3.3 operator=()

```
DynBodyInitNedTransState& jeod::DynBodyInitNedTransState::operator= (
    const DynBodyInitNedTransState & ) [delete]
```

The documentation for this class was generated from the following files:

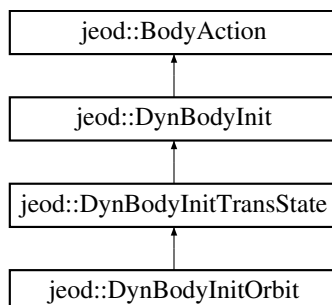
- [dyn_body_init_ned_trans_state.hh](#)
- [dyn_body_init_ned_trans_state.cc](#)

8.17 jeod::DynBodyInitOrbit Class Reference

Initialize a vehicle's translational state given an orbital specification.

```
#include <dyn_body_init_orbit.hh>
```

Inheritance diagram for `jeod::DynBodyInitOrbit`:



Public Types

- enum `OrbitalSet` {
`InvalidSet` = 0 , `SmaEcclncAscnodeArgperTimeperi` = 1 , `SmaEcclncAscnodeArgperManom` = 2 ,
`SlrEcclncAscnodeArgperTanom` = 3 ,
`IncAscnodeAltperAltapoArgperTanom` = 4 , `IncAscnodeAltperAltapoArgperTimeperi` = 5 , `SmaIncAscnodeArglatRadRadvel`
= 6 , `SmaEcclncAscnodeArgperTanom` = 10 ,
`CaseEleven` = 11 }

Identifies which orbital elements define the orbit.

Public Member Functions

- `DynBodyInitOrbit` ()=default
- `~DynBodyInitOrbit` () override=default
- `DynBodyInitOrbit` (const `DynBodyInitOrbit` &)=delete
- `DynBodyInitOrbit` & `operator=` (const `DynBodyInitOrbit` &)=delete
- void `initialize` (DynManager &dyn_manager) override

Initialize the initializer.

- void `apply` (DynManager &dyn_manager) override

Apply the initializer.

Data Fields

- std::string `planet_name`
The name of the planet around which the orbit is to be established.
- std::string `orbit_frame_name`
Planet reference frame name, optionally dot-prefixed with the planet name.
- `OrbitalSet` `set` {`InvalidSet`}
Specifies which set of orbital elements specify the orbit.
- double `semi_major_axis` {}
Semi-major axis.
- double `semi_latus_rectum` {}
Semi-latus rectum.
- double `alt_periapsis` {}
Periapsis altitude.
- double `alt_apoapsis` {}
Apoapsis altitude.
- double `orb_radius` {}
Distance from center of planet.
- double `radial_vel` {}
Time derivative of the orbital radius.
- double `eccentricity` {}
Eccentricity.
- double `inclination` {}
Inclination.
- double `ascending_node` {}
Longitude (or right ascension) of ascending node.
- double `arg_periapsis` {}
Argument of periapsis.
- double `arg_latitude` {}

- *Argument of latitude.*
double [time_periapsis](#) {}
- *Time since periapsis passage.*
double [mean_anomaly](#) {}
- *Mean anomaly.*
double [true_anomaly](#) {}
- *True anomaly.*

Protected Attributes

- Planet * [planet](#) {}
The planet.
- EphemerisRefFrame * [orbit_frame](#) {}
The orbit reference frame (ignoring rotation)

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitOrbit](#))

Additional Inherited Members

8.17.1 Detailed Description

Initialize a vehicle's translational state given an orbital specification.

Definition at line 82 of file `dyn_body_init_orbit.hh`.

8.17.2 Member Enumeration Documentation

8.17.2.1 OrbitalSet

```
enum jeod::DynBodyInitOrbit::OrbitalSet
```

Identifies which orbital elements define the orbit.

The goofy numbering scheme here is intentional. The numbers map directly to the corresponding `orbital_set` number in JEOD 1.4 / 1.5. NOTE: Orbital sets 4 and 11 are the same options.

Enumerator

InvalidSet	
SmaEccIncAscnodeArgperTimeperi	
SmaEccIncAscnodeArgperManom	
SlrEccIncAscnodeArgperTanom	
IncAscnodeAltperAltapoArgperTanom	
IncAscnodeAltperAltapoArgperTimeperi	
SmaIncAscnodeArglatRadRadvel	
SmaEccIncAscnodeArgperTanom	
CaseEleven	

Definition at line 94 of file `dyn_body_init_orbit.hh`.

8.17.3 Constructor & Destructor Documentation

8.17.3.1 DynBodyInitOrbit() [1/2]

```
jeod::DynBodyInitOrbit::DynBodyInitOrbit ( ) [default]
```

8.17.3.2 ~DynBodyInitOrbit()

```
jeod::DynBodyInitOrbit::~~DynBodyInitOrbit ( ) [override], [default]
```

8.17.3.3 DynBodyInitOrbit() [2/2]

```
jeod::DynBodyInitOrbit::DynBodyInitOrbit (
    const DynBodyInitOrbit & ) [delete]
```

8.17.4 Member Function Documentation

8.17.4.1 apply()

```
void jeod::DynBodyInitOrbit::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Apply the initializer.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Dynamics manager
----------------------	---------------------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 157 of file `dyn_body_init_orbit.cc`.

References `jeod::BodyAction::action_identifier`, `alt_apoapsis`, `alt_periapsis`, `jeod::DynBodyInitTransState::apply()`, `arg_latitude`, `arg_periapsis`, `ascending_node`, `CaseEleven`, `eccentricity`, `jeod::BodyActionMessages::illegal_value`,

IncAscnodeAltperAltapoArgperTanom, IncAscnodeAltperAltapoArgperTimeperi, inclination, InvalidSet, mean_↵ anomaly, orb_radius, orbit_frame, planet, jeod::DynBodyInit::position, radial_vel, semi_latus_rectum, semi_major_↵ _axis, set, SlrEcclncAscnodeArgperTanom, SmaEcclncAscnodeArgperManom, SmaEcclncAscnodeArgperTanom, SmaEcclncAscnodeArgperTimeperi, SmalncAscnodeArglatRadRadvel, time_periapsis, true_anomaly, and jeod::↵ DynBodyInit::velocity.

8.17.4.2 initialize()

```
void jeod::DynBodyInitOrbit::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

in, out	<i>dyn_manager</i>	Dynamics manager
---------	--------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 60 of file `dyn_body_init_orbit.cc`.

References [jeod::BodyAction::action_identifier](#), [CaseEleven](#), [jeod::DynBodyInit::find_planet\(\)](#), [jeod::BodyAction↵ Messages::illegal_value](#), [IncAscnodeAltperAltapoArgperTanom](#), [IncAscnodeAltperAltapoArgperTimeperi](#), [jeod↵ ::DynBodyInitTransState::initialize\(\)](#), [jeod::BodyActionMessages::invalid_name](#), [jeod::BodyActionMessages↵ ::invalid_object](#), [InvalidSet](#), [orbit_frame](#), [orbit_frame_name](#), [planet](#), [planet_name](#), [jeod::DynBodyInit::reference_↵ ref_frame](#), [set](#), [SlrEcclncAscnodeArgperTanom](#), [SmaEcclncAscnodeArgperManom](#), [SmaEcclncAscnodeArgper↵ Tanom](#), [SmaEcclncAscnodeArgperTimeperi](#), [SmalncAscnodeArglatRadRadvel](#), and [jeod::BodyAction::validate_↵ name\(\)](#).

8.17.4.3 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitOrbit::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitOrbit ) [private]
```

8.17.4.4 operator=()

```
DynBodyInitOrbit& jeod::DynBodyInitOrbit::operator= (
    const DynBodyInitOrbit & ) [delete]
```

8.17.5 Field Documentation

8.17.5.1 alt_apoapsis

```
double jeod::DynBodyInitOrbit::alt_apoapsis {}
```

Apoapsis altitude.

trick_units(m)

Definition at line 204 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.2 alt_periapsis

```
double jeod::DynBodyInitOrbit::alt_periapsis {}
```

Periapsis altitude.

trick_units(m)

Definition at line 199 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.3 arg_latitude

```
double jeod::DynBodyInitOrbit::arg_latitude {}
```

Argument of latitude.

trick_units(rad)

Definition at line 239 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.4 arg_periapsis

```
double jeod::DynBodyInitOrbit::arg_periapsis {}
```

Argument of periapsis.

trick_units(rad)

Definition at line 234 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.5 ascending_node

```
double jeod::DynBodyInitOrbit::ascending_node {}
```

Longitude (or right ascension) of ascending node.

trick_units(rad)

Definition at line 229 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.6 eccentricity

```
double jeod::DynBodyInitOrbit::eccentricity {}
```

Eccentricity.

trick_units(-)

Definition at line 219 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.7 inclination

```
double jeod::DynBodyInitOrbit::inclination {}
```

Inclination.

trick_units(rad)

Definition at line 224 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.8 mean_anomaly

```
double jeod::DynBodyInitOrbit::mean_anomaly {}
```

Mean anomaly.

trick_units(rad)

Definition at line 249 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.9 orb_radius

```
double jeod::DynBodyInitOrbit::orb_radius {}
```

Distance from center of planet.

trick_units(m)

Definition at line 209 of file dyn_body_init_orbit.hh.

Referenced by apply().

8.17.5.10 orbit_frame

```
EphemerisRefFrame* jeod::DynBodyInitOrbit::orbit_frame {} [protected]
```

The orbit reference frame (ignoring rotation)

trick_io(**)

Definition at line 265 of file dyn_body_init_orbit.hh.

Referenced by apply(), and initialize().

8.17.5.11 orbit_frame_name

```
std::string jeod::DynBodyInitOrbit::orbit_frame_name
```

Planet reference frame name, optionally dot-prefixed with the planet name.

If this specifies a rotating frame, a non-rotating frame instantaneously co-aligned with the rotating frame is assumed.

trick_units(-)

Definition at line 179 of file dyn_body_init_orbit.hh.

Referenced by initialize().

8.17.5.12 planet

```
Planet* jeod::DynBodyInitOrbit::planet {} [protected]
```

The planet.

trick_io(**)

Definition at line 260 of file dyn_body_init_orbit.hh.

Referenced by apply(), and initialize().

8.17.5.13 planet_name

```
std::string jeod::DynBodyInitOrbit::planet_name
```

The name of the planet around which the orbit is to be established.

This must be supplied, must name a planet, and the planet must have a gravity model. `trick_units(-)`

Definition at line 172 of file `dyn_body_init_orbit.hh`.

Referenced by `initialize()`.

8.17.5.14 radial_vel

```
double jeod::DynBodyInitOrbit::radial_vel {}
```

Time derivative of the orbital radius.

`trick_units(m/s)`

Definition at line 214 of file `dyn_body_init_orbit.hh`.

Referenced by `apply()`.

8.17.5.15 semi_latus_rectum

```
double jeod::DynBodyInitOrbit::semi_latus_rectum {}
```

Semi-latus rectum.

`trick_units(m)`

Definition at line 194 of file `dyn_body_init_orbit.hh`.

Referenced by `apply()`.

8.17.5.16 semi_major_axis

```
double jeod::DynBodyInitOrbit::semi_major_axis {}
```

Semi-major axis.

`trick_units(m)`

Definition at line 189 of file `dyn_body_init_orbit.hh`.

Referenced by `apply()`.

8.17.5.17 set

```
OrbitalSet jeod::DynBodyInitOrbit::set {InvalidSet}
```

Specifies which set of orbital elements specify the orbit.

trick_units(—)

Definition at line 184 of file dyn_body_init_orbit.hh.

Referenced by `apply()`, and `initialize()`.

8.17.5.18 time_periapsis

```
double jeod::DynBodyInitOrbit::time_periapsis {}
```

Time since periapsis passage.

trick_units(s)

Definition at line 244 of file dyn_body_init_orbit.hh.

Referenced by `apply()`.

8.17.5.19 true_anomaly

```
double jeod::DynBodyInitOrbit::true_anomaly {}
```

True anomaly.

trick_units(rad)

Definition at line 254 of file dyn_body_init_orbit.hh.

Referenced by `apply()`.

The documentation for this class was generated from the following files:

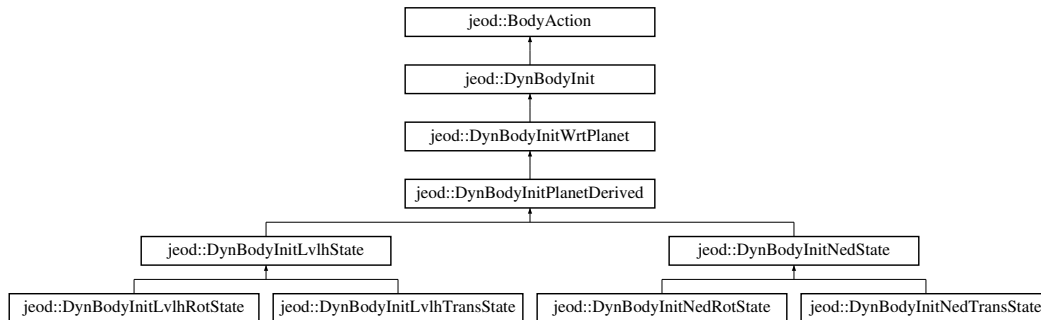
- [dyn_body_init_orbit.hh](#)
- [dyn_body_init_orbit.cc](#)

8.18 jeod::DynBodyInitPlanetDerived Class Reference

(Initialize selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet.

```
#include <dyn_body_init_planet_derived.hh>
```

Inheritance diagram for jeod::DynBodyInitPlanetDerived:



Public Member Functions

- [DynBodyInitPlanetDerived](#) ()=default
- [~DynBodyInitPlanetDerived](#) () override=default
- [DynBodyInitPlanetDerived](#) (const [DynBodyInitPlanetDerived](#) &)=delete
- [DynBodyInitPlanetDerived](#) & operator= (const [DynBodyInitPlanetDerived](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize the initializer.
- bool [is_ready](#) () override
Indicate whether the initializer is ready to run.
- void [apply](#) (DynManager &dyn_manager) override
Apply the initializer: This is just a pass through.

Data Fields

- std::string [ref_body_name](#) {""}
The name of the vehicle whose composite body frame is used to build the derived state with respect to which the vehicle initialization data are referenced.

Protected Attributes

- DynBody * [ref_body](#) {}
The vehicle corresponding to the ref_body_name.
- RefFrameItems::Items [required_items](#) {RefFrameItems::Pos_Vel_Att_Rate}
The state elements in the reference body's composite body frame that must be set before this initializer can proceed.
- bool [body_is_required](#) {true}
If true (default), the ref_body cannot be NULL.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitPlanetDerived](#))

Additional Inherited Members

8.18.1 Detailed Description

(Initialize selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet.

Definition at line 85 of file `dyn_body_init_planet_derived.hh`.

8.18.2 Constructor & Destructor Documentation

8.18.2.1 `DynBodyInitPlanetDerived()` [1/2]

```
jeod::DynBodyInitPlanetDerived::DynBodyInitPlanetDerived ( ) [default]
```

8.18.2.2 `~DynBodyInitPlanetDerived()`

```
jeod::DynBodyInitPlanetDerived::~~DynBodyInitPlanetDerived ( ) [override], [default]
```

8.18.2.3 `DynBodyInitPlanetDerived()` [2/2]

```
jeod::DynBodyInitPlanetDerived::DynBodyInitPlanetDerived (
    const DynBodyInitPlanetDerived & ) [delete]
```

8.18.3 Member Function Documentation

8.18.3.1 `apply()`

```
void jeod::DynBodyInitPlanetDerived::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Apply the initializer: This is just a pass through.

A derived class is responsible for setting the state that the [DynBodyInit](#) uses to initialize the state.

Parameters

<i>in</i> , <i>out</i>	<i>dyn_manager</i>	Dynamics manager
------------------------	--------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 90 of file `dyn_body_init_planet_derived.cc`.

References `jeod::DynBodyInitWrtPlanet::apply()`.

Referenced by `jeod::DynBodyInitLvlhState::apply()`, and `jeod::DynBodyInitNedState::apply()`.

8.18.3.2 initialize()

```
void jeod::DynBodyInitPlanetDerived::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

<i>in</i> , <i>out</i>	<i>dyn_manager</i>	Dynamics manager
------------------------	--------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 51 of file `dyn_body_init_planet_derived.cc`.

References `body_is_required`, `jeod::DynBodyInit::find_dyn_body()`, `jeod::DynBodyInitWrtPlanet::initialize()`, `ref_↔body`, and `ref_body_name`.

Referenced by `jeod::DynBodyInitLvlhState::initialize()`, and `jeod::DynBodyInitNedState::initialize()`.

8.18.3.3 is_ready()

```
bool jeod::DynBodyInitPlanetDerived::is_ready ( ) [override], [virtual]
```

Indicate whether the initializer is ready to run.

When the state is based on some reference body, that reference vehicle's composite body frame must contain the specified required items before the initializer can run.

Returns

Is initializer ready?

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 73 of file `dyn_body_init_planet_derived.cc`.

References `jeod::DynBodyInitWrtPlanet::is_ready()`, `ref_body`, and `required_items`.

8.18.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitPlanetDerived::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitPlanetDerived ) [private]
```

8.18.3.5 operator=()

```
DynBodyInitPlanetDerived& jeod::DynBodyInitPlanetDerived::operator= (
    const DynBodyInitPlanetDerived & ) [delete]
```

8.18.4 Field Documentation

8.18.4.1 body_is_required

```
bool jeod::DynBodyInitPlanetDerived::body_is_required {true} [protected]
```

If true (default), the ref_body cannot be NULL.

If false, the derived class must provide some means other than using a derived state to set the reference RefFrame.
trick_io(**)

Definition at line 117 of file dyn_body_init_planet_derived.hh.

Referenced by jeod::DynBodyInitNedState::DynBodyInitNedState(), jeod::DynBodyInitLvlhState::initialize(), jeod↔
::DynBodyInitNedState::initialize(), and initialize().

8.18.4.2 ref_body

```
DynBody* jeod::DynBodyInitPlanetDerived::ref_body {} [protected]
```

The vehicle corresponding to the ref_body_name.

Note that this is not a user-inputtable item. trick_io(**)

Definition at line 102 of file dyn_body_init_planet_derived.hh.

Referenced by jeod::DynBodyInitLvlhState::apply(), jeod::DynBodyInitNedState::apply(), jeod::DynBodyInitLvlh↔
RotState::initialize(), initialize(), and is_ready().

8.18.4.3 ref_body_name

```
std::string jeod::DynBodyInitPlanetDerived::ref_body_name { "" }
```

The name of the vehicle whose composite body frame is used to build the derived state with respect to which the vehicle initialization data are referenced.

trick_units(—)

Definition at line 95 of file dyn_body_init_planet_derived.hh.

Referenced by jeod::DynBodyInitLvHRotState::initialize(), jeod::DynBodyInitNedState::initialize(), and initialize().

8.18.4.4 required_items

```
RefFrameItems::Items jeod::DynBodyInitPlanetDerived::required_items {RefFrameItems::Pos_Vel_↔  
Att_Rate} [protected]
```

The state elements in the reference body's composite body frame that must be set before this initializer can proceed.

This is not user-inputtable; derived classes should set this item. The default is to require the full state to be set. trick_io(**)

Definition at line 110 of file dyn_body_init_planet_derived.hh.

Referenced by jeod::DynBodyInitLvHState::DynBodyInitLvHState(), jeod::DynBodyInitNedState::DynBodyInit↔NedState(), and is_ready().

The documentation for this class was generated from the following files:

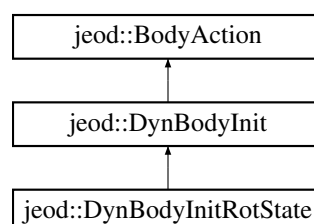
- [dyn_body_init_planet_derived.hh](#)
- [dyn_body_init_planet_derived.cc](#)

8.19 jeod::DynBodyInitRotState Class Reference

Initialize aspects of a vehicle's rotational state.

```
#include <dyn_body_init_rot_state.hh>
```

Inheritance diagram for jeod::DynBodyInitRotState:



Public Types

- enum [StateItems](#) { [Both](#) = 0 , [Attitude](#) = 1 , [Rate](#) = 2 }
- Identify which of attitude/rate is to be initialized.*

Public Member Functions

- [DynBodyInitRotState](#) ()=default
 - [~DynBodyInitRotState](#) () override=default
 - [DynBodyInitRotState](#) (const [DynBodyInitRotState](#) &)=delete
 - [DynBodyInitRotState](#) & [operator=](#) (const [DynBodyInitRotState](#) &)=delete
 - void [initialize](#) (DynManager &dyn_manager) override
- Initialize aspects of this object and forward the initializer to the immediate parent class.*
- void [apply](#) (DynManager &dyn_manager) override
- Apply the initializer.*
- RefFrameItems::Items [initializes_what](#) () override
- Indicate what parts of the vehicle state this object initializes.*
- bool [is_ready](#) () override
- Indicate whether this initializer is ready to be applied.*

Data Fields

- [StateItems](#) [state_items](#) {[Both](#)}
- State items to be initialized – attitude, rate, or both.*

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitRotState](#))

Additional Inherited Members

8.19.1 Detailed Description

Initialize aspects of a vehicle's rotational state.

Definition at line 81 of file `dyn_body_init_rot_state.hh`.

8.19.2 Member Enumeration Documentation

8.19.2.1 StateItems

```
enum jeod::DynBodyInitRotState::StateItems
```

Identify which of attitude/rate is to be initialized.

Enumerator

Both	Initialize both attitude and rate.
Attitude	Initialize attitude only.
Rate	Initialize rate only.

Definition at line 89 of file `dyn_body_init_rot_state.hh`.

8.19.3 Constructor & Destructor Documentation

8.19.3.1 DynBodyInitRotState() [1/2]

```
jeod::DynBodyInitRotState::DynBodyInitRotState ( ) [default]
```

8.19.3.2 ~DynBodyInitRotState()

```
jeod::DynBodyInitRotState::~~DynBodyInitRotState ( ) [override], [default]
```

8.19.3.3 DynBodyInitRotState() [2/2]

```
jeod::DynBodyInitRotState::DynBodyInitRotState (
    const DynBodyInitRotState & ) [delete]
```

8.19.4 Member Function Documentation

8.19.4.1 apply()

```
void jeod::DynBodyInitRotState::apply (
    DynamicsManager & dyn_manager ) [override], [virtual]
```

Apply the initializer.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Dynamics manager
----------------------	---------------------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 157 of file `dyn_body_init_rot_state.cc`.

References `jeod::DynBodyInit::apply()`, and `jeod::DynBodyInit::apply_user_inputs()`.

8.19.4.2 initialize()

```
void jeod::DynBodyInitRotState::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize aspects of this object and forward the initializer to the immediate parent class.

This class needs no initialization per se.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Dynamics manager
----------------------	---------------------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 135 of file `dyn_body_init_rot_state.cc`.

References `jeod::BodyAction::action_identifier`, `Attitude`, `Both`, `jeod::BodyActionMessages::illegal_value`, `jeod::DynBodyInit::initialize()`, `Rate`, and `state_items`.

8.19.4.3 initializes_what()

```
RefFrameItems::Items jeod::DynBodyInitRotState::initializes_what ( ) [override], [virtual]
```

Indicate what parts of the vehicle state this object initializes.

This is depends on the state specified by the user: `Both=attitude and rate`, `Attitude=attitude`, `Rate=rate`.

Returns

States initialized

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 56 of file `dyn_body_init_rot_state.cc`.

References `Attitude`, `Both`, `Rate`, and `state_items`.

Referenced by `is_ready()`.

8.19.4.4 is_ready()

```
bool jeod::DynBodyInitRotState::is_ready ( ) [override], [virtual]
```

Indicate whether this initializer is ready to be applied.

The full rotational state of the reference reference frame must be known to compute the subject reference frame's rotational state.

Returns

Is initializer ready?

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 82 of file `dyn_body_init_rot_state.cc`.

References [jeod::BodyAction::action_identifier](#), [initializes_what\(\)](#), [jeod::BodyActionMessages::invalid_object](#), [jeod::DynBodyInit::is_ready\(\)](#), and [jeod::DynBodyInit::reference_ref_frame](#).

8.19.4.5 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitRotState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitRotState ) [private]
```

8.19.4.6 operator=()

```
DynBodyInitRotState& jeod::DynBodyInitRotState::operator= (
    const DynBodyInitRotState & ) [delete]
```

8.19.5 Field Documentation

8.19.5.1 state_items

```
StateItems jeod::DynBodyInitRotState::state_items {Both}
```

State items to be initialized – attitude, rate, or both.

`trick_units(-)`

Definition at line 102 of file `dyn_body_init_rot_state.hh`.

Referenced by [initialize\(\)](#), and [initializes_what\(\)](#).

The documentation for this class was generated from the following files:

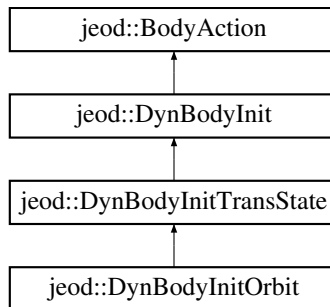
- [dyn_body_init_rot_state.hh](#)
- [dyn_body_init_rot_state.cc](#)

8.20 jeod::DynBodyInitTransState Class Reference

Initialize aspects of a vehicle's translational state.

```
#include <dyn_body_init_trans_state.hh>
```

Inheritance diagram for jeod::DynBodyInitTransState:



Public Types

- enum [StateItems](#) { [Both](#) = 0 , [Position](#) = 1 , [Velocity](#) = 2 }

Identify which of position/velocity is to be initialized.

Public Member Functions

- [DynBodyInitTransState](#) ()=default
- [~DynBodyInitTransState](#) () override=default
- [DynBodyInitTransState](#) (const [DynBodyInitTransState](#) &)=delete
- [DynBodyInitTransState](#) & operator= (const [DynBodyInitTransState](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize aspects of this object and forward the initializer to the immediate parent class.
- void [apply](#) (DynManager &dyn_manager) override
Apply the initializer.
- RefFrameItems::Items [initializes_what](#) () override
Indicate what parts of the vehicle state this object initializes.
- bool [is_ready](#) () override
Indicate whether this initializer is ready to be applied.

Data Fields

- [StateItems](#) [state_items](#) {[Both](#)}

State items to be initialized – position, velocity, or both.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitTransState](#))

Additional Inherited Members

8.20.1 Detailed Description

Initialize aspects of a vehicle's translational state.

Definition at line 81 of file dyn_body_init_trans_state.hh.

8.20.2 Member Enumeration Documentation

8.20.2.1 StateItems

```
enum jeod::DynBodyInitTransState::StateItems
```

Identify which of position/velocity is to be initialized.

Enumerator

Both	Initialize both position and velocity.
Position	Initialize position only.
Velocity	Initialize velocity only.

Definition at line 88 of file dyn_body_init_trans_state.hh.

8.20.3 Constructor & Destructor Documentation

8.20.3.1 DynBodyInitTransState() [1/2]

```
jeod::DynBodyInitTransState::DynBodyInitTransState ( ) [default]
```

8.20.3.2 ~DynBodyInitTransState()

```
jeod::DynBodyInitTransState::~~DynBodyInitTransState ( ) [override], [default]
```

8.20.3.3 DynBodyInitTransState() [2/2]

```
jeod::DynBodyInitTransState::DynBodyInitTransState (
    const DynBodyInitTransState & ) [delete]
```

8.20.4 Member Function Documentation

8.20.4.1 `apply()`

```
void jeod::DynBodyInitTransState::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Apply the initializer.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Dynamics manager
----------------------	---------------------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 162 of file `dyn_body_init_trans_state.cc`.

References [jeod::DynBodyInit::apply\(\)](#), and [jeod::DynBodyInit::apply_user_inputs\(\)](#).

Referenced by [jeod::DynBodyInitOrbit::apply\(\)](#).

8.20.4.2 `initialize()`

```
void jeod::DynBodyInitTransState::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize aspects of this object and forward the initializer to the immediate parent class.

This class needs no initialization per se.

Parameters

<code>in, out</code>	<code><i>dyn_manager</i></code>	Dynamics manager
----------------------	---------------------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 140 of file `dyn_body_init_trans_state.cc`.

References [jeod::BodyAction::action_identifier](#), [Both](#), [jeod::BodyActionMessages::illegal_value](#), [jeod::DynBodyInit::initialize\(\)](#), [Position](#), [state_items](#), and [Velocity](#).

Referenced by [jeod::DynBodyInitOrbit::initialize\(\)](#).

8.20.4.3 initializes_what()

```
RefFrameItems::Items jeod::DynBodyInitTransState::initializes_what ( ) [override], [virtual]
```

Indicate what parts of the vehicle state this object initializes.

This is depends on the state specified by the user: Both=position and velocity, Position=position, Velocity=velocity.

Returns

States initialized

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 55 of file `dyn_body_init_trans_state.cc`.

References Both, Position, `state_items`, and Velocity.

Referenced by `is_ready()`.

8.20.4.4 is_ready()

```
bool jeod::DynBodyInitTransState::is_ready ( ) [override], [virtual]
```

Indicate whether this initializer is ready to be applied.

The full state of the reference reference frame must be known to compute the position and velocity of the subject reference frame.

Returns

Is initializer ready?

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 81 of file `dyn_body_init_trans_state.cc`.

References `jeod::BodyAction::action_identifier`, `initializes_what()`, `jeod::BodyActionMessages::invalid_object`, `jeod::DynBodyInit::is_ready()`, and `jeod::DynBodyInit::reference_ref_frame`.

8.20.4.5 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitTransState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitTransState ) [private]
```

8.20.4.6 operator=()

```
DynBodyInitTransState& jeod::DynBodyInitTransState::operator= (
    const DynBodyInitTransState & ) [delete]
```

8.20.5 Field Documentation

8.20.5.1 state_items

```
StateItems jeod::DynBodyInitTransState::state_items {Both}
```

State items to be initialized – position, velocity, or both.

trick_units(–)

Definition at line 101 of file dyn_body_init_trans_state.hh.

Referenced by initialize(), and initializes_what().

The documentation for this class was generated from the following files:

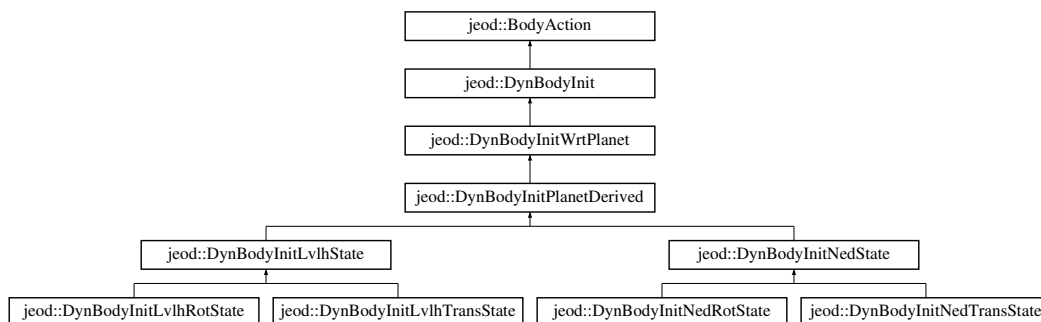
- [dyn_body_init_trans_state.hh](#)
- [dyn_body_init_trans_state.cc](#)

8.21 jeod::DynBodyInitWrtPlanet Class Reference

Initialize selected aspects of a vehicle's state with respect to some frame based on the planet.

```
#include <dyn_body_init_wrt_planet.hh>
```

Inheritance diagram for jeod::DynBodyInitWrtPlanet:



Public Member Functions

- [DynBodyInitWrtPlanet](#) ()=default
- [~DynBodyInitWrtPlanet](#) () override=default
- [DynBodyInitWrtPlanet](#) (const [DynBodyInitWrtPlanet](#) &)=delete
- [DynBodyInitWrtPlanet](#) & [operator=](#) (const [DynBodyInitWrtPlanet](#) &)=delete
- void [initialize](#) (DynManager &dyn_manager) override
Initialize the initializer.
- RefFrameItems::Items [initializes_what](#) () override
Indicate what parts of the vehicle state this object initializes.
- bool [is_ready](#) () override
Indicate whether the initializer is ready to run.
- void [apply](#) (DynManager &dyn_manager) override
Apply the initializer.

Data Fields

- std::string [planet_name](#) {""}
The name of the planet about which the reference body's LVLH frame is to be computed.
- RefFrameItems::Items [set_items](#) {RefFrameItems::Pos_Vel_Att_Rate}
The state elements to be set by this initializer.

Protected Attributes

- Planet * [planet](#) {}
The planet corresponding to the planet_name.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [DynBodyInitWrtPlanet](#))

Additional Inherited Members

8.21.1 Detailed Description

Initialize selected aspects of a vehicle's state with respect to some frame based on the planet.

Definition at line 85 of file `dyn_body_init_wrt_planet.hh`.

8.21.2 Constructor & Destructor Documentation

8.21.2.1 DynBodyInitWrtPlanet() [1/2]

```
jeod::DynBodyInitWrtPlanet::DynBodyInitWrtPlanet ( ) [default]
```

8.21.2.2 ~DynBodyInitWrtPlanet()

```
jeod::DynBodyInitWrtPlanet::~~DynBodyInitWrtPlanet ( ) [override], [default]
```

8.21.2.3 DynBodyInitWrtPlanet() [2/2]

```
jeod::DynBodyInitWrtPlanet::~DynBodyInitWrtPlanet (
    const DynBodyInitWrtPlanet & ) [delete]
```

8.21.3 Member Function Documentation

8.21.3.1 apply()

```
void jeod::DynBodyInitWrtPlanet::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Apply the initializer.

This is just a pass-through. Some derived class must do the actual work.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 88 of file `dyn_body_init_wrt_planet.cc`.

References [jeod::DynBodyInit::apply\(\)](#).

Referenced by [jeod::DynBodyInitPlanetDerived::apply\(\)](#).

8.21.3.2 initialize()

```
void jeod::DynBodyInitWrtPlanet::initialize (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the initializer.

Parameters

<i>in, out</i>	<i>dyn_manager</i>	Dynamics manager
----------------	--------------------	------------------

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 48 of file dyn_body_init_wrt_planet.cc.

References [jeod::DynBodyInit::find_planet\(\)](#), [jeod::DynBodyInit::initialize\(\)](#), [planet](#), [planet_name](#), and [jeod::DynBodyInit::reference_ref_frame](#).

Referenced by [jeod::DynBodyInitPlanetDerived::initialize\(\)](#).

8.21.3.3 initializes_what()

```
RefFrameItems::Items jeod::DynBodyInitWrtPlanet::initializes_what ( ) [override], [virtual]
```

Indicate what parts of the vehicle state this object initializes.

Returns

States initialized

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 66 of file dyn_body_init_wrt_planet.cc.

References [set_items](#).

8.21.3.4 is_ready()

```
bool jeod::DynBodyInitWrtPlanet::is_ready ( ) [override], [virtual]
```

Indicate whether the initializer is ready to run.

This particular implementation is just a pass-through.

Returns

Is initializer ready?

Reimplemented from [jeod::DynBodyInit](#).

Definition at line 76 of file dyn_body_init_wrt_planet.cc.

References [jeod::DynBodyInit::is_ready\(\)](#).

Referenced by [jeod::DynBodyInitPlanetDerived::is_ready\(\)](#).

8.21.3.5 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::DynBodyInitWrtPlanet::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    DynBodyInitWrtPlanet ) [private]
```

8.21.3.6 operator=()

```
DynBodyInitWrtPlanet& jeod::DynBodyInitWrtPlanet::operator= (
    const DynBodyInitWrtPlanet & ) [delete]
```

8.21.4 Field Documentation

8.21.4.1 planet

```
Planet* jeod::DynBodyInitWrtPlanet::planet {} [protected]
```

The planet corresponding to the planet_name.

Note that this is not a user inputtable item. trick_io(**)

Definition at line 106 of file dyn_body_init_wrt_planet.hh.

Referenced by jeod::DynBodyInitLvlhState::apply(), jeod::DynBodyInitNedState::apply(), jeod::DynBodyInitNedState::initialize(), and initialize().

8.21.4.2 planet_name

```
std::string jeod::DynBodyInitWrtPlanet::planet_name {""}
```

The name of the planet about which the reference body's LVLH frame is to be computed.

trick_units(-)

Definition at line 94 of file dyn_body_init_wrt_planet.hh.

Referenced by jeod::DynBodyInitNedState::apply(), and initialize().

8.21.4.3 set_items

```
RefFrameItems::Items jeod::DynBodyInitWrtPlanet::set_items {RefFrameItems::Pos_Vel_Att_Rate}
```

The state elements to be set by this initializer.

trick_units(−)

Definition at line 99 of file dyn_body_init_wrt_planet.hh.

Referenced by `jeod::DynBodyInitNedState::apply()`, `jeod::DynBodyInitLvHRotState::DynBodyInitLvHRotState()`, `jeod::DynBodyInitLvHTransState::DynBodyInitLvHTransState()`, `jeod::DynBodyInitNedRotState::DynBodyInitNedRotState()`, `jeod::DynBodyInitNedTransState::DynBodyInitNedTransState()`, `jeod::DynBodyInitLvHRotState::initialize()`, `jeod::DynBodyInitLvHTransState::initialize()`, `jeod::DynBodyInitNedRotState::initialize()`, `jeod::DynBodyInitNedTransState::initialize()`, and `initializes_what()`.

The documentation for this class was generated from the following files:

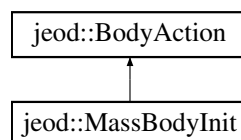
- [dyn_body_init_wrt_planet.hh](#)
- [dyn_body_init_wrt_planet.cc](#)

8.22 jeod::MassBodyInit Class Reference

Base class for initializing a MassBody.

```
#include <mass_body_init.hh>
```

Inheritance diagram for `jeod::MassBodyInit`:



Public Member Functions

- [MassBodyInit](#) ()=default
- [~MassBodyInit](#) () override
- *Destructor.*
- [MassBodyInit](#) (const [MassBodyInit](#) &)=delete
- [MassBodyInit](#) & [operator=](#) (const [MassBodyInit](#) &)=delete
- void [apply](#) (DynManager &dyn_manager) override
- *Initialize the core mass properties of the subject MassBody.*
- void [allocate_points](#) (size_t num_points)
- *Allocate points.*
- MassPointInit * [get_mass_point](#) (size_t index)
- *Access a point in the vector.*

Data Fields

- `MassPropertiesInit` [properties](#)
Specifications for the subject mass body's core mass properties.
- `std::vector< MassPointInit * >` [points](#)
Specifications for the subject mass body's mass points.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (`jeod`, `MassBodyInit`)

Additional Inherited Members

8.22.1 Detailed Description

Base class for initializing a `MassBody`.

Items initialized by this action are

- The body's core mass properties
- The body's mass points.

Definition at line 88 of file `mass_body_init.hh`.

8.22.2 Constructor & Destructor Documentation

8.22.2.1 `MassBodyInit()` [1/2]

```
jeod::MassBodyInit::MassBodyInit ( ) [default]
```

8.22.2.2 `~MassBodyInit()`

```
jeod::MassBodyInit::~~MassBodyInit ( ) [override]
```

Destructor.

Definition at line 87 of file `mass_body_init.cc`.

References `points`.

8.22.2.3 MassBodyInit() [2/2]

```
jeod::MassBodyInit::MassBodyInit (
    const MassBodyInit & ) [delete]
```

8.22.3 Member Function Documentation

8.22.3.1 allocate_points()

```
void jeod::MassBodyInit::allocate_points (
    size_t num_points )
```

Allocate points.

Parameters

in	<i>num_points</i>	number of additional points to be allocated.
----	-------------------	--

Definition at line 76 of file `mass_body_init.cc`.

References `points`.

8.22.3.2 apply()

```
void jeod::MassBodyInit::apply (
    DynManager & dyn_manager ) [override], [virtual]
```

Initialize the core mass properties of the subject `MassBody`.

Parameters

in, out	<i>dyn_manager</i>	Jeod manager
---------	--------------------	--------------

Reimplemented from [jeod::BodyAction](#).

Definition at line 55 of file `mass_body_init.cc`.

References `jeod::BodyAction::action_identifier`, `jeod::BodyAction::apply()`, `jeod::BodyAction::mass_subject`, `points`, `properties`, and `jeod::BodyActionMessages::trace`.

8.22.3.3 get_mass_point()

```
MassPointInit * jeod::MassBodyInit::get_mass_point (
    size_t index )
```

Access a point in the vector.

Parameters

in	index	the index of the point in the vector.
----	-------	---------------------------------------

Definition at line 100 of file mass_body_init.cc.

References points.

8.22.3.4 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::MassBodyInit::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    MassBodyInit ) [private]
```

8.22.3.5 operator=()

```
MassBodyInit& jeod::MassBodyInit::operator= (
    const MassBodyInit & ) [delete]
```

8.22.4 Field Documentation

8.22.4.1 points

```
std::vector<MassPointInit *> jeod::MassBodyInit::points
```

Specifications for the subject mass body's mass points.

trick_units(-)

Definition at line 101 of file mass_body_init.hh.

Referenced by allocate_points(), apply(), get_mass_point(), and ~MassBodyInit().

8.22.4.2 properties

MassPropertiesInit jeod::MassBodyInit::properties

Specifications for the subject mass body's core mass properties.

trick_units(—)

Definition at line 96 of file mass_body_init.hh.

Referenced by apply().

The documentation for this class was generated from the following files:

- [mass_body_init.hh](#)
- [mass_body_init.cc](#)

Chapter 9

File Documentation

9.1 `body_action.cc` File Reference

Define methods for the BodyAction class.

```
#include <cstdint>
#include <stdlib>
#include <string>
#include <typeinfo>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "../include/body_action.hh"
#include "../include/body_action_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.1.1 Detailed Description

Define methods for the BodyAction class.

9.2 `body_action.hh` File Reference

Define the class BodyAction, the base class used for performing actions on a MassBody or DynBody object.

```
#include <string>
#include "dynamics/dyn_body/include/class_declarations.hh"
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "dynamics/mass/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "dynamics/mass/include/mass.hh"
```

Data Structures

- class [jeod::BodyAction](#)
BodyAction is the base class for the [BodyAction](#) model.

Namespaces

- [jeod](#)
Namespace jeod.

9.2.1 Detailed Description

Define the class BodyAction, the base class used for performing actions on a MassBody or DynBody object.

9.3 body_action_messages.cc File Reference

Implement the class BodyActionMessages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/body_action_messages.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [MAKE_BODYACTION_MESSAGE_CODE](#)(id) JEOD_MAKE_MESSAGE_CODE(BodyActionMessages, "dynamics/body_action/", id)

9.3.1 Detailed Description

Implement the class BodyActionMessages.

9.3.2 Macro Definition Documentation

9.3.2.1 MAKE_BODYACTION_MESSAGE_CODE

```
#define MAKE_BODYACTION_MESSAGE_CODE(  
    id ) JEOD_MAKE_MESSAGE_CODE(BodyActionMessages, "dynamics/body_action/", id)
```

Definition at line 37 of file body_action_messages.cc.

9.4 body_action_messages.hh File Reference

Define the class BodyActionMessages, the class that specifies the message IDs used in the BodyAction model.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::BodyActionMessages](#)
Specifies the message IDs used in the [BodyAction](#) model.

Namespaces

- [jeod](#)
Namespace jeod.

9.4.1 Detailed Description

Define the class BodyActionMessages, the class that specifies the message IDs used in the BodyAction model.

9.5 body_attach.cc File Reference

Define methods for the mass body initialization class.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "dynamics/mass/include/mass.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/body_action_messages.hh"
#include "../include/body_attach.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.5.1 Detailed Description

Define methods for the mass body initialization class.

9.6 body_attach.hh File Reference

Define the class `MassBodyAttach`, the base class used for attaching a pair of `MassBody` objects to one another.

```
#include "dynamics/dyn_body/include/class_declarations.hh"
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "dynamics/mass/include/class_declarations.hh"
#include "utils/ref_frames/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_action.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::BodyAttach](#)
Provides the basic ability to attach one `MassBody` to another.

Namespaces

- [jeod](#)
Namespace `jeod`.

9.6.1 Detailed Description

Define the class `MassBodyAttach`, the base class used for attaching a pair of `MassBody` objects to one another.

9.7 body_attach_aligned.cc File Reference

Define methods for the mass body initialization class.

```
#include <cstddef>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "dynamics/mass/include/mass.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/body_action_messages.hh"
#include "../include/body_attach_aligned.hh"
```

Namespaces

- [jeod](#)
Namespace `jeod`.

9.7.1 Detailed Description

Define methods for the mass body initialization class.

9.8 `body_attach_aligned.hh` File Reference

Define the class `BodyAttachAligned`, which causes one `MassBody` to be attached to another at a pair of `MassPoints`.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_attach.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::BodyAttachAligned](#)
Attaches a pair of `MassBody` objects at a pair of `MassPoints`.

Namespaces

- [jeod](#)
Namespace `jeod`.

9.8.1 Detailed Description

Define the class `BodyAttachAligned`, which causes one `MassBody` to be attached to another at a pair of `MassPoints`.

9.9 `body_attach_matrix.cc` File Reference

Define methods for the mass body initialization class.

```
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "dynamics/mass/include/mass.hh"
#include "../include/body_attach_matrix.hh"
```

Namespaces

- [jeod](#)
Namespace `jeod`.

9.9.1 Detailed Description

Define methods for the mass body initialization class.

9.10 body_attach_matrix.hh File Reference

Define the class MassBodyAttachMatrix, which causes one MassBody to be attached given a transformation.

```
#include "utils/orientation/include/orientation.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_attach.hh"
```

Data Structures

- class [jeod::BodyAttachMatrix](#)

Attaches a pair of MassBody objects using the offset+matrix attach mechanism.

Namespaces

- [jeod](#)

Namespace jeod.

9.10.1 Detailed Description

Define the class MassBodyAttachMatrix, which causes one MassBody to be attached given a transformation.

9.11 body_detach.cc File Reference

Define methods for the MassBodyDetach class.

```
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "dynamics/mass/include/mass.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/body_action_messages.hh"
#include "../include/body_detach.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.11.1 Detailed Description

Define methods for the MassBodyDetach class.

9.12 body_detach.hh File Reference

Define the class MassBodyDetach, the base class used for detaching one MassBody object from one another.

```
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_action.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::BodyDetach](#)

Provides the basic ability to detach one MassBody from another.

Namespaces

- [jeod](#)

Namespace jeod.

9.12.1 Detailed Description

Define the class MassBodyDetach, the base class used for detaching one MassBody object from one another.

9.13 body_detach_specific.cc File Reference

Define methods for the BodyDetachSpecific class.

```
#include <cstdint>
#include <string>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "dynamics/mass/include/mass.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/body_action_messages.hh"
#include "../include/body_detach_specific.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.13.1 Detailed Description

Define methods for the BodyDetachSpecific class.

9.14 body_detach_specific.hh File Reference

Define the class `MassBodyDetachSpecific`, the class used for detaching one `MassBody` object from another specified `MassBody`.

```
#include "dynamics/mass/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_action.hh"
#include "class_declarations.hh"
```

Data Structures

- class `jeod::BodyDetachSpecific`

Causes the subject body to detach from a specific body by severing the link immediately spawning from the detach←→_from body.

Namespaces

- `jeod`

Namespace jeod.

9.14.1 Detailed Description

Define the class `MassBodyDetachSpecific`, the class used for detaching one `MassBody` object from another specified `MassBody`.

9.15 body_reattach.cc File Reference

Define methods for the mass body initialization class.

```
#include "dynamics/mass/include/mass.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/body_action_messages.hh"
#include "../include/body_reattach.hh"
```

Namespaces

- `jeod`

Namespace jeod.

9.15.1 Detailed Description

Define methods for the mass body initialization class.

9.16 body_reattach.hh File Reference

Define the class MassBodyReattach, which causes one MassBody to be reattached given a transformation.

```
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "utils/orientation/include/orientation.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_action.hh"
```

Data Structures

- class [jeod::BodyReattach](#)
Alters the nature of an existing attachment.

Namespaces

- [jeod](#)
Namespace jeod.

9.16.1 Detailed Description

Define the class MassBodyReattach, which causes one MassBody to be reattached given a transformation.

9.17 class_declarations.hh File Reference

Forward declarations of classes defined in dyn_body_init_XXX.hh files.

Namespaces

- [jeod](#)
Namespace jeod.

9.17.1 Detailed Description

Forward declarations of classes defined in dyn_body_init_XXX.hh files.

9.18 dyn_body_frame_switch.cc File Reference

Define methods for the class DynBodyFrameSwitch.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "environment/gravity/include/gravity_controls.hh"
#include "environment/gravity/include/gravity_interaction.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_frame_switch.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.18.1 Detailed Description

Define methods for the class DynBodyFrameSwitch.

9.19 dyn_body_frame_switch.hh File Reference

Define the class DynBodyFrameSwitch, the BodyAction derived class used for switch a DynBody's integration frame.

```
#include "dynamics/dyn_body/include/class_declarations.hh"
#include "utils/ref_frames/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_action.hh"
#include "class_declarations.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
```

Data Structures

- class [jeod::DynBodyFrameSwitch](#)

Switch a DynBody's integration frame to a specified frame when the body switches to that integration frame's sphere of influence.

Namespaces

- [jeod](#)

Namespace jeod.

9.19.1 Detailed Description

Define the class DynBodyFrameSwitch, the BodyAction derived class used for switch a DynBody's integration frame.

9.20 dyn_body_init.cc File Reference

Define methods for the base body initialization class.

```
#include <cstdint>
#include <typeinfo>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.20.1 Detailed Description

Define methods for the base body initialization class.

9.21 dyn_body_init.hh File Reference

Define the class DynBodyInit, the base class used for initializing the state of a DynBody object.

```
#include "dynamics/dyn_body/include/class_declarations.hh"
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "environment/planet/include/class_declarations.hh"
#include "utils/orientation/include/orientation.hh"
#include "utils/ref_frames/include/class_declarations.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "utils/ref_frames/include/ref_frame_state.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_action.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::DynBodyInit](#)

Base class for initialize the state of a DynBody.

Namespaces

- [jeod](#)

Namespace jeod.

9.21.1 Detailed Description

Define the class DynBodyInit, the base class used for initializing the state of a DynBody object.

9.22 dyn_body_init_lvlh_rot_state.cc File Reference

Define methods for DynBodyInitLvLhRotState.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_lvlh_rot_state.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.22.1 Detailed Description

Define methods for DynBodyInitLvLhRotState.

9.23 dyn_body_init_lvlh_rot_state.hh File Reference

Define the class DynBodyInitLvLhRotState, which initialize a vehicle's rotational state with respect to some vehicle's LVLH frame.

```
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init_lvlh_state.hh"
```

Data Structures

- class [jeod::DynBodyInitLvlhRotState](#)
Initialize a vehicle's rotational state with respect to some vehicle's LVLH frame.

Namespaces

- [jeod](#)
Namespace jeod.

9.23.1 Detailed Description

Define the class `DynBodyInitLvlhRotState`, which initialize a vehicle's rotational state with respect to some vehicle's LVLH frame.

9.24 dyn_body_init_lvlh_state.cc File Reference

Define methods for the `DynBodyInitLvlhState` class.

```
#include <cstdint>
#include "dynamics/derived_state/include/lvlh_relative_derived_state.hh"
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_lvlh_state.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

9.24.1 Detailed Description

Define methods for the `DynBodyInitLvlhState` class.

9.25 dyn_body_init_lvlh_state.hh File Reference

Define the class `DynBodyInitLvlhState`, the base class for initializing selected aspects of a vehicle's state with respect to some vehicle's LVLH frame.

```
#include "utils/lvlh_frame/include/lvlh_frame.hh"
#include "utils/lvlh_frame/include/lvlh_type.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init_planet_derived.hh"
```

Data Structures

- class [jeod::DynBodyInitLvlhState](#)

Initialize selected aspects of a vehicle's state with respect to some vehicle's LVLH frame.

Namespaces

- [jeod](#)

Namespace jeod.

9.25.1 Detailed Description

Define the class `DynBodyInitLvlhState`, the base class for initializing selected aspects of a vehicle's state with respect to some vehicle's LVLH frame.

9.26 `dyn_body_init_lvlh_trans_state.cc` File Reference

Define methods for `DynBodyInitLvlhTransState`.

```
#include <cstdint>
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_lvlh_trans_state.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.26.1 Detailed Description

Define methods for `DynBodyInitLvlhTransState`.

9.27 `dyn_body_init_lvlh_trans_state.hh` File Reference

Define the class `DynBodyInitLvlhTransState`, which initialize a vehicle's translational state with respect to some other vehicle's LVLH frame.

```
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init_lvlh_state.hh"
```

Data Structures

- class [jeod::DynBodyInitLvlhTransState](#)

initialize a vehicle's translational state with respect to some other vehicle's LVLH frame.

Namespaces

- [jeod](#)

Namespace jeod.

9.27.1 Detailed Description

Define the class DynBodyInitLvlhTransState, which initialize a vehicle's translational state with respect to some other vehicle's LVLH frame.

9.28 dyn_body_init_ned_rot_state.cc File Reference

Define methods for DynBodyInitNedRotState.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_ned_rot_state.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.28.1 Detailed Description

Define methods for DynBodyInitNedRotState.

9.29 dyn_body_init_ned_rot_state.hh File Reference

Define the class DynBodyInitNedRotState, which initialize a vehicle's rotational state wrt some other vehicle's North-East-Down frame.

```
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init_ned_state.hh"
```

Data Structures

- class [jeod::DynBodyInitNedRotState](#)

Initialize a vehicle's rotational state wrt some vehicle's North-East-Down frame.

Namespaces

- [jeod](#)

Namespace jeod.

9.29.1 Detailed Description

Define the class `DynBodyInitNedRotState`, which initialize a vehicle's rotational state wrt some other vehicle's North-East-Down frame.

9.30 `dyn_body_init_ned_state.cc` File Reference

Define methods for `DynBodyInitNedState`.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "environment/planet/include/planet.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/planet_fixed/north_east_down/include/north_east_down.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "utils/ref_frames/include/ref_frame_state.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_ned_state.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.30.1 Detailed Description

Define methods for `DynBodyInitNedState`.

9.31 dyn_body_init_ned_state.hh File Reference

Define the class DynBodyInitNedState, the base class for initializing selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet.

```
#include "environment/ephemerides/ephem_interface/include/class_declarations.↵
hh"
#include "utils/planet_fixed/north_east_down/include/north_east_down.hh"
#include "utils/planet_fixed/planet_fixed_posn/include/alt_lat_long_state.↵
hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init_planet_derived.hh"
```

Data Structures

- class [jeod::DynBodyInitNedState](#)

Initialize selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet.

Namespaces

- [jeod](#)

Namespace jeod.

9.31.1 Detailed Description

Define the class DynBodyInitNedState, the base class for initializing selected aspects of a vehicle's state with respect to either some vehicle's North-East-Down frame or the North-East-Down frame for a specified location on the planet.

9.32 dyn_body_init_ned_trans_state.cc File Reference

Define methods for DynBodyInitNedTransState.

```
#include <cstdint>
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_ned_trans_state.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.32.1 Detailed Description

Define methods for DynBodyInitNedTransState.

9.33 dyn_body_init_ned_trans_state.hh File Reference

Define the class DynBodyInitNedTransState, which initialize a vehicle's translational state wrt some other vehicle's North-East-Down frame.

```
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init_ned_state.hh"
```

Data Structures

- class [jeod::DynBodyInitNedTransState](#)
Initialize a vehicle's translational state wrt some vehicle's North-East-Down frame.

Namespaces

- [jeod](#)
Namespace jeod.

9.33.1 Detailed Description

Define the class DynBodyInitNedTransState, which initialize a vehicle's translational state wrt some other vehicle's North-East-Down frame.

9.34 dyn_body_init_orbit.cc File Reference

Define classes for items represented in some ephemeris model.

```
#include <cmath>
#include <cstdlib>
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "environment/planet/include/planet.hh"
#include "utils/math/include/vector3.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/named_item/include/named_item.hh"
#include "utils/orbital_elements/include/orbital_elements.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_orbit.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.34.1 Detailed Description

Define classes for items represented in some ephemeris model.

9.35 dyn_body_init_orbit.hh File Reference

Define the class DynBodyInitOrbit, which initializes a vehicle in in some orbit.

```
#include "environment/ephemerides/ephem_interface/include/class_declarations.↵
hh"
#include "environment/planet/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init_trans_state.hh"
```

Data Structures

- class [jeod::DynBodyInitOrbit](#)

Initialize a vehicle's translational state given an orbital specification.

Namespaces

- [jeod](#)

Namespace jeod.

9.35.1 Detailed Description

Define the class DynBodyInitOrbit, which initializes a vehicle in in some orbit.

9.36 dyn_body_init_planet_derived.cc File Reference

Define methods for the DynBodyInitPlanetDerived class.

```
#include <cstdlib>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_planet_derived.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.36.1 Detailed Description

Define methods for the DynBodyInitPlanetDerived class.

9.37 dyn_body_init_planet_derived.hh File Reference

Define the class DynBodyInitPlanetDerived, the base class for initializing selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet.

```
#include "dynamics/dyn_body/include/class_declarations.hh"
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init_wrt_planet.hh"
```

Data Structures

- class [jeod::DynBodyInitPlanetDerived](#)

(Initialize selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet.

Namespaces

- [jeod](#)

Namespace jeod.

9.37.1 Detailed Description

Define the class DynBodyInitPlanetDerived, the base class for initializing selected aspects of a vehicle's state with respect to some state that is derived from some vehicle's state in conjunction with a planet.

9.38 dyn_body_init_rot_state.cc File Reference

Define methods for DynBodyInitRotState.

```
#include <cstddef>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_rot_state.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.38.1 Detailed Description

Define methods for DynBodyInitRotState.

9.39 dyn_body_init_rot_state.hh File Reference

Define the class DynBodyInitRotState that initialize aspects of a vehicle's rotational state.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "dyn_body_init.hh"
```

Data Structures

- class [jeod::DynBodyInitRotState](#)

Initialize aspects of a vehicle's rotational state.

Namespaces

- [jeod](#)

Namespace jeod.

9.39.1 Detailed Description

Define the class DynBodyInitRotState that initialize aspects of a vehicle's rotational state.

9.40 dyn_body_init_trans_state.cc File Reference

Define methods for DynBodyInitTransState.

```
#include <cstdint>
#include "dynamics/dyn_body/include/dyn_body.hh"
#include "environment/ephemerides/ephem_interface/include/ephem_ref_frame.↵
hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_trans_state.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.40.1 Detailed Description

Define methods for DynBodyInitTransState.

9.41 dyn_body_init_trans_state.hh File Reference

Define the class DynBodyInitTransState that initialize aspects of a vehicle's translational state.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "class_declarations.hh"
#include "dyn_body_init.hh"
```

Data Structures

- class [jeod::DynBodyInitTransState](#)

Initialize aspects of a vehicle's translational state.

Namespaces

- [jeod](#)

Namespace jeod.

9.41.1 Detailed Description

Define the class DynBodyInitTransState that initialize aspects of a vehicle's translational state.

9.42 dyn_body_init_wrt_planet.cc File Reference

Define methods for the DynBodyInitWrtPlanet class.

```
#include <cstdint>
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "environment/planet/include/planet.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "../include/body_action_messages.hh"
#include "../include/dyn_body_init_wrt_planet.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.42.1 Detailed Description

Define methods for the DynBodyInitWrtPlanet class.

9.43 dyn_body_init_wrt_planet.hh File Reference

Define the class DynBodyInitWrtPlanet, the base class for initializing selected aspects of a vehicle's state with respect to some state that is connected to a planet in some way.

```
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "environment/planet/include/class_declarations.hh"
#include "utils/ref_frames/include/ref_frame_items.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "dyn_body_init.hh"
```

Data Structures

- class [jeod::DynBodyInitWrtPlanet](#)

Initialize selected aspects of a vehicle's state with respect to some frame based on the planet.

Namespaces

- [jeod](#)

Namespace jeod.

9.43.1 Detailed Description

Define the class DynBodyInitWrtPlanet, the base class for initializing selected aspects of a vehicle's state with respect to some state that is connected to a planet in some way.

9.44 mass_body_init.cc File Reference

Define methods for the mass body initialization class.

```
#include <cstddef>
#include "dynamics/dyn_manager/include/dyn_manager.hh"
#include "dynamics/mass/include/mass.hh"
#include "utils/math/include/matrix3x3.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/body_action_messages.hh"
#include "../include/mass_body_init.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.44.1 Detailed Description

Define methods for the mass body initialization class.

9.45 mass_body_init.hh File Reference

Define the class MassBodyInit, the base class used for initializing the core mass properties of a MassBody object.

```
#include <vector>
#include "dynamics/dyn_manager/include/class_declarations.hh"
#include "dynamics/mass/include/class_declarations.hh"
#include "dynamics/mass/include/mass_properties_init.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "body_action.hh"
#include "class_declarations.hh"
```

Data Structures

- class [jeod::MassBodyInit](#)

Base class for initializing a MassBody.

Namespaces

- [jeod](#)

Namespace jeod.

9.45.1 Detailed Description

Define the class MassBodyInit, the base class used for initializing the core mass properties of a MassBody object.

Index

- ~BodyAction
 - jeod::BodyAction, [21](#)
- ~BodyAttach
 - jeod::BodyAttach, [33](#)
- ~BodyAttachAligned
 - jeod::BodyAttachAligned, [38](#)
- ~BodyAttachMatrix
 - jeod::BodyAttachMatrix, [41](#)
- ~BodyDetach
 - jeod::BodyDetach, [44](#)
- ~BodyDetachSpecific
 - jeod::BodyDetachSpecific, [47](#)
- ~BodyReattach
 - jeod::BodyReattach, [52](#)
- ~DynBodyFrameSwitch
 - jeod::DynBodyFrameSwitch, [55](#)
- ~DynBodyInit
 - jeod::DynBodyInit, [61](#)
- ~DynBodyInitLvLhRotState
 - jeod::DynBodyInitLvLhRotState, [72](#)
- ~DynBodyInitLvLhState
 - jeod::DynBodyInitLvLhState, [74](#)
- ~DynBodyInitLvLhTransState
 - jeod::DynBodyInitLvLhTransState, [78](#)
- ~DynBodyInitNedRotState
 - jeod::DynBodyInitNedRotState, [80](#)
- ~DynBodyInitNedState
 - jeod::DynBodyInitNedState, [84](#)
- ~DynBodyInitNedTransState
 - jeod::DynBodyInitNedTransState, [88](#)
- ~DynBodyInitOrbit
 - jeod::DynBodyInitOrbit, [92](#)
- ~DynBodyInitPlanetDerived
 - jeod::DynBodyInitPlanetDerived, [100](#)
- ~DynBodyInitRotState
 - jeod::DynBodyInitRotState, [105](#)
- ~DynBodyInitTransState
 - jeod::DynBodyInitTransState, [109](#)
- ~DynBodyInitWrtPlanet
 - jeod::DynBodyInitWrtPlanet, [113](#)
- ~MassBodyInit
 - jeod::MassBodyInit, [118](#)
- action_identifier
 - jeod::BodyAction, [25](#)
- action_name
 - jeod::BodyAction, [26](#)
- active
 - jeod::BodyAction, [26](#)
- allocate_points
 - jeod::MassBodyInit, [119](#)
- alt_apoapsis
 - jeod::DynBodyInitOrbit, [93](#)
- alt_periapsis
 - jeod::DynBodyInitOrbit, [94](#)
- altlatlong_type
 - jeod::DynBodyInitNedState, [86](#)
- ang_velocity
 - jeod::DynBodyInit, [67](#)
- apply
 - jeod::BodyAction, [21](#)
 - jeod::BodyAttach, [33](#)
 - jeod::BodyAttachAligned, [38](#)
 - jeod::BodyAttachMatrix, [42](#)
 - jeod::BodyDetach, [45](#)
 - jeod::BodyDetachSpecific, [48](#)
 - jeod::BodyReattach, [52](#)
 - jeod::DynBodyFrameSwitch, [56](#)
 - jeod::DynBodyInit, [61](#)
 - jeod::DynBodyInitLvLhState, [75](#)
 - jeod::DynBodyInitNedState, [84](#)
 - jeod::DynBodyInitOrbit, [92](#)
 - jeod::DynBodyInitPlanetDerived, [100](#)
 - jeod::DynBodyInitRotState, [105](#)
 - jeod::DynBodyInitTransState, [110](#)
 - jeod::DynBodyInitWrtPlanet, [114](#)
 - jeod::MassBodyInit, [119](#)
- apply_user_inputs
 - jeod::DynBodyInit, [61](#)
- arg_latitude
 - jeod::DynBodyInitOrbit, [94](#)
- arg_periapsis
 - jeod::DynBodyInitOrbit, [94](#)
- ascending_node
 - jeod::DynBodyInitOrbit, [94](#)
- Attitude
 - jeod::DynBodyInitRotState, [105](#)
- body_action.cc, [123](#)
- body_action.hh, [123](#)
- body_action_messages.cc, [124](#)
 - MAKE_BODYACTION_MESSAGE_CODE, [124](#)
- body_action_messages.hh, [125](#)
- body_attach.cc, [125](#)
- body_attach.hh, [126](#)
- body_attach_aligned.cc, [126](#)
- body_attach_aligned.hh, [127](#)
- body_attach_matrix.cc, [127](#)
- body_attach_matrix.hh, [128](#)
- body_detach.cc, [128](#)

- body_detach.hh, 129
- body_detach_specific.cc, 129
- body_detach_specific.hh, 130
- body_frame_id
 - jeod::DynBodyInit, 67
- body_is_required
 - jeod::DynBodyInitPlanetDerived, 102
- body_reattach.cc, 130
- body_reattach.hh, 131
- body_ref_frame
 - jeod::DynBodyInit, 67
- BodyAction, 13
 - jeod::BodyAction, 21
- BodyActionMessages
 - jeod::BodyActionMessages, 28, 29
- BodyAttach
 - jeod::BodyAttach, 33
- BodyAttachAligned
 - jeod::BodyAttachAligned, 38
- BodyAttachMatrix
 - jeod::BodyAttachMatrix, 41, 42
- BodyDetach
 - jeod::BodyDetach, 44, 45
- BodyDetachSpecific
 - jeod::BodyDetachSpecific, 47
- BodyReattach
 - jeod::BodyReattach, 52
- Both
 - jeod::DynBodyInitRotState, 105
 - jeod::DynBodyInitTransState, 109
- CaseEleven
 - jeod::DynBodyInitOrbit, 91
- class_declarations.hh, 131
- compute_rotational_state
 - jeod::DynBodyInit, 62
- compute_translational_state
 - jeod::DynBodyInit, 62
- dyn_body_frame_switch.cc, 132
- dyn_body_frame_switch.hh, 132
- dyn_body_init.cc, 133
- dyn_body_init.hh, 133
- dyn_body_init_lvih_rot_state.cc, 134
- dyn_body_init_lvih_rot_state.hh, 134
- dyn_body_init_lvih_state.cc, 135
- dyn_body_init_lvih_state.hh, 135
- dyn_body_init_lvih_trans_state.cc, 136
- dyn_body_init_lvih_trans_state.hh, 136
- dyn_body_init_ned_rot_state.cc, 137
- dyn_body_init_ned_rot_state.hh, 137
- dyn_body_init_ned_state.cc, 138
- dyn_body_init_ned_state.hh, 139
- dyn_body_init_ned_trans_state.cc, 139
- dyn_body_init_ned_trans_state.hh, 140
- dyn_body_init_orbit.cc, 140
- dyn_body_init_orbit.hh, 141
- dyn_body_init_planet_derived.cc, 141
- dyn_body_init_planet_derived.hh, 142
- dyn_body_init_rot_state.cc, 142
- dyn_body_init_rot_state.hh, 143
- dyn_body_init_trans_state.cc, 143
- dyn_body_init_trans_state.hh, 144
- dyn_body_init_wrt_planet.cc, 144
- dyn_body_init_wrt_planet.hh, 145
- dyn_detach_from
 - jeod::BodyDetachSpecific, 50
- dyn_parent
 - jeod::BodyAttach, 36
- dyn_subject
 - jeod::BodyAction, 26
- Dynamics, 13
- DynBodyFrameSwitch
 - jeod::DynBodyFrameSwitch, 55
- DynBodyInit
 - jeod::DynBodyInit, 60, 61
- DynBodyInitLvihRotState
 - jeod::DynBodyInitLvihRotState, 71, 72
- DynBodyInitLvihState
 - jeod::DynBodyInitLvihState, 74
- DynBodyInitLvihTransState
 - jeod::DynBodyInitLvihTransState, 78
- DynBodyInitNedRotState
 - jeod::DynBodyInitNedRotState, 80
- DynBodyInitNedState
 - jeod::DynBodyInitNedState, 83, 84
- DynBodyInitNedTransState
 - jeod::DynBodyInitNedTransState, 88
- DynBodyInitOrbit
 - jeod::DynBodyInitOrbit, 92
- DynBodyInitPlanetDerived
 - jeod::DynBodyInitPlanetDerived, 100
- DynBodyInitRotState
 - jeod::DynBodyInitRotState, 105
- DynBodyInitTransState
 - jeod::DynBodyInitTransState, 109
- DynBodyInitWrtPlanet
 - jeod::DynBodyInitWrtPlanet, 113, 114
- eccentricity
 - jeod::DynBodyInitOrbit, 95
- fatal_error
 - jeod::BodyActionMessages, 29
- find_body_frame
 - jeod::DynBodyInit, 62
- find_dyn_body
 - jeod::DynBodyInit, 63
- find_planet
 - jeod::DynBodyInit, 63
- find_ref_frame
 - jeod::DynBodyInit, 64
- get_identifier
 - jeod::BodyAction, 22
- get_mass_point
 - jeod::MassBodyInit, 119
- get_subject_dyn_body

- jeod::BodyAction, 22
- illegal_value
 - jeod::BodyActionMessages, 29
- IncAscnodeAltperAltapoArgperTanom
 - jeod::DynBodyInitOrbit, 91
- IncAscnodeAltperAltapoArgperTimeperi
 - jeod::DynBodyInitOrbit, 91
- inclination
 - jeod::DynBodyInitOrbit, 95
- initialize
 - jeod::BodyAction, 22
 - jeod::BodyAttach, 34
 - jeod::BodyAttachAligned, 39
 - jeod::BodyDetachSpecific, 48
 - jeod::DynBodyFrameSwitch, 56
 - jeod::DynBodyInit, 64
 - jeod::DynBodyInitLv1hRotState, 72
 - jeod::DynBodyInitLv1hState, 75
 - jeod::DynBodyInitLv1hTransState, 78
 - jeod::DynBodyInitNedRotState, 81
 - jeod::DynBodyInitNedState, 84
 - jeod::DynBodyInitNedTransState, 88
 - jeod::DynBodyInitOrbit, 93
 - jeod::DynBodyInitPlanetDerived, 101
 - jeod::DynBodyInitRotState, 106
 - jeod::DynBodyInitTransState, 110
 - jeod::DynBodyInitWrtPlanet, 114
- initializes_what
 - jeod::DynBodyInit, 65
 - jeod::DynBodyInitRotState, 106
 - jeod::DynBodyInitTransState, 110
 - jeod::DynBodyInitWrtPlanet, 115
- integ_frame
 - jeod::DynBodyFrameSwitch, 57
- integ_frame_name
 - jeod::DynBodyFrameSwitch, 57
- invalid_name
 - jeod::BodyActionMessages, 30
- invalid_object
 - jeod::BodyActionMessages, 30
- InvalidSet
 - jeod::DynBodyInitOrbit, 91
- is_ready
 - jeod::BodyAction, 23
 - jeod::BodyDetach, 45
 - jeod::BodyDetachSpecific, 48
 - jeod::DynBodyFrameSwitch, 56
 - jeod::DynBodyInit, 65
 - jeod::DynBodyInitPlanetDerived, 101
 - jeod::DynBodyInitRotState, 106
 - jeod::DynBodyInitTransState, 111
 - jeod::DynBodyInitWrtPlanet, 115
- is_same_subject_body
 - jeod::BodyAction, 23
- is_subject_dyn_body
 - jeod::BodyAction, 23
- jeod, 17
- jeod::BodyAction, 19
 - ~BodyAction, 21
 - action_identifier, 25
 - action_name, 26
 - active, 26
 - apply, 21
 - BodyAction, 21
 - dyn_subject, 26
 - get_identifier, 22
 - get_subject_dyn_body, 22
 - initialize, 22
 - is_ready, 23
 - is_same_subject_body, 23
 - is_subject_dyn_body, 23
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 23
 - mass_subject, 27
 - operator=, 24
 - set_subject_body, 24
 - shutdown, 24
 - terminate_on_error, 27
 - validate_body_inputs, 24
 - validate_name, 25
- jeod::BodyActionMessages, 28
 - BodyActionMessages, 28, 29
 - fatal_error, 29
 - illegal_value, 29
 - invalid_name, 30
 - invalid_object, 30
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 29
 - not_performed, 30
 - null_pointer, 31
 - operator=, 29
 - trace, 31
- jeod::BodyAttach, 32
 - ~BodyAttach, 33
 - apply, 33
 - BodyAttach, 33
 - dyn_parent, 36
 - initialize, 34
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 34
 - mass_parent, 36
 - operator=, 34
 - ref_parent, 36
 - set_parent_body, 35
 - set_parent_frame, 35
 - succeeded, 36
- jeod::BodyAttachAligned, 37
 - ~BodyAttachAligned, 38
 - apply, 38
 - BodyAttachAligned, 38
 - initialize, 39
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 39
 - operator=, 39
 - parent_point_name, 39
 - subject_point_name, 40
- jeod::BodyAttachMatrix, 40
 - ~BodyAttachMatrix, 41
 - apply, 42

- BodyAttachMatrix, [41](#), [42](#)
- JEOD_CLASS_ESTABLISH_FRIENDS2, [42](#)
- offset_pstr_cstr_pstr, [43](#)
- operator=, [42](#)
- pstr_cstr, [43](#)
- jeod::BodyDetach, [43](#)
 - ~BodyDetach, [44](#)
 - apply, [45](#)
 - BodyDetach, [44](#), [45](#)
 - is_ready, [45](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [45](#)
 - operator=, [46](#)
- jeod::BodyDetachSpecific, [46](#)
 - ~BodyDetachSpecific, [47](#)
 - apply, [48](#)
 - BodyDetachSpecific, [47](#)
 - dyn_detach_from, [50](#)
 - initialize, [48](#)
 - is_ready, [48](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [49](#)
 - mass_detach_from, [50](#)
 - operator=, [49](#)
 - set_detach_from_body, [49](#)
- jeod::BodyReattach, [51](#)
 - ~BodyReattach, [52](#)
 - apply, [52](#)
 - BodyReattach, [52](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [53](#)
 - offset_pstr_cstr_pstr, [53](#)
 - operator=, [53](#)
 - pstr_cstr, [53](#)
- jeod::DynBodyFrameSwitch, [54](#)
 - ~DynBodyFrameSwitch, [55](#)
 - apply, [56](#)
 - DynBodyFrameSwitch, [55](#)
 - initialize, [56](#)
 - integ_frame, [57](#)
 - integ_frame_name, [57](#)
 - is_ready, [56](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [57](#)
 - operator=, [57](#)
 - sort_grav_controls, [58](#)
 - switch_distance, [58](#)
 - switch_sense, [58](#)
 - SwitchOnApproach, [55](#)
 - SwitchOnDeparture, [55](#)
 - SwitchSense, [55](#)
- jeod::DynBodyInit, [59](#)
 - ~DynBodyInit, [61](#)
 - ang_velocity, [67](#)
 - apply, [61](#)
 - apply_user_inputs, [61](#)
 - body_frame_id, [67](#)
 - body_ref_frame, [67](#)
 - compute_rotational_state, [62](#)
 - compute_translational_state, [62](#)
 - DynBodyInit, [60](#), [61](#)
 - find_body_frame, [62](#)
 - find_dyn_body, [63](#)
 - find_planet, [63](#)
 - find_ref_frame, [64](#)
 - initialize, [64](#)
 - initializes_what, [65](#)
 - is_ready, [65](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [66](#)
 - operator=, [66](#)
 - orientation, [68](#)
 - position, [68](#)
 - rate_in_parent, [68](#)
 - reference_ref_frame, [68](#)
 - reference_ref_frame_name, [69](#)
 - report_failure, [66](#)
 - reverse_sense, [69](#)
 - state, [69](#)
 - subscribed_frame, [70](#)
 - velocity, [70](#)
- jeod::DynBodyInitLvlhRotState, [71](#)
 - ~DynBodyInitLvlhRotState, [72](#)
 - DynBodyInitLvlhRotState, [71](#), [72](#)
 - initialize, [72](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [72](#)
 - operator=, [73](#)
- jeod::DynBodyInitLvlhState, [73](#)
 - ~DynBodyInitLvlhState, [74](#)
 - apply, [75](#)
 - DynBodyInitLvlhState, [74](#)
 - initialize, [75](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [76](#)
 - lvlh_object_ptr, [76](#)
 - lvlh_type, [76](#)
 - operator=, [76](#)
 - set_lvlh_frame_object, [76](#)
- jeod::DynBodyInitLvlhTransState, [77](#)
 - ~DynBodyInitLvlhTransState, [78](#)
 - DynBodyInitLvlhTransState, [78](#)
 - initialize, [78](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [79](#)
 - operator=, [79](#)
- jeod::DynBodyInitNedRotState, [79](#)
 - ~DynBodyInitNedRotState, [80](#)
 - DynBodyInitNedRotState, [80](#)
 - initialize, [81](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [82](#)
 - operator=, [82](#)
- jeod::DynBodyInitNedState, [82](#)
 - ~DynBodyInitNedState, [84](#)
 - altlatlong_type, [86](#)
 - apply, [84](#)
 - DynBodyInitNedState, [83](#), [84](#)
 - initialize, [84](#)
 - JEOD_CLASS_ESTABLISH_FRIENDS2, [85](#)
 - operator=, [85](#)
 - pflix_ptr, [86](#)
 - ref_point, [86](#)
 - set_use_alt_pfix, [85](#)
 - use_alt_pfix, [86](#)

- jeod::DynBodyInitNedTransState, 87
 - ~DynBodyInitNedTransState, 88
 - DynBodyInitNedTransState, 88
 - initialize, 88
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 89
 - operator=, 89
- jeod::DynBodyInitOrbit, 89
 - ~DynBodyInitOrbit, 92
 - alt_apoapsis, 93
 - alt_periapsis, 94
 - apply, 92
 - arg_latitude, 94
 - arg_periapsis, 94
 - ascending_node, 94
 - CaseEleven, 91
 - DynBodyInitOrbit, 92
 - eccentricity, 95
 - IncAscnodeAltperAltapoArgperTanom, 91
 - IncAscnodeAltperAltapoArgperTimeperi, 91
 - inclination, 95
 - initialize, 93
 - InvalidSet, 91
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 93
 - mean_anomaly, 95
 - operator=, 93
 - orb_radius, 95
 - orbit_frame, 96
 - orbit_frame_name, 96
 - OrbitalSet, 91
 - planet, 96
 - planet_name, 96
 - radial_vel, 97
 - semi_latus_rectum, 97
 - semi_major_axis, 97
 - set, 97
 - SlrEccIncAscnodeArgperTanom, 91
 - SmaEccIncAscnodeArgperManom, 91
 - SmaEccIncAscnodeArgperTanom, 91
 - SmaEccIncAscnodeArgperTimeperi, 91
 - SmaIncAscnodeArglatRadRadvel, 91
 - time_periapsis, 98
 - true_anomaly, 98
- jeod::DynBodyInitPlanetDerived, 99
 - ~DynBodyInitPlanetDerived, 100
 - apply, 100
 - body_is_required, 102
 - DynBodyInitPlanetDerived, 100
 - initialize, 101
 - is_ready, 101
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 101
 - operator=, 102
 - ref_body, 102
 - ref_body_name, 102
 - required_items, 103
- jeod::DynBodyInitRotState, 103
 - ~DynBodyInitRotState, 105
 - apply, 105
 - Attitude, 105
 - Both, 105
 - DynBodyInitRotState, 105
 - initialize, 106
 - initializes_what, 106
 - is_ready, 106
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 107
 - operator=, 107
 - Rate, 105
 - state_items, 107
 - StateItems, 104
- jeod::DynBodyInitTransState, 108
 - ~DynBodyInitTransState, 109
 - apply, 110
 - Both, 109
 - DynBodyInitTransState, 109
 - initialize, 110
 - initializes_what, 110
 - is_ready, 111
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 111
 - operator=, 111
 - Position, 109
 - state_items, 112
 - StateItems, 109
 - Velocity, 109
- jeod::DynBodyInitWrtPlanet, 112
 - ~DynBodyInitWrtPlanet, 113
 - apply, 114
 - DynBodyInitWrtPlanet, 113, 114
 - initialize, 114
 - initializes_what, 115
 - is_ready, 115
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 115
 - operator=, 116
 - planet, 116
 - planet_name, 116
 - set_items, 116
- jeod::MassBodyInit, 117
 - ~MassBodyInit, 118
 - allocate_points, 119
 - apply, 119
 - get_mass_point, 119
 - JEOD_CLASS_ESTABLISH_FRIENDS2, 120
 - MassBodyInit, 118
 - operator=, 120
 - points, 120
 - properties, 120
- JEOD_CLASS_ESTABLISH_FRIENDS2
 - jeod::BodyAction, 23
 - jeod::BodyActionMessages, 29
 - jeod::BodyAttach, 34
 - jeod::BodyAttachAligned, 39
 - jeod::BodyAttachMatrix, 42
 - jeod::BodyDetach, 45
 - jeod::BodyDetachSpecific, 49
 - jeod::BodyReattach, 53
 - jeod::DynBodyFrameSwitch, 57
 - jeod::DynBodyInit, 66
 - jeod::DynBodyInitLvHRotState, 72

- jeod::DynBodyInitLvHState, 76
- jeod::DynBodyInitLvHTransState, 79
- jeod::DynBodyInitNedRotState, 82
- jeod::DynBodyInitNedState, 85
- jeod::DynBodyInitNedTransState, 89
- jeod::DynBodyInitOrbit, 93
- jeod::DynBodyInitPlanetDerived, 101
- jeod::DynBodyInitRotState, 107
- jeod::DynBodyInitTransState, 111
- jeod::DynBodyInitWrtPlanet, 115
- jeod::MassBodyInit, 120
- lvh_object_ptr
 - jeod::DynBodyInitLvHState, 76
- lvh_type
 - jeod::DynBodyInitLvHState, 76
- MAKE_BODYACTION_MESSAGE_CODE
 - body_action_messages.cc, 124
- mass_body_init.cc, 145
- mass_body_init.hh, 146
- mass_detach_from
 - jeod::BodyDetachSpecific, 50
- mass_parent
 - jeod::BodyAttach, 36
- mass_subject
 - jeod::BodyAction, 27
- MassBodyInit
 - jeod::MassBodyInit, 118
- mean_anomaly
 - jeod::DynBodyInitOrbit, 95
- Models, 13
- not_performed
 - jeod::BodyActionMessages, 30
- null_pointer
 - jeod::BodyActionMessages, 31
- offset_pstr_cstr_pstr
 - jeod::BodyAttachMatrix, 43
 - jeod::BodyReattach, 53
- operator=
 - jeod::BodyAction, 24
 - jeod::BodyActionMessages, 29
 - jeod::BodyAttach, 34
 - jeod::BodyAttachAligned, 39
 - jeod::BodyAttachMatrix, 42
 - jeod::BodyDetach, 46
 - jeod::BodyDetachSpecific, 49
 - jeod::BodyReattach, 53
 - jeod::DynBodyFrameSwitch, 57
 - jeod::DynBodyInit, 66
 - jeod::DynBodyInitLvHRotState, 73
 - jeod::DynBodyInitLvHState, 76
 - jeod::DynBodyInitLvHTransState, 79
 - jeod::DynBodyInitNedRotState, 82
 - jeod::DynBodyInitNedState, 85
 - jeod::DynBodyInitNedTransState, 89
 - jeod::DynBodyInitOrbit, 93
 - jeod::DynBodyInitPlanetDerived, 102
 - jeod::DynBodyInitRotState, 107
 - jeod::DynBodyInitTransState, 111
 - jeod::DynBodyInitWrtPlanet, 116
 - jeod::MassBodyInit, 120
- orb_radius
 - jeod::DynBodyInitOrbit, 95
- orbit_frame
 - jeod::DynBodyInitOrbit, 96
- orbit_frame_name
 - jeod::DynBodyInitOrbit, 96
- OrbitalSet
 - jeod::DynBodyInitOrbit, 91
- orientation
 - jeod::DynBodyInit, 68
- parent_point_name
 - jeod::BodyAttachAligned, 39
- pfix_ptr
 - jeod::DynBodyInitNedState, 86
- planet
 - jeod::DynBodyInitOrbit, 96
 - jeod::DynBodyInitWrtPlanet, 116
- planet_name
 - jeod::DynBodyInitOrbit, 96
 - jeod::DynBodyInitWrtPlanet, 116
- points
 - jeod::MassBodyInit, 120
- Position
 - jeod::DynBodyInitTransState, 109
- position
 - jeod::DynBodyInit, 68
- properties
 - jeod::MassBodyInit, 120
- pstr_cstr
 - jeod::BodyAttachMatrix, 43
 - jeod::BodyReattach, 53
- radial_vel
 - jeod::DynBodyInitOrbit, 97
- Rate
 - jeod::DynBodyInitRotState, 105
- rate_in_parent
 - jeod::DynBodyInit, 68
- ref_body
 - jeod::DynBodyInitPlanetDerived, 102
- ref_body_name
 - jeod::DynBodyInitPlanetDerived, 102
- ref_parent
 - jeod::BodyAttach, 36
- ref_point
 - jeod::DynBodyInitNedState, 86
- reference_ref_frame
 - jeod::DynBodyInit, 68
- reference_ref_frame_name
 - jeod::DynBodyInit, 69
- report_failure
 - jeod::DynBodyInit, 66
- required_items

- jeod::DynBodyInitPlanetDerived, 103
- reverse_sense
 - jeod::DynBodyInit, 69
- semi_latus_rectum
 - jeod::DynBodyInitOrbit, 97
- semi_major_axis
 - jeod::DynBodyInitOrbit, 97
- set
 - jeod::DynBodyInitOrbit, 97
- set_detach_from_body
 - jeod::BodyDetachSpecific, 49
- set_items
 - jeod::DynBodyInitWrtPlanet, 116
- set_lvlh_frame_object
 - jeod::DynBodyInitLvlhState, 76
- set_parent_body
 - jeod::BodyAttach, 35
- set_parent_frame
 - jeod::BodyAttach, 35
- set_subject_body
 - jeod::BodyAction, 24
- set_use_alt_pfix
 - jeod::DynBodyInitNedState, 85
- shutdown
 - jeod::BodyAction, 24
- SlrEcclIncAscnodeArgperTanom
 - jeod::DynBodyInitOrbit, 91
- SmaEcclIncAscnodeArgperManom
 - jeod::DynBodyInitOrbit, 91
- SmaEcclIncAscnodeArgperTanom
 - jeod::DynBodyInitOrbit, 91
- SmaEcclIncAscnodeArgperTimeperi
 - jeod::DynBodyInitOrbit, 91
- SmaIncAscnodeArglatRadRadvel
 - jeod::DynBodyInitOrbit, 91
- sort_grav_controls
 - jeod::DynBodyFrameSwitch, 58
- state
 - jeod::DynBodyInit, 69
- state_items
 - jeod::DynBodyInitRotState, 107
 - jeod::DynBodyInitTransState, 112
- StateItems
 - jeod::DynBodyInitRotState, 104
 - jeod::DynBodyInitTransState, 109
- subject_point_name
 - jeod::BodyAttachAligned, 40
- subscribed_frame
 - jeod::DynBodyInit, 70
- succeeded
 - jeod::BodyAttach, 36
- switch_distance
 - jeod::DynBodyFrameSwitch, 58
- switch_sense
 - jeod::DynBodyFrameSwitch, 58
- SwitchOnApproach
 - jeod::DynBodyFrameSwitch, 55
- SwitchOnDeparture
 - jeod::DynBodyFrameSwitch, 55
- SwitchSense
 - jeod::DynBodyFrameSwitch, 55
- terminate_on_error
 - jeod::BodyAction, 27
- time_periapsis
 - jeod::DynBodyInitOrbit, 98
- trace
 - jeod::BodyActionMessages, 31
- true_anomaly
 - jeod::DynBodyInitOrbit, 98
- use_alt_pfix
 - jeod::DynBodyInitNedState, 86
- validate_body_inputs
 - jeod::BodyAction, 24
- validate_name
 - jeod::BodyAction, 25
- Velocity
 - jeod::DynBodyInitTransState, 109
- velocity
 - jeod::DynBodyInit, 70